

Seeing Risk: From Judgement to Model

A computer-vision approach to expert visual safety impression and intersection crash prediction in Rotterdam

G. van Wijnen

Technische Universiteit Delft

Seeing Risk: From Judgement to Model

A computer-vision approach to expert visual safety impression and intersection crash prediction in Rotterdam

by

G. van Wijnen

to obtain the degree of Master of Science
at the Delft University of Technology,
to be defended publicly on Monday August 24, 2026 at 14:00.

Student number:	5309778	
Project duration:	February 12, 2026 – August 24, 2026	
Thesis committee:	Dr. ir. S. (Sander) van Cranenburgh, Dr. A. P. (Amir Pooyan) Afghari, Dr. F. (Francisco) Garrido-Valenzuela, L. P. (Lennert) Kuip, Ir. K. W. (Kevin Wang) Su,	TU Delft, first supervisor TU Delft, second supervisor TU Delft, third supervisor Gemeente Rotterdam, supervisor Gemeente Rotterdam, supervisor

An electronic version of this thesis is available at <http://repository.tudelft.nl/>.

Preface

Since secondary school I have had a strong interest in broad societal issues as well as in computer-science-like challenges. At the time I long doubted between Technology, Policy and Management and Computer Science, and eventually chose the former, with the Information and Communication track, hoping to keep a foot in both worlds. Now, at the end of this academic journey, I find it a fitting conclusion that this thesis has allowed me to bring together computer vision and computer-science components with a complex TPM-like problem. In that sense, this feels like it has come full circle.

My thanks go first to my supervisory committee, from both TU Delft and the Municipality of Rotterdam, who supported me throughout this entire process. Sander van Cranenburgh, my first supervisor, made an enormous contribution to this research. Our conversations were a pleasure, and alongside valuable academic advice you also regularly gave me encouragement when it was needed. Amir Pooyan Afghari, my second supervisor, meant just as much to me. You were consistently the most critical reader of my report, and it was precisely because of that that I enjoyed the discussions that followed so much. I began this project with little knowledge of road safety research, and feel that I have since been able to glimpse at least a small part of the field. Francisco Garrido-Valenzuela was my third supervisor, and supported me in a way I greatly appreciate. I enjoyed our technical discussions and your enthusiasm, and I genuinely value how much time you always made to help me.

From the Municipality of Rotterdam, Lennert Kuip and Kevin Su were likewise part of my supervisory committee, and I owe them a great deal of thanks as well. Lennert, thank you for your attention and positive guidance throughout the process. I am grateful to have had a glimpse into such a fine organisation, and especially enjoyed attending the National Road Safety Congress together with you, where I saw how much activity there is in this field. Kevin, thank you for your excellent substantive and procedural guidance. You asked at nearly every meeting how I was doing personally, not only how the research was progressing, and I greatly valued your involvement and enthusiasm for the project. Finally, thanks to all the other colleagues at the municipality who contributed in their own way, from interesting discussions to filling in my survey.

I would also like to thank all the friends I made during my time as a student. Without the people around me, I genuinely would not have been able to reach where I stand today.

Finally, a word for my family. I thank my parents for the unconditional care and love with which you raised me, throughout my entire life. You have always believed in me and given me the space to find my own way, and I realise how much that has meant to me, including during this project. Together with them, I also thank my younger brother and sister.

With this I close this preface, looking back with great pleasure on this special period. I hope you enjoy reading this thesis as much as I enjoyed writing it!

Thijs van Wijnen
Rotterdam, augustus 2026

Contents

List of Abbreviations	iv
Executive Summary	1
1 Introduction	3
1.1 Background	3
1.2 Research objective and research questions	4
1.3 Key construct and scope	4
1.4 Thesis outline	5
2 Literature review	6
2.1 Network screening and safety performance functions	6
2.2 Intersection safety: risk indicators and infrastructure characteristics	7
2.3 Visual safety assessment: attribute checklists and their configural limits	8
2.4 Expert elicitation	9
2.5 Computer vision for urban safety analysis	10
2.6 Knowledge gap	11
3 Method	13
3.1 Research approach	13
3.2 Research design overview	14
3.3 Phase 1: Intersection operationalisation (RQ1)	14
3.3.1 Population definition and geometric processing	14
3.3.2 Stratification.	16
3.3.3 Photo selection and reprojection.	16
3.4 Phase 2: Expert elicitation and scoring (RQ2)	18
3.4.1 Preparatory interview.	18
3.4.2 Pairwise comparison design and survey protocol.	18
3.4.3 Bradley-Terry model specification	20
3.4.4 Consistency and reliability assessment	21
3.5 Phase 3: Score generalisation (RQ3)	22
3.5.1 Architecture selection and transfer learning.	22
3.5.2 Aggregation and feature compression.	23
3.5.3 Training procedure and evaluation	23
3.6 Phase 4: Crash model and enrichment (RQ4)	24
3.6.1 Negative Binomial model specification	24
3.6.2 Model enrichment and nested model comparison	25
3.6.3 Robustness and measurement-error checks	26
3.7 Phase 5: Validation analysis (RQ5)	28
3.7.1 Empirical Bayes estimation and ranking comparison	28
3.7.2 Signal decomposition: traffic volume and speed entanglement	28
3.7.3 Attention map analysis	29
3.7.4 Expert validation session	30
4 Data sources and availability	31
4.1 Overview of data sources	31
4.2 Quantitative characterisation of the data sources.	32
5 Results	35
5.1 Intersection operationalisation (RQ1)	35
5.1.1 Population definition and stratification outcomes	35

5.1.2	Photo matching outcomes	35
5.2	Expert elicitation and scoring (RQ2)	36
5.2.1	Preparatory interview outcomes	36
5.2.2	Graph coverage	36
5.2.3	Bradley-Terry scores and score distribution	37
5.2.4	Score reliability and coverage	38
5.3	Score generalisation (RQ3)	38
5.3.1	Architecture comparison	38
5.3.2	Model performance on the validation set	39
5.3.3	Predicted score distribution across the full network	39
5.4	Crash model and enrichment (RQ4)	41
5.4.1	Negative Binomial baseline: model fit and variable significance	41
5.4.2	Model enrichment: likelihood-ratio test and fit statistics	41
5.4.3	Coefficient comparison and the CV-score effect	42
5.4.4	Robustness and measurement-error checks	43
5.5	Validation analysis (RQ5)	44
5.5.1	Empirical Bayes-based ranking comparison	44
5.5.2	Signal decomposition results: traffic volume entanglement and speed proxy	45
5.5.3	Explainability: CLS-to-patch attention analysis	46
5.5.4	Expert validation session	47
6	Discussion	48
6.1	Interpretation of the main finding	48
6.2	Implications for network screening practice and policy	50
6.3	Limitations and mitigations	51
6.4	Directions for future research	53
7	Conclusion	55
	AI Statement	56
	References	57
A	Junction deduplication	62
B	Approach leg photo selection	64
C	Preparatory interview	66
C.1	Interview guide	66
C.2	Thematic summary of interview outcomes	66
D	Survey task interface	68
E	Rater agreement	69
E.1	Percentage agreement	69
F	Full computer vision results	71
G	Baseline crash model, predictor selection and diagnostics	73
G.1	Candidate predictors and collinearity	73
G.2	Incremental specification	73
G.3	Residual diagnostics	75
H	Coefficient robustness and signal decomposition	77
H.1	Coefficient robustness detail	77
H.2	Signal decomposition detail	78
I	CLS attention maps	80
J	Expert validation session	82
J.1	Session guide	82
J.2	Thematic summary of session outcomes	82

List of Abbreviations

Abbreviation	Definition
AADT	Annual Average Daily Traffic
AIC	Akaike Information Criterion
BIC	Bayesian Information Criterion
BRON	Bestand geRegistreerde Ongevallen in Nederland
CI	Confidence Interval
CNN	Convolutional Neural Network
CV	Computer Vision
EB	Empirical Bayesian
FOV	Field of View
FOW	Function of Way
ILSR	Iterative Luce Spectral Ranking
iRAP	International Road Assessment Programme
IRR	Incidence Rate Ratio
LL	Log-Likelihood
LR	Likelihood Ratio
MAE	Mean Absolute Error
MSE	Mean Squared Error
NB	Negative Binomial
NDW	Nationale Databank Wegverkeersgegevens
NWB	Nationaal Wegenbestand
PCA	Principal Component Analysis
PLS	Partial Least Squares
RMSE	Root Mean Squared Error
SD	Standard Deviation
SE	Standard Error
SIMEX	Simulation-Extrapolation
SPF	Safety Performance Function
SPV	Strategisch Plan Verkeersveiligheid
SVI	Street-View Imagery
VRI	Verkeersregelinstallatie

Executive Summary

Road safety in the Netherlands has improved substantially over recent decades, yet progress has stalled, and traffic crashes continue to cause substantial annual casualties. The Strategic Plan for Road Safety 2030 (SPV 2030) sets out a shift toward risk-driven policy, asking road authorities to identify and address the most significant risks in their networks proactively, rather than reacting to locations where crashes have already occurred. Intersections are central to this shift. Roughly one third of all road fatalities in the Netherlands occur at intersections, rising to nearly half within built-up areas. For a densely populated municipality such as Rotterdam, reliably identifying dangerous intersections in advance allows limited infrastructure resources to be allocated where they yield the greatest safety benefit. An unreliable prioritisation model, conversely, both leaves genuine risk unaddressed and diverts scarce engineering capacity to locations that only appear hazardous by chance.

Standard network screening models are constrained by what is systematically recorded in national and municipal databases, such as crash records, traffic volumes, and infrastructure characteristics like those held in the NWB. The traffic safety model the Municipality of Rotterdam operates is a case in point, since none of its predictors encode the visual or spatial qualities of an intersection's layout. Rotterdam's traffic engineers do incorporate this visual dimension when they judge a location in the field, assessing whether conflict points are well-separated, whether sightlines are adequate, and whether the design communicates safe behaviour. This judgement is not currently part of the model that drives citywide prioritisation. Beyond this practical relevance, the question is also of academic interest, since integrated analyses of expert visual judgement in road safety modelling remain scarce, and few datasets link intersections to a holistic, expert-derived visual safety impression. It remained unknown whether this visual information, once systematically captured, carries crash-relevant signal beyond what conventional infrastructure and traffic variables already encode.

The objective of this study is to understand whether expert visual knowledge of intersection safety contributes to explaining intersection safety once made explicit and tested empirically alongside conventional infrastructure and traffic variables. The central research question asks to what extent incorporating expert-assessed visual safety information improves crash prediction at Rotterdam intersections, compared to a model based exclusively on conventional variables. Answering this required building the predictor itself, since no prior study had produced it in usable form: intersections first had to be operationalised as comparable units, the expert visual impression elicited as a quantified score, that score generalised across the full network, its incremental value tested in a crash model, and the result validated against established traffic engineering knowledge and practice.

The study followed a five-phase measurement chain, each phase producing the input the next required. The Rotterdam intersection population was first operationalised from the NWB and stratified by junction type and traffic control regime, yielding a citywide population of 4.715 intersections, most of which were represented by a single standardised street-level photograph. Twenty-six traffic professionals then compared 145 of these intersections pairwise on assessed design risk, alongside a traffic volume indicator so raters could weigh design separately from exposure; the resulting 678 comparisons were converted into continuous Bradley-Terry safety scores. A computer vision model, comparing four candidate backbone architectures and ultimately combining a DINOv2-Small backbone with principal component analysis and Ridge regression, was trained on these scores to generalise them across the network, reaching a modest validation fit ($R^2 = 0.33$). This CV-derived score was then added as a predictor to a Negative Binomial safety performance function fitted network-wide, alongside the conventional infrastructure and traffic variables, with measurement-error correction applied to account for the score's own prediction uncertainty. A final validation phase compared this result against established traffic engineering knowledge and network-wide ranking practice.

Adding the CV-derived score does not significantly improve crash prediction. Under the primary specification, its coefficient corresponds to an incidence rate ratio of 0.593 for predicted crash counts, im-

precise and short of statistical significance ($p = 0.083$). AIC improves only marginally, below the conventional threshold for a meaningful difference, while BIC favours the baseline model. This result held under nearly all robustness checks. Accounting for weak spatial dependence between nearby intersections widened the coefficient's standard error without shifting the estimate, and re-expressing the score as a rank or a binary indicator weakened it further. The same absence of incremental value also appeared when the raw human judgements, rather than the CV-generalised score, were tested directly on the 145 expert-scored intersections. Excluding major-road traffic volume as a diagnostic reversed the coefficient's sign to positive, though still short of significance, pointing to traffic volume as the specific source of the null result explored below.

This absence of value in the raw human judgements points to the underlying expert construct as the source of the limitation, not the computer vision step. The CV score correlates strongly with traffic volume ($\rho \approx 0.64$), but once traffic volume is accounted for in the crash model, its association with crashes drops to near zero. Two mechanisms could explain this pattern. Danger and traffic volume are visually intertwined in the photograph itself, so a rater judging how dangerous an intersection looks cannot help but also register how busy it is, and the survey's traffic volume indicator, intended to help raters set exposure aside, did not fully separate the two. Bradley-Terry aggregation could compound this. Reducing every pairwise judgement to a single scalar per intersection means a coarse busy-versus-quiet contrast dominates, and any finer, volume-independent signal is largely absorbed. This entanglement was visible even in the model's attention patterns. At high-scoring, dangerous-looking intersections, attention concentrated on the open road toward the horizon. At low-scoring intersections, it fell instead on building facades, roadside vegetation, and parked vehicles, markers of a visually enclosed, lower-volume setting. Taken together, these patterns suggest that an integrated visual impression functions less as an additional judgement of intersection design and more as an intuitive readout of the same traffic volume and speed cues the conventional model already captures.

For Rotterdam, this finding gives clear but bounded direction for safety investment. As operationalised and tested here, the holistic visual construct does not significantly improve the conventional crash-and-exposure-based model, because it already strongly overlaps with the conventional variables that model relies on. On this evidence, a citywide CV-based scoring layer built on this same construct is not the most effective next investment for ranking purposes, and scarce safety budget is better directed at those existing indicators instead. This is not a general verdict on whether visual information can add value. It reflects what this specific construct and this specific test showed, not a general claim that photographs or visual assessment cannot contribute under a different operationalisation.

Beyond ranking, the pipeline and the scores it produced deliver value in their own right. The predicted safety scores are, in themselves, a new visually grounded data source that can support internal municipal discussion and communication. More speculatively, because the CV score is computed from imagery alone and correlates strongly with traffic volume, it may serve as a possible proxy for traffic exposure in settings where volume data is unavailable, such as low- and middle-income contexts lacking systematic traffic registration. Within Rotterdam itself, it could also serve as an interim risk indicator for newly built or recently modified intersections, where the data the conventional model requires is not yet available. This is a use this thesis identifies but does not test, and it would require dedicated validation against measured traffic volume before informing practice.

Methodologically, the study contributes a replicable end-to-end pipeline for testing whether an expert visual construct adds incremental predictive value over a conventional crash model. This pipeline serves as a template other researchers can apply to evaluate similar visual predictors against a genuine baseline. A further, more technical contribution is the intersection-consolidation methodology developed to define the population, reusable by any study that needs a clean, one-node-per-intersection population from road network data.

Several limitations qualify these findings. The direct construct test on the 145 expert-scored intersections was underpowered, so its null result is not strong evidence that no effect exists, it may simply reflect insufficient power to detect one. Visual danger and traffic volume are inherently intertwined in a photograph, so the null result should be read in that light rather than as ruling out the underlying construct. The analysis is further limited to Rotterdam's intra-ring network, and no claim is made about transferability to other cities or network types.

Introduction

1.1. Background

Road safety in the Netherlands improved substantially over recent decades, yet progress has stalled (SWOV, 2020, 2023). Each year, road traffic crashes continue to result in substantial annual casualties. Within this context, the Strategic Plan for Road Safety 2030 (SPV 2030) sets out a shared course for road authorities and other traffic professionals to drastically reduce the number of road casualties (Ministerie van Infrastructuur en Waterstaat, 2018). Central to this ambition is a shift towards risk-driven policy. This policy prevents crashes by proactively addressing the most significant risks in the traffic system, rather than reacting to locations where crashes have already occurred (Ministerie van Infrastructuur en Waterstaat, 2018). Concretely, this means that road authorities are expected to conduct risk analyses of their own road networks and translate these, together with regional partners, into joint implementation programmes with concrete measures to address the most important risks (Kennisnetwerk SPV, 2023a). This policy shift is consequential. It asks all road authorities to invest in tools and datasets that allow them to act on predicted risk, not only on observed harm.

Intersections occupy a central position in this shift. Approximately one third of all road fatalities in the Netherlands occur at intersections, and within built-up areas this share rises to nearly half (Kennisnetwerk SPV, 2023b). For a densely populated urban municipality such as Rotterdam, this concentration of risk at a limited number of locations represents both a structural safety problem and a concrete policy opportunity. If dangerous intersections can be identified reliably and in advance, the limited public resources available for infrastructure investment can be allocated where they yield the greatest safety benefit. Conversely, if the prioritisation model is unreliable, two distinct failures occur: genuinely dangerous intersections go undetected and remain untreated, while locations that appear statistically hazardous by chance absorb scarce engineering capacity that could have been directed elsewhere. Both errors carry real costs, one in unaddressed risk, the other in wasted resources, and reducing them simultaneously requires a model that captures as much safety-relevant information as possible.

The municipality of Rotterdam operates a traffic safety model that classifies locations into risk categories based on recorded injury crashes, traffic volumes, and infrastructure characteristics drawn from systematic registrations such as the NWB (Nationaal Wegenbestand) (Gemeente Rotterdam, 2019). The model achieves good statistical accuracy, yet none of its predictors encode the visual or spatial qualities of an intersection's layout. In practice, Rotterdam's traffic engineers do incorporate this visual dimension. When they visit a location, they judge whether conflict points are well-separated, whether sightlines are adequate, and whether the layout communicates safe behaviour to road users. This assessment is forward-looking, identifying risk embedded in the physical design regardless of whether it has already materialised in recorded crashes. This judgement, however, is not incorporated into the model that drives network-wide prioritisation.

Beyond its policy relevance, the topic is also of academic interest. Few datasets link intersections to a holistic, expert-derived visual safety impression, and integrated analyses of expert visual judgement in road safety modelling remain scarce in an urban setting. Addressing this requires both a structured way

to incorporate expert visual assessment into quantitative crash prediction and substantive evidence on whether street-level features carry crash-relevant information in such a setting.

It remains unknown, however, whether this visual information, once systematically captured, carries crash-relevant signal beyond what conventional infrastructure and traffic variables already encode, or whether it largely duplicates what those variables already contain. Rotterdam, with an established intersection risk model and an active black spot practice, offers a representative setting in which this question is concrete and policy-relevant.

1.2. Research objective and research questions

The objective of this study is to understand whether expert visual knowledge of intersection safety contributes to explaining intersection safety once made explicit and, alongside conventional infrastructure and traffic variables, tested empirically.

To address this objective, the study poses the following central research question:

To what extent does incorporating expert-assessed visual safety information improve crash prediction at intersections in Rotterdam, compared to a model based exclusively on conventional infrastructure and traffic variables?

To answer this question in a structured manner, five sub-questions guide the research process, each producing the input for the next:

RQ1: *How can road intersections be spatially defined and visually represented as consistent, comparable units for expert-based safety assessment?*

RQ2: *How can expert safety judgements about road intersections be elicited and converted into a continuous, comparable safety score?*

RQ3: *To what extent can the elicited safety score be generalised to unlabelled intersections across the full urban network?*

RQ4: *To what extent does the addition of the generalised safety score improve a crash model for Rotterdam intersections based on conventional infrastructure and traffic variables?*

RQ5: *To what extent are the generalised safety scores coherent with established traffic engineering knowledge, and to what extent does incorporating them into the crash model change network-wide prioritisation?*

1.3. Key construct and scope

This study operationalises a central construct that requires definition before the scope is specified. Throughout this thesis, *visual safety impression* refers to the holistic judgement that a trained traffic professional forms when viewing an intersection's physical layout. This is the same layout that the recorded attributes of Section 2.2 also describe. It captures an overall sense of whether the design communicates clear priority relationships and adequate sightlines, and whether its spatial organisation reads as legible overall. This impression is conceptually distinct from *subjective safety perception*, a road user's personal, emotional experience of a location that correlates only weakly with objective safety and that road safety measures frequently affect in different, sometimes opposite, directions (Sørensen & Mosslemi, 2009; SWOV, 2012), and from *objective safety*, operationalised in this study as the observed crash count at an intersection, derived from historical BRON (Bestand geRegistreerde Ongevallen in Nederland) records.

The study covers intersections within the Rotterdam municipal road network where at least two conflicting traffic streams, involving motorised vehicles, cyclists, or pedestrians, converge. The scope is limited to locations that function as junction points in the urban traffic network: T-junctions, four-leg intersections, and comparable configurations, whether signalised or otherwise-regulated. Locations operating under a fundamentally different conflict geometry or traffic function were excluded. Roundabouts were excluded because their design principle separates conflicts rather than managing them at a shared point; highway interchanges were excluded because their operating speeds and design standards are not comparable to urban intersections; and dedicated cyclist-only or pedestrian-only

crossings were excluded because they involve a structurally different and less severe conflict pattern. The geographic scope was restricted to intersections within the Rotterdam ring road, which ensured internal homogeneity of urban morphology and design standards.

1.4. Thesis outline

The remainder of this thesis is structured as follows: Chapter 2 reviews the literature across five thematic strands: network screening and SPFs, intersection safety and risk indicators, Visual safety assessment: attribute checklists and their configural limits, expert elicitation, and computer vision for urban safety analysis, followed by the knowledge gap this study addresses. Chapter 3 presents the research design across five sequential phases. Chapter 4 describes the data sources and their availability. Chapter 5 reports the results for each research question. Chapter 6 discusses the findings, their implications, limitations, and directions for future research. Chapter 7 concludes the thesis.

Literature review

2.1. Network screening and safety performance functions

Network screening is the process by which road authorities review a network to identify and prioritise the locations most likely to benefit from a safety intervention (American Association of State Highway and Transportation Officials, 2010; Odijk & Van Petegem, 2024). Within that process, detecting the specific high-risk locations, historically termed hotspot or black spot identification, is a well-established methodological task (Cheng & Washington, 2005). Washington et al. (2018) group the available detection methods into six broad categories: naïve methods based on raw crash counts, safety performance functions (SPFs) estimating expected crash frequencies, Empirical Bayesian and full Bayesian methods correcting for regression to the mean, severity-based approaches, spatial methods, and Continuous Risk Profile approaches. The standard modelling instrument for the most rigorous of these approaches is the SPF, a regression model that relates expected crash frequency to explanatory variables such as traffic volume and road characteristics (American Association of State Highway and Transportation Officials, 2010). In the Dutch and broader European literature, including SWOV's own usage, the same type of model is sometimes called a crash prediction model (CPM), a related but broader term under which the SPF is a narrower case restricted to a single standardised road profile, whose deviations are corrected through a separate crash modification factor (CMF) and calibration factor (Caliendo et al., 2007; Odijk & Van Petegem, 2024; SWOV, 2025). The SPF used in this thesis instead enters road and traffic characteristics directly as covariates, without this separate CMF layer. Because crash counts in aggregate exhibit overdispersion relative to the Poisson distribution, the Negative Binomial (NB) regression model has become the default SPF specification (Lord & Mannering, 2010; Washington et al., 2018). The SPF is thus the statistical instrument through which network screening is operationalised. Identifying high-risk locations in practice means estimating an expected crash frequency from an SPF and comparing it against observed counts.

In practice, the NB model is most commonly combined with an Empirical Bayesian (EB) procedure, which corrects for regression to the mean by weighting observed counts against model predictions and substantially outperforms simpler ranking methods (Ghadi & Török, 2019; Montella, 2010; Persaud & Lyon, 2007; Washington et al., 2018). The quality of the EB estimate depends directly on the quality of the underlying SPF, and therefore on the completeness of the predictor set. Lord and Mannering (2010) identify omitted-variable bias as one of the most persistent challenges in crash-frequency modelling. When safety-relevant site characteristics are absent, the SPF misattributes their effect to the error term or to correlated included variables.

The predictor set of a standard NB model is constrained by what is systematically recorded in national databases. In the Dutch context, this means traffic volumes from the NWB (Nationaal Wegenbestand) and geometric and regulatory characteristics from municipal and national registrations. The visual and spatial qualities of intersections at street level, qualities that experienced practitioners integrate effortlessly into their safety judgements, remain absent from these databases. Intini et al. (2021) provide direct empirical evidence for the resulting omission. Adding visually inspected variables, such as

pavement condition and sight distance, to NB models for urban intersections in Bari, Italy that already included standard traffic, geometric, and control variables, produced statistically significant improvements confirmed through likelihood-ratio tests against the same models without these variables, indicating that visually assessed characteristics carry explanatory power that standard databases do not capture.

A second, distinct limitation concerns the dependent variable rather than the predictors. Network screening models are fundamentally reactive. They rely on historical crash records, and "proactive road safety" refers to acting on model outputs before the next crash occurs at a location, not to the model capturing risk independently of crash history (Washington et al., 2018). The recorded crash data on which these models depend suffer from two distinct measurement biases, underreporting of single-vehicle and cyclist crashes and spatial imprecision in geocoding, compounded by a third and structurally different problem, regression to the mean, which arises not from the crash count itself but from selecting sites on the basis of a short observation period (Lord & Mannering, 2010). This limitation bounds what any SPF can achieve regardless of its predictor set, and constrains the interpretation of all crash-based ranking. The incomplete predictor set connects to the expert assessment literature reviewed later in this chapter. Before turning to that literature, the next section first examines which infrastructure and risk-indicator characteristics an SPF's predictor set should include in the first place.

2.2. Intersection safety: risk indicators and infrastructure characteristics

An intersection is not a self-evident unit in digitised road network data. Divided carriageways are commonly stored as parallel segments, so a single physical junction registers as multiple graph nodes, and administrative boundaries introduce further splits that have no counterpart in the physical road. Boeing (2017) addressed this discrepancy in OSMnx, an open-source toolkit for street network analysis, by algorithmically correcting network topology, removing nodes that represent neither a genuine junction nor a dead end and consolidating the resulting edges into a simplified graph. The problem proved more persistent than a single fix could resolve. Boeing (2025) later named it explicitly as the street intersection miscount problem and proposed refined graph simplification solutions. Fleischmann et al. (2026) independently developed an adaptive algorithm built around the detection of topological artefacts, arguing that existing tools still fall short of a fully automated, reproducible procedure. Read together, this sequence of contributions suggests that translating raw network data into a defensible set of intersections is an unsettled methodological question in its own right, not a preliminary step that can be assumed away.

Which characteristics of an intersection, once defined, carry predictive weight is a separate question. Khattak et al. (2021) compared several functional forms of the negative binomial model against a generalised Poisson specification for urban intersections, and found that both model fit and the relative importance of predictors shifted depending on whether an intersection was signalised, treating control type as a substantive distinction rather than a matter of data organisation. Elvik et al. (2009) reach a comparable conclusion from an entirely different angle in their synthesis of more than a century of evaluation studies. Converting a four-leg intersection into two three-leg junctions is catalogued as a measure with a quantified, positive safety effect, precisely because it reduces the number of conflict points a driver must resolve. Janssen (2004) grounds this pattern in the Dutch network specifically. The study compares crash rates per million motor vehicles across three-leg and four-leg junctions under priority-to-right, special priority regulation, and signal control. These categories remain distinguishable once traffic volume is taken into account. The convergence of an international meta-analysis, a model-comparison study, and a national empirical study on the same two characteristics, leg count and control regime, suggests that their status as risk indicators is not an artefact of any single methodology.

Speed also affects intersection risk, through a mechanism distinct from the discrete design features discussed above. L. Aarts and van Schagen (2006) reviewed the relationship between driving speed and crash involvement at both the aggregate and the individual level, concluding that higher speeds are associated with a greater likelihood and severity of crashes independently of road layout. Elvik et al. (2019) updated this evidence base with a meta-analysis restricted to studies published after 2000, and found that the Power Model and an Exponential Model of the speed-safety relationship both remain

well supported, with the strength of the relationship largely unchanged despite improvements in vehicle safety technology over the intervening decades. Because this mechanism, impact energy and the time available to detect and react to a hazard, operates independently of an intersection's geometry or right-of-way regime, speed constitutes a risk dimension of its own rather than a proxy for the characteristics already discussed.

These internationally derived indicators have a direct counterpart in Dutch road safety policy. The Kennisnetwerk SPV (2023a) factsheet on risk indicators groups intersection type, speed, and priority regulation among the recognised indicators for risk-driven safety monitoring, and the Kennisnetwerk SPV (2023b) definition of a sufficiently safe intersection is constructed from the same characteristics. L. T. Aarts (2018) traces how this indicator set emerged from a broader Safety Performance Indicator movement, one that reframed Dutch road safety governance around measurable, design-based risk factors rather than retrospective crash counts alone. The same characteristics that international crash-frequency research identifies as statistically significant predictors thus also function, within the Netherlands, as the categories through which road authorities are expected to monitor and act on risk.

These indicators are necessary but not sufficient. They describe individual risk factors rather than how those factors combine into an intersection's overall impression. The literature on expert visual assessment, reviewed in the following section, addresses precisely this composite dimension, providing both the empirical foundation and the theoretical rationale for treating it as a distinct construct.

2.3. Visual safety assessment: attribute checklists and their configurational limits

Where human expertise is applied systematically to visual road safety assessment, it takes the form of checklist-based protocols. The International Road Assessment Programme (iRAP) methodology uses trained coders to score attributes such as lane markings, sight distance restriction and traffic control type from photographic or video surveys into a star rating (iRAP, 2013; Kačan et al., 2020), and instruments including EuroRAP, ProMeV and CycleRAP apply comparable logic to particular road types or user groups (Bax et al., 2017; iRAP, 2023, 2025). Ratings have been produced in more than 140 countries, and their relation to crash outcomes has been tested against accident costs per kilometre in Australia (McInerney & Fletcher, 2013) and with an empirical Bayes approach on the Hungarian rural network (Ambros et al., 2017). Road safety audits draw more openly on the experience of the audit team, but they too examine a road against a structured list of concerns (Asian Development Bank, 2022). Section 2.2 asked which attributes belong in a statistical model. This section asks how human judgement enters the assessment of them, and in both traditions it enters attribute by attribute. This section examines what such decomposition leaves out, and why an integrated judgement might capture it.

This question, whether decomposed or integrated judgement better captures road safety, is not new to the field. Hammond et al. (1987) had experienced highway engineers judge road sections against an accident rate criterion, either as an immediate overall impression or as a checklist of cues the engineers themselves had identified as essential. That the cues themselves originated from the engineers' own professional judgement is telling, since the checklist condition did not draw on some independent, external measure of the road, but on the same expertise, decomposed into separate items. The road sections, the assessors and the underlying cues stayed the same. Only the judgement procedure varied, an immediate overall impression in one condition against explicitly scoring the same cues one by one in the other. Intuitive judgement matched or outperformed the checklist condition in empirical accuracy. Shbeeb (2022) finds a similar pattern more recently. A 184-element checklist applied to 105 streets in Amman predicted crashes poorly when its elements were weighted equally, and only produced valid estimates, at a reported correlation of 69 per cent, once expert judgement of each element's relative safety impact was introduced.

Why weighting and combination would make such a difference is not obvious from these two studies alone, but cognitive science offers a general reason. A list and a diagram can contain exactly the same information, yet still not be equally easy to reason with, because a diagram lays that information out by location while a list lays it out in sequence (Larkin & Simon, 1987). A diagram can therefore show a relation directly, where a list only implies it and leaves the reader to reconstruct it. Vision research calls

such a relation configural. It is a property of how elements relate to each other, not of the elements themselves, and one that is often easier to spot as a whole than by inspecting each element in turn (Pomerantz & Portillo, 2011). Road safety research has long recognised a domain-specific version of this. The self-explaining roads concept holds that a layout should allow road users to infer immediately how to behave and what to expect (Theeuwes & Godthelp, 1995), a proposition since grounded in statistical learning and subjective road categorisation (Theeuwes, 2021) and confirmed empirically in urban settings, where drivers learn to expect a given speed from the visual configuration of the road rather than from posted limits (Charlton & Starkey, 2017). Categorisation and expectation are properties of the whole. A layout either matches a learned category or it does not, and no single attribute value settles this. What this strand identifies is a property of the layout, legibility, not a prediction of behaviour. Counting or scoring features individually does not straightforwardly indicate risk, and a specification that enters such counts as separate additive terms does not represent the relations that do.

Sight obstruction is the case where the limits of decomposition can be stated precisely. The iRAP methodology codes sight distance restriction explicitly, alongside intersection type, intersection quality and intersection channelisation, and its Star Rating Score combines risk factors multiplicatively rather than treating them as independent (iRAP, 2013). The constraint lies elsewhere. Coding assigns a categorical value to a road segment, whereas the adequacy of a sight line is a relation between the position of an approaching road user, the object obstructing the view, and the path of the conflicting stream. A single urban intersection contains many such relations, one for each pairing of approach and conflicting movement. An obstruction immaterial to one movement may conceal exactly the stream another must yield to, and LiDAR-based visibility assessments confirm that blockage percentages differ substantially between approaches at the same intersection (Kilani et al., 2021). A segment-level attribute does not distinguish these cases. This reflects the resolution at which network-wide screening is designed to operate rather than an oversight in the instruments themselves.

Taken together, these strands indicate what attribute-based scoring leaves unrepresented. It is not further attributes, but the configural relations between them, and the sight relations that segment-level coding does not express. Both are claims about what a representational format can carry, how completely a list-based or segment-level format can express relations that already exist in the physical layout, not about what a human assessor additionally perceives. Two questions follow. Section 2.4 addresses how such judgement can be elicited reliably and at the scale a predictive model requires. Section 2.6 addresses the second. The studies reviewed above establish predictive validity for visual assessment, but typically without conditioning on the kind of full traffic-exposure baseline discussed there.

2.4. Expert elicitation

Expert elicitation is the field concerned with extracting and formalising professional judgement into a structured, quantifiable form suitable for downstream modelling. Cooke (1991) established the methodological foundations of the field, arguing that expert judgement constitutes a legitimate scientific knowledge source only when the elicitation satisfies a set of principled conditions: the procedure must be structured and reproducible, neutral with respect to the outcome, and open to peer scrutiny. This ensures that elicited values reflect the target construct rather than individual raters' personal biases or the framing of the task. The target construct these conditions aim to capture is often a form of tacit pattern recognition. Klein (1998) shows that experts in safety-critical domains develop reliable pattern recognition from perceptual cues anchored in domain knowledge. Colson and Cooke (2018) review three decades of applications across domains including nuclear safety, climate science, and public health, documenting how these principles were operationalised and validated as the approach spread from its origins in risk analysis. A recurring theme across this literature is that the appropriate elicitation format depends on the nature of the target construct: single measurable quantities, multi-dimensional constructs, and holistic impressions each impose different demands on how judgement can reliably be captured (O'Hagan, 2019). The visual safety impression of an intersection falls into the last of these categories, which frames the elicitation-format question addressed in the remainder of this section.

Within this field, two elicitation formats dominate the collection of judgements on continuous constructs: direct rating on a numerical scale and comparative judgement through pairwise comparison. Harpe (2015) documents the principal weakness of the first. Direct rating on Likert-type scales produces

systematic differences across raters in both scale use and discriminability. These differences fail to average out in small panels, and a single outlier can dominate the aggregate. The comparative alternative traces to Thurstone (1927), whose law of comparative judgement established that people are more reliable and consistent when comparing two items than when assigning absolute scores, and who showed that such pairwise data can be scaled to an interval measure of a latent construct. Pollitt (2012) revived and extended this principle for modern assessment, arguing that comparative judgement outperforms direct estimation precisely when the construct is continuous and multi-dimensional, because the relative judgement ("A is safer than B") is cognitively simpler and removes the need for a shared numerical reference frame. Verhavert et al. (2019) substantiate this claim at scale, reporting in a meta-analysis across a broad range of assessment contexts that comparative judgement achieves consistently high reliability, and Phelps et al. (2015) demonstrate the same advantage specifically for image-based tasks, where raters cannot easily maintain a common numerical anchor. This convergence of evidence identifies comparative judgement as the better-suited format for eliciting holistic, multi-dimensional constructs.

The Bradley-Terry model (Bradley & Terry, 1952) provides the statistical foundation for converting pairwise comparison data into continuous scores. It is the pairwise case of the broader Plackett-Luce family, itself grounded in Luce's choice axiom (Luce, 1977), whose independence from irrelevant alternatives property holds that the relative odds of choosing one item over another depend only on those two items and not on the other items in the comparison set. This property places items compared against different opponents on a common latent scale, which is what makes pairwise comparisons from an incomplete set of pairings scalable to a single interval measure. The model estimates a latent strength parameter λ_i for each item i , where the probability that item i is preferred over item j is modelled as $P(i > j) = \lambda_i / (\lambda_i + \lambda_j)$. The structural analogy to stated choice experiments is direct. Each pairwise comparison reveals an ordinal preference between two alternatives, from which interval-scale parameters are estimated (Thurstone, 1927). In the urban perception literature, the MIT Place Pulse project (Salesses et al., 2013) demonstrated that pairwise comparison produces robust and replicable perception scores across large samples, and subsequent work trained computer vision (CV) models directly on these scores (Dubey et al., 2016; Naik et al., 2014).

2.5. Computer vision for urban safety analysis

Computer vision models automatically learn hierarchical visual representations and can be adapted to smaller datasets through transfer learning (He et al., 2016; Tan & Le, 2019). The application of street-view imagery (SVI) to road safety research has matured rapidly, as documented by two recent systematic reviews: Ye et al. (2025) catalogue computer vision models used in crash and safety studies across a broad range of geographical contexts, while Zhang et al. (2024) frame the broader paradigm of urban visual intelligence, demonstrating that SVI enables large-scale safety inference that previously required costly field surveys. Both reviews draw predominantly on peer-reviewed empirical studies published between 2015 and 2024, with a concentration of methodologically mature work appearing after 2019 coinciding with the widespread availability of large-scale pre-trained vision models.

A first cluster of studies establishes that visual features extracted from SVI carry crash-predictive information at various spatial scales. These studies predominantly decompose the street scene into discrete segmented features, and relate those individual elements to crash outcomes, rather than modelling the holistic impression a scene conveys as a whole. Yao et al. (2019) link Street View features to urban perception scores, Li and Chen (2025) predict car crash frequencies from street-level imagery, and Perez and Fusco (2025) show that AI can convert street images into numerical safety indicators. At the intersection level specifically, Yue (2024) applies semantic segmentation and object detection to predict pedestrian crashes, Kwon and Cho (2020) show that CV-derived features are significantly associated with perceived crash risk, and Han et al. (2025) integrate SVI features into a random-parameters Negative Binomial-Lindley intersection model, reporting improvement over a baseline of traffic volume and geometry. Chen et al. (2025) extend this segmentation-based approach with causal inference, finding that features such as visual openness causally raise crash risk through higher speeds, while noting that the absence of fine-grained design and speed controls remains a principal limitation of such work. A recurring property of this cluster is that the visual features are often evaluated on their own predictive merit, and in several cases without a conventional baseline of infrastructure and traffic variables against

which their added value could be judged.

A second cluster shows that safety judgements extracted from imagery, whether by trained human raters or by language models, align with predefined categorisations or human annotations, a methodological precedent for extracting safety information from images rather than a validation against crash outcomes. Hamim and Ukkusuri (2024) find that perceived safety, rated from street-view imagery, differs significantly between locations pre-classified as crash-prone and non-crash-prone. Hwang et al. (2024) demonstrate that large multimodal models can generate safety scores and natural language scene descriptions for street-crossing contexts, using visual and textual input to support structured safety judgements aligned with human annotations.

The direct methodological precedents for the convolutional neural network (CNN)-based safety score generalisation in this study are Kačan et al. (2020), who trained a multi-task CNN on 52 iRAP attributes using expert-annotated road video, and Kačan et al. (2024), who extended this with dynamic loss weighting and sequential processing via Bi-LSTM, confirming the maturity of the approach. This study builds on Kačan et al. (2020) by modelling a holistic safety judgement rather than discrete attributes, and by linking the resulting score to an SPF. The integration follows the architectural principle proposed by Van Cranenburgh and Garrido-Valenzuela (2025), in which CNN-extracted image features enter a statistical model as a predictor while preserving the interpretability of the tabular coefficients.

With the individual literature strands reviewed, the following section draws them together to articulate the knowledge gap this study addressed.

2.6. Knowledge gap

The literature reviewed in the preceding sections reveals three fields that are each well developed in isolation. The road safety literature provides robust methods for crash prediction at intersections, with Negative Binomial regression and Empirical Bayesian estimation as the methodological standard. The computer vision literature demonstrates that CV models can effectively learn human perceptual judgements about urban environments from pairwise comparison data and generalise them at network scale. The expert elicitation literature establishes that pairwise comparison in combination with Bradley-Terry estimation yields reliable, interval-scale assessments of multi-dimensional constructs, including visual safety, that can serve as training labels for downstream models.

Combining these three elements would let an SPF draw directly on a professional visual judgement of intersection safety, alongside the infrastructure and traffic variables it already contains, and would show whether that judgement adds explanatory power once the conventional infrastructure and traffic variables are already in the model. No existing study combines all three elements: holistic expert visual safety scores collected via pairwise comparison and Bradley-Terry estimation, a CV-model trained to generalise those scores to unlabelled intersections, and the introduction of that resulting score as an additional predictor in a NB SPF. Table 2.1 positions the precedents against these three elements.

Table 2.1: Positioning of the present study against the precedents

Study	Holistic expert visual score	CV-model generalising the score to unlabelled intersections	Score introduced as an additional predictor in NB SPF
Kačan et al. (2020, 2024)	Partial	Yes	No
Han et al. (2025)	No	Yes	Yes
Salesses et al. (2013); Dubey et al. (2016) and Naik et al. (2014); Yao et al. (2019)	Partial	Yes	No
Lord and Mannering (2010); Washington et al. (2018)	No	No	No
This study	Yes	Yes	Yes

As the table shows, each precedent covers only part of the required combination: the strongest CV precedents do not rest on holistic expert elicitation, the strongest elicitation precedents are not linked to crash prediction, and the crash-frequency literature doesn't include the CV-score. None of them therefore tests whether such a judgement can sharpen an SPF beyond its existing predictors.

3

Method

3.1. Research approach

Answering the central question requires building the predictor it concerns, the expert visual impression of intersection safety, since no prior study has produced it in usable form. The visual impression is not directly observable, it is not recorded at scale, and it has no representation that a crash model can ingest. Establishing its incremental value therefore requires four things to be constructed in sequence before any test is possible: the intersection itself must first be operationalised as a standardised, comparable unit of analysis, the resulting impression must be elicited as a quantified score, that score must be generalised to every intersection in the network, and only then can it be entered alongside the conventional variables in a crash model. The research design takes the form of a measurement chain for exactly this reason, with each phase producing the input the next phase requires.

Each phase in the chain addresses what the previous one leaves open. Expert elicitation converts a holistic professional judgement into a continuous safety score, but only for the limited set of intersections that experts can realistically compare. A computer vision model generalises that score to every intersection in the network, while inheriting the noise the elicited labels may carry. The Negative Binomial crash model then tests whether the resulting score, available network-wide, adds predictive value over the conventional infrastructure and traffic variables. The central question becomes answerable at this final step, where the visual signal is weighed against the established baseline rather than on its own.

This final step defines the character of the test. The question is evaluative. It concerns the added predictive value of a new predictor type relative to an existing baseline, not whether visual features correlate with crashes in isolation. Entering the visual score alongside the full conventional baseline tests for incremental value, separating a genuine contribution from an association the standard variables already capture. Figure 3.1 illustrates this comparison conceptually. The five phases that follow operationalise this chain in sequence, from defining the unit of analysis through to the incremental crash-model test.

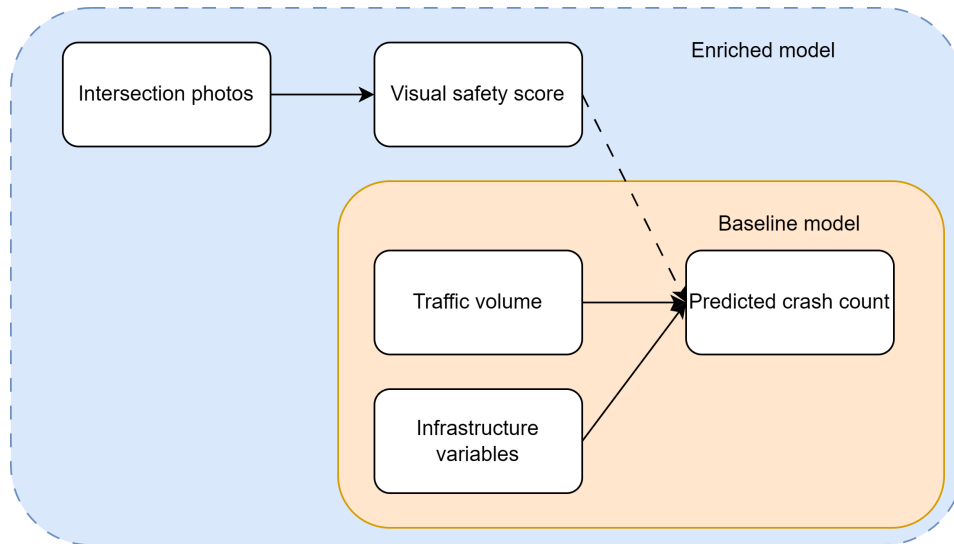


Figure 3.1: Conceptual illustration of the study's central test. The baseline model predicts crash count from individually coded attributes such as traffic volume and junction type; the enriched model adds the CV-derived visual safety score, capturing the configural relations between those same elements.

3.2. Research design overview

As stated above, the study was structured around five sequential phases, each of which produced the input for the next. Figure 3.2 presents the overall design: data sources enter each phase on the left, while the right-hand boxes name the phase, its associated research question, and the methods applied, with the artefact passed downstream labelled on each connecting arrow. The data sources, methods and analytical procedures are detailed per phase below.

3.3. Phase 1: Intersection operationalisation (RQ1)

3.3.1. Population definition and geometric processing

The initial junction population was derived from the NWB and its junction layer, in the coordinate reference system EPSG:28992 (RD New). Consistent with the scope defined in Section 1.3, the spatial extent covers all junctions within the Rotterdam ring road, and roundabouts and highway interchanges were excluded on the same grounds. Junctions were further filtered on road type to retain only motorised-traffic-carrying roads, which excludes junctions on dedicated cycle paths, footpaths, and pedestrian zones. Junctions with fewer than three connected street segments were additionally excluded, as these represent geometric artefacts rather than genuine multi-approach junctions. The ring-road boundary also aligns with the geographic coverage of the Cyclomedia street-level imagery secured through the municipal licence, a further practical consideration in fixing this boundary.

Two geometric cleaning steps were then applied, because standard road network databases encode some junctions as multiple topology nodes at a single physical conflict point, which inflates the count if left unresolved (Boeing, 2025). First, dual-carriageway junctions within 25 metres of each other were merged into a single node, as these represent a single physical intersection encoded as multiple topology nodes due to the separate storage of each carriageway direction. Second, minor connector stubs shorter than 6 metres were removed, as these represent geometric artefacts rather than genuine approach legs. The 25-metre threshold was determined empirically after testing a range of values for consolidating the intersections. The 6-metre threshold follows the NDW specification that roads less than 6 metres apart overlap (CROW, n.d.; NDW, n.d.). Figure 3.3 illustrates the merging procedure for one cluster from the Rotterdam network. Appendix A further illustrates the effect of both procedures on example locations from the Rotterdam network.

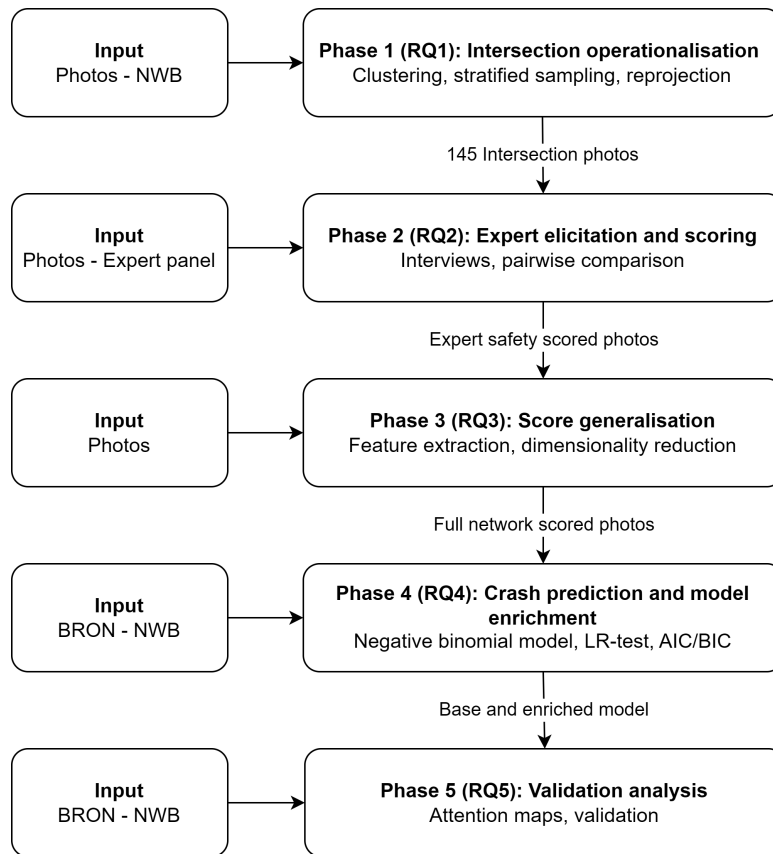


Figure 3.2: Research design overview

Note. NWB = Nationaal Wegenbestand (national road network register); BRON = Bestand geRegistreerde Ongevallen in Nederland (national crash registration database); RQ = research question; LR-test = likelihood-ratio test; AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion.

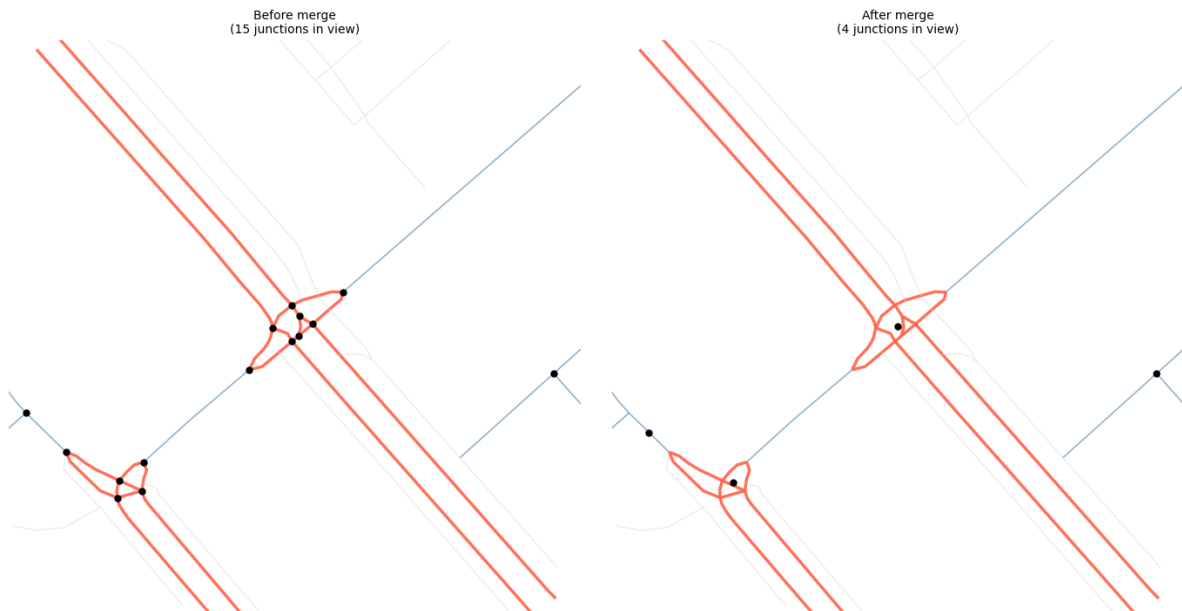


Figure 3.3: Dual-carriageway junction merge for one cluster in the Rotterdam network (cluster size 8); see Appendix A for the corresponding bayonet-merge example

3.3.2. Stratification

Intersections were stratified on two dimensions: junction type (T-junction or 4-leg+-junction) and traffic control regime (signalised, priority road, or uncontrolled), yielding six strata. These two dimensions reflect the primary typological distinctions by which intersections are classified in the Dutch context (SWOV, 2022). Stratification ensured that the sampled intersections covered the full range of intersection types present in the network, which is a necessary condition for the CV model trained in Phase 3 to generalise across that range.

Sampling was restricted to the 4,615 intersections with usable panorama coverage. A random draw with a fixed seed was applied within each stratum, with a practical cap of 70 intersections per stratum, comfortably above the number later required per stratum for CV model training. The signalised and T-junction stratum was sampled exhaustively because its total eligible population ($n = 18$) fell below the cap. Equal allocation across strata was preferred over proportional sampling because the CV model performs better with a more equal distribution of training labels. This yielded a stratified sample of 368 intersections across the six strata, as shown in Table 3.1.

Table 3.1: Survey sample per stratum (type $\times$ control regime)

Stratum	N sampled
Signalised, 4-leg	70
Priority-controlled, 4 leg	70
Uncontrolled, 4-leg	70
Signalised, T-junction	18 (exhausted)
Priority-controlled, T-junction	70
Uncontrolled, T-junction	70
Total before quality control	368

3.3.3. Photo selection and reprojection

Each intersection was represented by a single street-level image. This single-image format gave the expert comparison task and the CV-model training a shared, unambiguous basis for judgement. Each respondent evaluated one defined view rather than choosing among several, and the model was trained on one consistent input per label, allowing it to learn a clear correspondence between visual features and score. Multiple candidate approach-leg photographs were matched per intersection to provide a backup should the primary candidate fail quality control, but only one photograph ultimately represented each intersection. Each approach leg across the network was matched to a Cyclomedia photograph using a fixed protocol based on capture distance, alignment between the photograph's viewing direction and the leg's approach bearing, and containment within a leg-specific road corridor that excluded photographs on nearby parallel roads. The exact distance, bearing, and corridor thresholds governing this procedure are reported in Table B.1. Candidates satisfying all three criteria were ranked by proximity to a target capture distance of approximately 20 metres, a distance at which the full intersection geometry including all approach legs and the conflict area is generally visible within the selected field of view. Where two legs of the same intersection matched photographs looking at the intersection from near-identical directions, the less well-positioned of the two was treated as redundant and removed. Figure 3.4 illustrates this matching procedure for one representative intersection. The full technical description is given in Appendix B.

Photographs were originally captured as 360-degree panoramas and pre-processed by the data provider into four directional 90-degree frames per capture point before acquisition. Where the bearing of a matched leg fell between two adjacent frames, these were stitched into a synthetic view aligned with the leg bearing.

Manual quality control was conducted on the matched photographs. Photos were rejected if they showed severe occlusion of the intersection geometry, active construction works, or image artefacts, and replaced with the next-best candidate. All selected photos were reprojected to a standardised perspective with a 90° horizontal field of view. The 90-degree FOV was selected after testing a range of values and confirmation of an experienced traffic engineer. Output format was JPEG at quality 90,

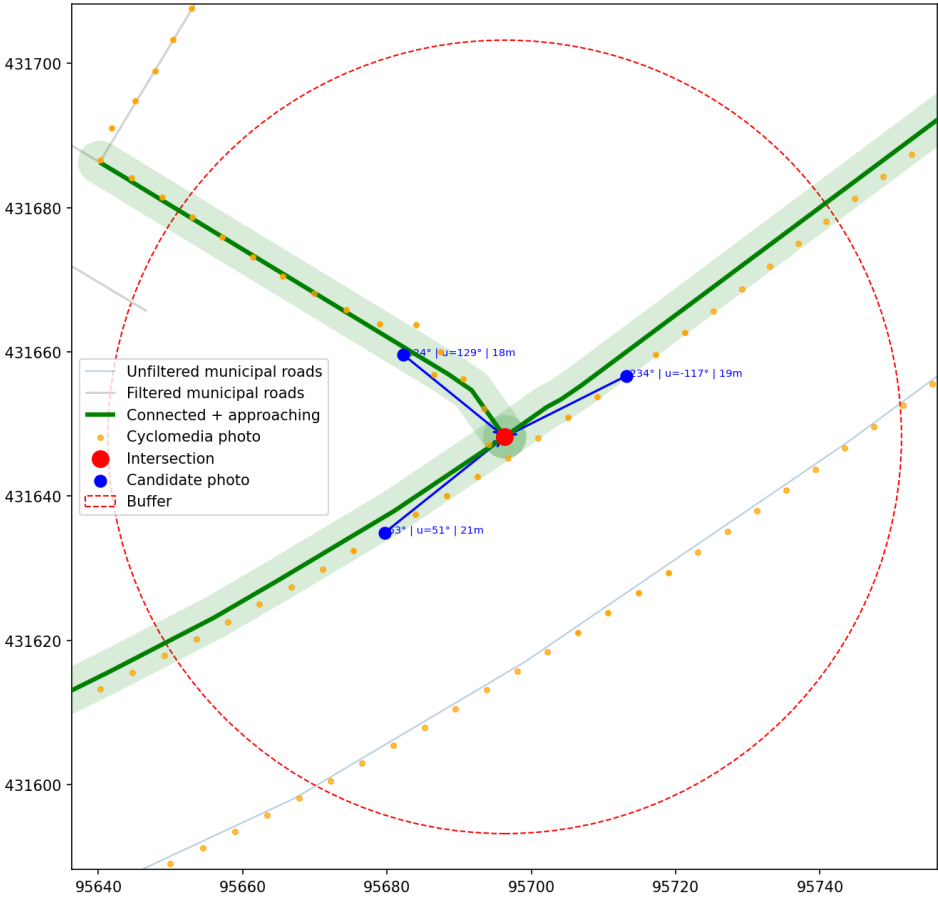


Figure 3.4: Approach-leg-to-photograph matching procedure for one representative intersection, showing the sequential application of the capture-distance, bearing-alignment, and road-corridor criteria

resolution 1500×880 pixels. Standardisation removed variation in perspective, zoom, and camera angle across capture locations, ensuring consistency of the photo set from which the expert survey and CV model inputs were later drawn. Figure 3.5 shows an example of the final reprojected output.



Figure 3.5: Example of a reprojected survey image, standardised to a 90° horizontal field of view and a capture distance of approximately 20 metres from the intersection centroid

3.4. Phase 2: Expert elicitation and scoring (RQ2)

Phase 1 produced a stratified population of intersections, each matched to a representative street-level photograph. Turning these photographs into a quantified safety judgement requires eliciting that judgement from domain experts in a structured, reproducible form. This phase establishes the elicitation protocol, collects expert comparisons across a subset of the network, and converts them into a continuous Bradley-Terry safety score for each compared intersection.

3.4.1. Preparatory interview

Before the pairwise comparison instrument was finalised, a semi-structured interview was conducted with an experienced traffic safety professional at the Municipality of Rotterdam. The interview followed a guide organised around five thematic clusters: current safety assessment practice, the role of behaviour in intersection design, the comparability of intersections from photographs, photograph framing and survey format, and practical survey logistics.

The interview served two functions. The first was to establish if a holistic visual impression, derived from a street-level photograph, carries relevant information that experts use in their judgement but is not encoded in current datasets. The second was to derive the constraints and confounders that a photograph-based elicitation method would have to accommodate. Interview responses were consolidated into a thematic summary. The interview guide and themes are provided in Appendix C.

3.4.2. Pairwise comparison design and survey protocol

Survey structure

Respondents were presented with pairs of intersection photographs and asked to indicate which intersection presents a higher overall design risk. The task required a holistic judgement. No checklist or scoring rubric was provided during the assessment. This format was preferred over direct rating on a numerical scale because comparative judgement tasks produce higher inter-rater reliability when the underlying construct is continuous and multi-dimensional, as raters do not need to maintain a shared numerical reference frame across assessments. Figure 3.6 shows the task interface as presented to respondents.

The comparison task was designed to cover the network as a whole, across all intersection types and traffic volumes, in line with the municipality's interest in a single citywide ranking rather than separate assessments per intersection type that would fragment the result. A design of this kind inevitably pairs intersections that differ in how busy they are, since limiting comparisons to similarly busy intersections would have ruled out exactly the network-wide comparison the design was meant to support.

In this setting, busier intersections risk being read as more dangerous by default, a possibility the preparatory interview pointed to. Each pair was therefore presented with an accompanying traffic volume class for both intersections, so respondents could weigh the design against its expected volume instead of reading volume itself as the risk signal. The traffic volumes were classified into three fixed bands: low (under 1.000 vehicles per day), moderate (1.000–6.000 vehicles per day), and high (over 6.000 vehicles per day).

This traffic volume class was a modelled annual average intensity estimate from the regional dataset described in Chapter 4, bucketed coarsely so it could sit alongside a photograph as a quick point of reference. Section 5.5.2 examines how far this anchor achieved its intended separation between design and exposure, and Section 6.3 discusses what that result means for the elicitation method.

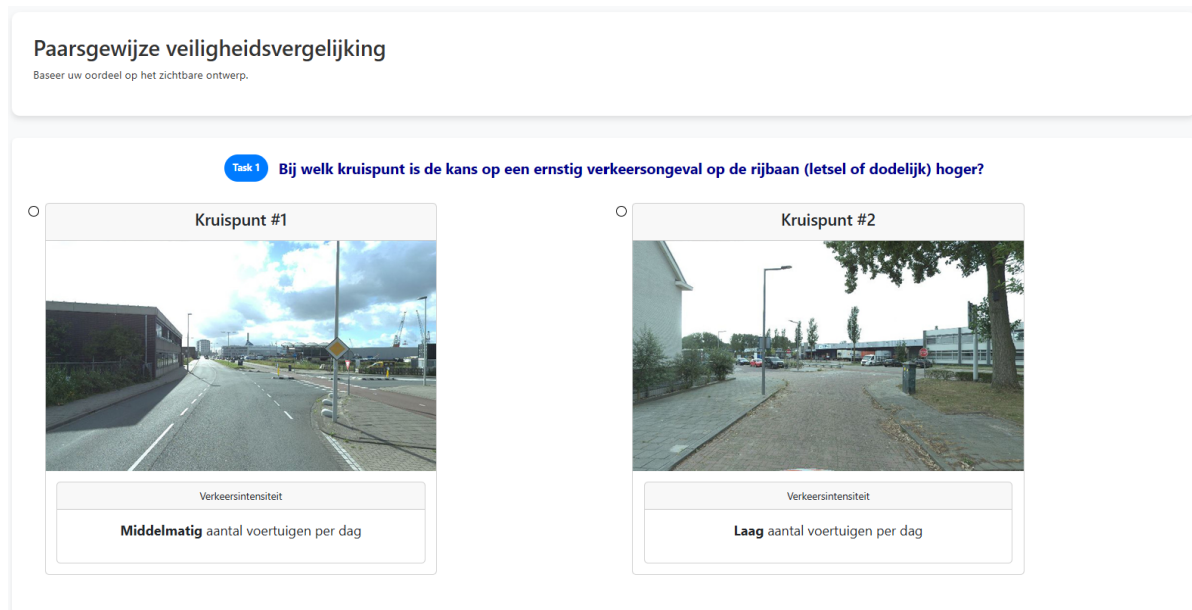


Figure 3.6: Screenshot of the pairwise comparison task as presented to respondents

Panel composition

The panel comprised 26 respondents who completed the comparison tasks for the analysis. Respondents were recruited from traffic engineering and road safety backgrounds. Inclusion required a professional role involving infrastructure design or safety assessment (traffic engineers, road designers, traffic safety specialists, or urban designers with a road safety focus). Respondents were primarily professionals at the Municipality of Rotterdam, supplemented by practitioners from related organisations.

Pair design

The survey covered 145 of the 368 stratified intersections. This number was determined because of two constraints: the minimum number of comparisons per intersection required for stable Bradley-Terry estimation, and the realistic comparison capacity per respondent session. Expanding the covered set beyond 145 would have required either more respondents or longer sessions than were feasible within the available expert panel.

The pairwise comparison schedule was built as a layered hub-spoke architecture over the six strata, covering 145 intersections (11 hubs and 134 spokes), as shown in Figure 3.7. It comprised 314 unique pairs across four layers, as shown in Table 3.2. The hub-spoke architecture was chosen over a fully random pair schedule because it guarantees a connected comparison graph regardless of which respondents complete the survey. The hub-hub backbone layer ensures that all strata are bridged, while the spoke layers provide the within-stratum comparisons needed for fine-grained score differentiation. Because the 13 hub-hub backbone pairs were shown to almost every respondent, they are referred to as anchor pairs throughout this thesis, and the 11 hub intersections as anchor intersections; Section 3.4.4 uses them to assess between-rater agreement. A fully random schedule of the same size

would risk disconnected components if participation was uneven across sessions, which would make Bradley-Terry estimation impossible for isolated nodes.

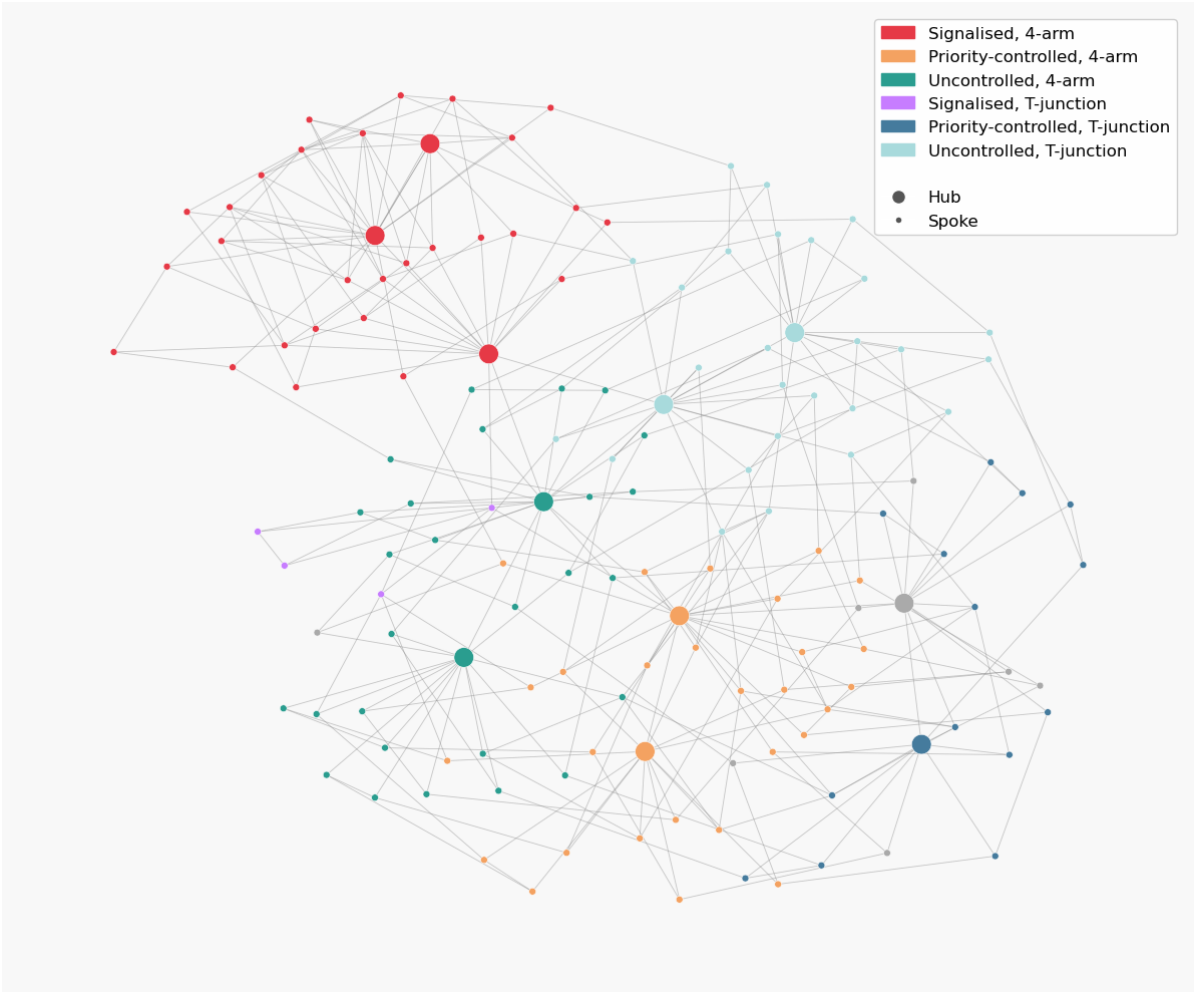


Figure 3.7: Layered hub-spoke pair design schedule across the six strata (11 hubs, 134 spokes; 314 unique pairs across four layers)

Table 3.2: Pair design: number of unique pairs per layer

Layer	Pairs
Backbone hub-hub	13
Hub-spoke	132
Diagonal cross-stratum spoke-spoke	77
Within-stratum spoke-spoke	92
Total unique pairs	314

3.4.3. Bradley-Terry model specification

A pairwise comparison indicates which of two intersections is judged riskier, but not by how much. Bradley-Terry estimation (Bradley & Terry, 1952) takes this ordinal signal and positions every intersection on a single continuous scale, the standard way of turning such comparisons into a usable measure. A categorical judgement, such as low, medium, or high risk, would still rest on the same ordinal information. Pairwise comparisons order intersections, but they do not mark where a boundary between categories should fall. The continuous scale also carries directly into the two steps that follow, serving as a graded regression label for the CV model in Phase 3 and as a continuous predictor in the Negative Binomial model in Phase 4.

Rather than scoring each intersection independently, the model treats the full set of comparisons as a single connected graph and jointly estimates one latent strength parameter per intersection, so that every comparison contributes simultaneously to all scores reachable through the graph. This joint estimation is what allows the model to correct for opponent strength, which is why it was preferred over simpler alternatives such as a raw win proportion. An intersection that repeatedly defeated strong opponents receives a higher score than one that accumulated the same number of wins against weak opponents. This correction is important in a sparse and unbalanced comparison graph, where different intersections face systematically different sets of opponents.

This connected graph draws on a small fraction of all possible pairs. With 145 scored intersections there are 10,440 possible unique pairs, and the design covers 314 of them, roughly 3%. What matters for the joint estimation above is that these comparisons still link every intersection to every other one through some chain, even where two intersections were never directly compared. This is the standard connectivity condition for Bradley-Terry estimation on incomplete designs (Hunter, 2004). A comparison between i and k , together with a separate comparison between k and j , already positions i and j relative to each other through k . The hub-spoke architecture from Section 3.4.2 provides this connectivity directly. Its hub-hub backbone connects all six strata into a single core, and every spoke intersection is tied into that core through at least one hub comparison, so every intersection remains part of the same graph regardless of which comparisons individual respondents happened to complete.

The model specifies the probability that intersection i is preferred over intersection j , that is, judged as presenting the higher design risk, as:

$$P(i > j) = \frac{\lambda_i}{\lambda_i + \lambda_j} \quad (3.1)$$

where λ_i is the latent safety strength of intersection i . The latent strength parameters were estimated by maximum likelihood. This estimate was computed with the ILSR (Iterative Luce Spectral Ranking) estimator (Maystre & Grossglauser, 2015), preferred over direct likelihood optimisation for its more stable convergence on the sparse comparison graph. A regularisation parameter of $\alpha = 0.01$ was applied to dampen score estimates for thinly compared intersections toward the population mean. Sensitivity analysis across $\alpha \in \{0.001, 0.01, 0.05, 0.1, 0.5\}$ confirmed that the resulting rank order is stable for $\alpha \geq 0.01$ without dampening the relative effects.

Before normalisation, the raw log-strength scores were checked for outliers using Tukey fences ($Q_1 - 1.5 \times \text{IQR}$ to $Q_3 + 1.5 \times \text{IQR}$). The resulting bounds of $[-1.671, 1.773]$ contained all 145 intersections, and a sorted bar chart and histogram of the raw scores showed no distinct outlying points either. This confirmed that the normalisation was not driven by isolated extreme values. The scores were then min-max normalised to the interval $[0, 1]$, where 1 denotes the intersection assessed as most dangerous and 0 the least dangerous. Min-max normalisation was preferred over z-score standardisation because the bounded $[0, 1]$ output aligns directly with the sigmoid activation used in the CV regression head in Phase 3, avoiding the need for an additional rescaling step between phases. Normalised scores served directly as continuous regression labels for the CV model in Phase 3.

3.4.4. Consistency and reliability assessment

Consistency was assessed at two levels using three complementary indicators. Within-rater consistency was measured through circular triads, which test whether individual respondents produce transitive orderings, a necessary condition for the Bradley-Terry transitivity assumption. Between-rater agreement was assessed through two statistics that capture different aspects of shared judgement: Fleiss' κ measures agreement corrected for chance across respondents, while Kendall's W measures the degree of rank consensus across the anchor intersections. Together the three indicators cover both the internal coherence of individual respondents and the extent to which the panel as a whole converges on a shared ordering.

Fleiss' κ must be interpreted with caution in a forced-choice pairwise design. Because each comparison has exactly two outcomes, the chance-expected agreement is fixed at 0.5 regardless of how strongly raters agree. A near-zero or negative κ can therefore result from near-equal pairs where genuine agreement exists but both options are chosen roughly equally, rather than from inconsistent judgement.

Percent agreement per anchor pair is reported alongside κ as a more direct measure of concordance in this setting, given that near-equal pairs are expected to produce low κ regardless of rater consistency.

Inter-rater agreement on anchor pairs functioned as a validity gate for the downstream phases. Insufficient agreement would indicate that experts do not hold shared safety judgements, which would undermine the labelling procedure and preclude meaningful CV model training.

3.5. Phase 3: Score generalisation (RQ3)

Phase 2 produced a continuous safety score for the 145 intersections experts could feasibly compare, a labelled sample far smaller than the network it is meant to represent. Extending that judgement across the rest of the network requires a model that can learn which visual pattern the label tracks from the labelled subset, and then read the same pattern out of an unseen photograph. Computer vision models pretrained on large image collections and adapted through transfer learning are suited to exactly this task, since they already encode general visual structure that would otherwise require far more labelled data than 145 intersections can provide. This phase compares candidate architectures for that role, before applying the selected model to score every intersection in the network.

3.5.1. Architecture selection and transfer learning

Four backbone architectures were compared to identify which most accurately generalises the expert-derived safety scores to unlabelled intersections, summarised in Table 3.3. Two of the four, DINOv2-Small and CLIP ViT-B/16, are built on the Vision Transformer architecture, which adapts the self-attention mechanism (Vaswani et al., 2017) to image patches (Dosovitskiy et al., 2021). The four were selected to cover three distinct pretraining paradigms: supervised object classification on ImageNet, represented by ResNet-50 and EfficientNet-B4; self-supervised scene-level learning on diverse unlabelled imagery, represented by DINOv2-Small; and contrastive multimodal learning on image-text pairs, represented by CLIP ViT-B/16 (Radford et al., 2021). This coverage allows the architecture comparison to test whether the pretraining objective, rather than model size or architecture family alone, determines how well the resulting features align with expert visual safety judgement.

Table 3.3: Backbone architectures evaluated as frozen feature extractors

Backbone	Type	Parameters	Feature dim	Pretraining
ResNet-50	Supervised CNN	25M	2.048	ImageNet
EfficientNet-B4	Supervised CNN	19M	1.792	ImageNet
DINOv2-Small (primary)	Self-supervised ViT	21M	384	142M unlabelled images
CLIP ViT-B/16	Contrastive ViT	86M	512	400M image-text pairs

DINOv2-Small was the a priori primary candidate on the grounds that self-supervised pretraining on diverse scenes is expected to generalise better to street-level imagery than supervised ImageNet pretraining, because the DINOv2 pretraining objective trains the model to represent spatial and structural scene properties rather than object identity categories (Oquab et al., 2024).

In every case the backbone was used as a frozen feature extractor rather than fine-tuned end-to-end. End-to-end fine-tuning would require updating tens of millions of parameters on 109 training intersections, a ratio that causes severe overfitting regardless of regularisation strength. Freezing the backbone constrains the trainable parameter count to the regression head alone, which is well-determined at this sample size.

Two families of regression heads were evaluated on the frozen features. The first was a neural head that maps the feature vector to a single safety score bounded to the unit interval by a sigmoid output: either a single linear layer, $\text{Linear}(\text{dim} \rightarrow 1) \rightarrow \text{Sigmoid}$, or a multilayer perceptron, $\text{Linear}(\text{dim} \rightarrow 128) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.3) \rightarrow \text{Linear}(128 \rightarrow 1) \rightarrow \text{Sigmoid}$, with narrower hidden widths of 32 and 2 units also tested. The second was a non-neural alternative that first reduced the feature vector with unsupervised principal component analysis and then fitted Ridge regression on the labelled subset. Because this

dimensionality reduction is fitted on the full city-wide feature set rather than on the labels alone, it is described in the next section. Backbone weights remained frozen throughout, so no end-to-end fine-tuning was performed.

3.5.2. Aggregation and feature compression

The backbone feature vector is high-dimensional (for example 384 values for DINOv2-Small) and cannot enter the Negative Binomial model directly without inflating the predictor-to-observation ratio. Each candidate pipeline therefore compressed the feature vector to a single scalar safety score, in one of the two ways corresponding to the head families introduced above.

The neural head performed this compression in a single learned step, mapping the full feature vector to a scalar. The principal component route compressed in two stages: the features were first standardised and reduced with unsupervised principal component analysis (PCA; Jolliffe & Cadima, 2016), a technique that re-expresses a set of correlated features as a smaller set of uncorrelated components ordered by the variance they capture, after which Ridge regression (Hoerl & Kennard, 1970) mapped the reduced representation to the safety score.

This principal component analysis was fitted on the full city-wide feature set of 4.615 vectors, covering both labelled and unlabelled intersections, rather than on the 109 labelled training intersections alone. Fitting the dimensionality reduction on this much larger unlabelled set gave it a more representative view of the feature space than the labels alone could provide. Because PCA is unsupervised and does not use the safety labels, fitting it on the full set introduces no label leakage from the unlabelled intersections. The reduction is exposed only to the distribution of the visual features, not to the target.

Within this route, the number of retained components and the regression penalty were treated as hyperparameters and selected by grid search. For ResNet-50 and DINOv2-Small, the two backbones considered from the outset of the comparison, the grid also varied the regression method itself, contrasting Ridge regression with Partial Least Squares (PLS) regression, a supervised alternative that selects components by their covariance with the safety label rather than by variance alone, and whether components were L2-normalised or whitened before regression. Including PLS in the grid guarded against a specific limitation of the variance-based route. A feature dimension carrying comparatively little variance could still be informative for the safety score, and PCA's variance ordering would discard it before Ridge ever saw it. PLS retains dimensions according to their covariance with the label rather than their variance, regardless of how much variance they explain. CLIP ViT-B/16, added later as a further architecture, was evaluated with a narrower Ridge-only grid over the same two core hyperparameters. EfficientNet-B4 was evaluated only through the neural-head route. The chosen configuration and the comparison between the two compression routes are reported in Section 5.3.1 and Appendix F.

Compressing to a single scalar served two purposes. It maintained a favourable predictor-to-observation ratio in the Negative Binomial model, where the visual score enters as one predictor among several, and it matched the holistic single-score safety construct elicited from the experts in Phase 2.

3.5.3. Training procedure and evaluation

Images were normalised using the ImageNet channel mean and standard deviation ([0.485, 0.456, 0.406] and [0.229, 0.224, 0.225]) for ResNet-50, EfficientNet-B4, and DINOv2-Small. CLIP ViT-B/16 used its own bundled channel statistics instead. Input resolution followed the backbone. ResNet-50, EfficientNet-B4, and DINOv2-Small each received a forced square resize (256, 400, and 560 pixels respectively) followed at training time by a random crop to the backbone's native resolution (224, 380, and 518 pixels), a translation augmentation that for DINOv2-Small also matches the patch size of 14. CLIP ViT-B/16 was resized to 224 x 224 through a two-stage pipeline: a resize to 248 x 248 and a random crop to 224 x 224 within this augmentation step, followed by CLIP's own preprocessing, which applies a further deterministic resize and centre crop at 224 x 224. Since the image already measured 224 x 224 at that point, this second step left it unchanged during training. Horizontal flip, colour jitter, and Gaussian blur were added to the random crop as further training-time augmentation. Vertical flips and large rotations were excluded because the upright orientation of street-level images carries semantic content that should be preserved. The validation transform applied a deterministic resize without cropping for ResNet-50, EfficientNet-B4, and DINOv2-Small. For CLIP ViT-B/16, validation applied the same bundled preprocessing directly to the unresized photograph, so its resize-then-centre-

crop step performed the only crop in that backbone's validation pipeline. The feature extraction used for the principal component route followed the same deterministic, uncropped transform as validation.

The data split was performed strictly at intersection level. Splitting at image level would have constituted data leakage, because approach-leg photos of the same intersection could otherwise appear in both the training and validation sets. Eight of the 145 surveyed intersections were added to the survey design at a later stage and were not carried into the intersection-level dataset from which the CV training and validation sets were drawn. The remaining 137 labelled intersections with a usable photo were split roughly 80/20 into a training set of 109 intersections and a validation set of 28 intersections, stratified on intersection type and centre-versus-periphery location. The selected configuration was additionally checked with five-fold cross-validation. Folds were grouped spatially so that adjacent, visually similar intersections tended to fall in the same fold, which limits performance inflation from spatial autocorrelation. This grouping is approximate, since a large geographic cluster can still span more than one fold.

The neural regression heads were trained with a combined objective that weights a mean squared error term against a RankNet pairwise ranking term. In the final configuration pure MSE was used, because the combined loss produced rankings equivalent to pure MSE on the validation set while adding training complexity without benefit. Validation and early stopping always used MSE for consistency. The principal component route did not use this objective. Its Ridge stage minimises a penalised least squares criterion, governed by a separate Ridge penalty.

Model performance was evaluated with four metrics: mean absolute error (MAE), root mean squared error (RMSE), R squared, and Spearman ρ . Selection combined two criteria, applied in sequence. A configuration first had to reach a statistically significant ρ , since the score's downstream role is to rank intersections by assessed safety, rank correlation is the metric most directly tied to that use, and a model without significant ρ was not considered regardless of its absolute error. Among configurations that cleared this threshold, the lowest validation MAE decided, preferred over RMSE because it is less sensitive to individual outliers in a validation set of 28 intersections. A single poorly predicted intersection can disproportionately inflate RMSE. This rule applied throughout the comparison, including the choice between Ridge and PLS for DINOv2-Small, where PLS reached a higher rank correlation ($\rho = 0.632$ against 0.554) but Ridge reached the lower MAE (0.156 against 0.170) and was retained. The selected model and its validation metrics are reported in Chapter 5 and carried forward to Phase 4, where the model scores every intersection in the network.

3.6. Phase 4: Crash model and enrichment (RQ4)

Phase 3 turned the expert-elicited safety score into a predictor available for every intersection in the network. Whether that score contributes anything beyond what conventional infrastructure and traffic variables already explain can only be established by testing it directly against those variables. This requires entering the score alongside those variables in a single model of crash counts, separating a genuine contribution from an association the conventional variables already capture. This phase specifies that baseline model, adds the visual score to it, and tests whether the addition improves the fit.

3.6.1. Negative Binomial model specification

Crash counts are non-negative integers, and counts of events as rare as crashes typically spread further than a Poisson model assumes. Negative Binomial regression, the standard specification for this kind of overdispersed count data, accommodates that extra spread and takes the form

$$\log(E[Y_i]) = \beta_0 + \beta_1 \cdot \log(V_{\text{major},i} + 1) + \beta_2 \cdot \log(V_{\text{minor},i} + 1) + \sum_j \beta_j \cdot C_{j,i} \quad (3.2)$$

where Y_i is the observed crash count at intersection i over the study period, $V_{\text{major},i}$ and $V_{\text{minor},i}$ are the AADT (Annual Average Daily Traffic) values on the major and minor approaches, and $C_{j,i}$ are categorical infrastructure terms. AADT was included as a predictor rather than as a fixed exposure offset. An offset would assume that crashes rise in exact proportion to traffic, that is, that the elasticity equals

one. The road safety literature finds that this elasticity is generally not one (Elvik & Bjørnskau, 2017), so estimating it as a free coefficient is more appropriate. The AADT terms were log-transformed, which is the standard form for traffic exposure in SPFs (Montella, 2010) and lets the coefficient be read as an elasticity. Treating AADT as an ordinary predictor also allows its inclusion or exclusion to be varied in the robustness analysis.

Overdispersion was tested empirically with a Cameron-Trivedi overdispersion test in its NB2 form, testing the one-sided hypothesis $\alpha > 0$. The result of this test determined whether the Negative Binomial (NB2) or the Poisson specification served as the primary model. The specification also treats intersections as independent observations, the standard starting point for safety performance functions (Lord & Mannering, 2010). This assumption is checked directly on the fitted residuals, reported in Section 3.6.3 as part of the robustness analysis.

The dependent variable was derived from the BRON crash registry for the period 2020 to 2024, a five-year window that follows the three-to-five-year accumulation period recommended for SPFs to dampen regression to the mean and average out incidental year-to-year fluctuations (Federal Highway Administration, 2017). Crashes were summed over the full period rather than averaged per year, which preserves the integer count structure the Negative Binomial model assumes. The primary outcome was restricted to injury or fatal crashes involving at least one motor vehicle, because BRON underregisters non-motor-vehicle and light-injury crashes. Two broader nested definitions (all registered crashes, and all injury crashes across road users) were retained for a robustness check. Crash records were spatially joined to NWB junction identifiers.

Candidate predictors, listed in Table 3.4, were drawn from intersection type, leg count, major-road and minor-road AADT, speed limit category and its continuous form, and priority and control type. Candidates were first screened for pairwise collinearity using a correlation threshold, after which the remaining set entered an incremental selection procedure. A variable was retained if it met two simultaneous criteria, statistical significance at $p < 0.05$, confirming that the association is unlikely to be a sampling artefact, and a reduction in the Akaike Information Criterion (AIC), confirming that the improvement in fit outweighs the cost of the additional parameter. Both criteria were required because significance alone does not penalise model complexity and AIC reduction alone does not guard against chance associations in a moderately sized predictor set, a combination confirmed by a likelihood ratio test. One exception was made: the priority regime terms did not meet this bar as a block ($\Delta\text{AIC} = +0.53$, LRT $p = 0.104$) but were retained for specification completeness rather than fit, detailed in Section G.2.

Table 3.4: Candidate predictors considered for the baseline Negative Binomial model

Predictor	Definition	Reference category
Intersection type	Four-or-more-leg vs T-junction	T-junction
Leg count	Continuous count of intersection legs	Continuous
log major-road AADT	$\log(V_{\text{major}} + 1)$	Continuous
log minor-road AADT	$\log(V_{\text{minor}} + 1)$	Continuous
Speed limit category	50 km/h vs 30 km/h	30 km/h
Speed limit (continuous)	Posted speed limit in km/h	Continuous
Signalised	Dummy: signalised priority	No priority
Priority-controlled	Dummy: priority-controlled	No priority

The model was fitted on 4.474 intersections, excluding the 137 intersections used to train and validate the CV model in Phase 3. For those intersections the CV-derived score is a fitted value rather than a genuine out-of-sample prediction. Including them would give the enriched model an artificially advantaged predictor at those locations, distorting the comparison between baseline and enriched model.

3.6.2. Model enrichment and nested model comparison

The CV-derived safety score from Phase 3 was added to the baseline model as a single extra predictor, in a similar spirit to Van Cranenburgh and Garrido-Valenzuela (2025), who added a computer-vision-

derived image representation to a conventional discrete choice model as an additional linear predictor rather than replacing the model altogether, giving the enriched model

$$\log(E[Y_i]) = \beta_0 + \beta_1 \cdot \log(V_{\text{major},i} + 1) + \beta_2 \cdot \log(V_{\text{minor},i} + 1) + \sum_j \beta_j \cdot C_{j,i} + \beta_z \cdot Z_i \quad (3.3)$$

where the first terms are the baseline linear predictor above, Z_i is the CV-derived safety score for intersection i , re-scaled to $[0, 1]$ via MinMax normalisation on the modelling set before entering the model, and β_z is its estimated coefficient. The baseline model is nested within the enriched model, since the baseline is the same model with β_z fixed at zero. This nesting means the gain in fit from the single extra predictor can be tested formally.

The enriched model was judged against the baseline on several complementary criteria. A likelihood ratio test gave a formal test of whether the added score improved fit. Two information criteria were computed alongside it: AIC and the Bayesian Information Criterion (BIC), which weigh fit against the cost of the extra parameter but penalise that parameter differently, so they can point to different conclusions. The coefficient β_z was then interpreted substantively, on its sign, its magnitude, and its confidence interval, rather than on statistical significance alone. Finally, the dispersion parameter was compared between the two models as a diagnostic check that adding the score did not distort the count process.

3.6.3. Robustness and measurement-error checks

Five procedures qualified the estimated CV-score coefficient before it was interpreted. The first checked how much measurement error in the score biases the coefficient, and produced a bias-adjusted estimate. The second tested whether the model's residuals carried spatial dependence between nearby intersections. The third tested how robust the coefficient is to reasonable modelling choices. The fourth tested whether the coefficient's significance depends on collapsing the visual signal to a single scalar rather than retaining it as a multi-dimensional block. The fifth compared the CV score directly against the raw human judgement it was trained to reproduce. The first four were applied to the same enriched model described in the previous section. The fifth instead re-estimated the model on the subset of intersections where both measures were available.

Measurement error correction

The predictor entering the model is not observed directly but predicted by the Phase 3 model, and that prediction deviates from the Bradley-Terry score it was trained to reproduce. From the crash model's perspective, the CV score is therefore an imperfectly measured predictor, not a directly observed one. This deviation, quantified by the Phase 3 validation root mean square error of about 0.18, functions as measurement error, which biases the coefficient toward zero. The SIMEX (Simulation-Extrapolation) method calculates this bias. It adds increasing artificial error to the score, tracks how the coefficient shifts, and extrapolates back to the no-error case, here with 100 runs and quadratic extrapolation (Cook & Stefanski, 1994; Hardin et al., 2003). The corrected estimate was compared with the uncorrected one and with a simpler correction to confirm the conclusion did not depend on the method (reported in Section G.3). Throughout, the correction is read as a check on the direction and sensitivity of the effect, not as a precise estimate of its size.

Spatial dependence check

The Negative Binomial model treats each intersection as an independent observation, an assumption that can fail when unmodelled local factors, such as a shared arterial corridor or a nearby cluster of trip generators, raise or lower crash risk at several intersections at once. A spatial weights matrix connected each intersection to its five nearest neighbours by Euclidean distance in projected coordinates, and Moran's I (Moran, 1950) was computed on the deviance residuals of both the baseline and the enriched model, with significance assessed by a permutation test (999 permutations). Where this indicated spatial dependence, the enriched model was additionally refitted with cluster-robust standard errors, using intersections grouped into a spatial grid as the clustering unit, as a further check on whether the CV-score coefficient's significance depended on treating observations as independent.

Specification robustness

Every specification embeds a series of choices, in how the score is coded, which controls are included, which data the model is fitted on, and how uncertainty is estimated. A series of checks tested whether the CV-score coefficient's sign and rough magnitude survived reasonable alternatives to each of these choices in turn. The form of the score was entered as a raw value, as a rank, and as a binary flag for the safest-looking quartile. The set of controls was varied by refitting the model with each control variable left out in turn, with particular attention to the two traffic-volume terms, given the entanglement between visual danger and traffic volume anticipated during elicitation (Section 3.4.2). The data the model was fitted on varied across the two broader crash definitions from the previous section. The uncertainty estimate was varied by using a bootstrap with 500 resamples rather than the model standard errors alone.

Block operationalisation and calibration check

This check tested whether the coefficient's significance depends on collapsing the visual signal to a single scalar, by retaining the same underlying feature representation from Phase 3 as an unreduced, multi-dimensional block instead. Two decompositions of the same standardised feature matrix were compared, a supervised route and an unsupervised route. Partial Least Squares (PLS) was fitted on the 137 labelled intersections against the Bradley-Terry score, using the same procedure considered as an alternative to Ridge regression in the Phase 3 grid search (Section 3.5.2). An unsupervised Principal Component Analysis (PCA) was fitted on the same feature matrix without reference to the label, for comparison. Testing both lets the result distinguish whether any effect depends on guidance by the expert label specifically, or is a property of the underlying feature space regardless of how it is decomposed.

Both retained five components, matching the component count already validated in the Phase 3 grid search, and each entered the enriched specification as a block of five linear terms in place of the single CV-score term, tested against the baseline on the same 4,474-intersection modelling set with five degrees of freedom instead of one.

These five terms are constructed by optimising over a much larger feature space, for variance in the case of PCA or for covariance with the label in the case of PLS, rather than fixed in advance like the baseline predictors. Under this kind of data-dependent construction, the conventional asymptotic χ^2 reference for the resulting likelihood-ratio test may not describe its true sampling distribution, in the same way that coefficients from stepwise or best-subset variable selection warrant caution beyond their nominal p-value. This was checked directly rather than assumed, using two hundred alternative five-dimensional projections of the same feature matrix, each an orthonormal rotation independent of both the label and the feature variance structure, entered into the identical baseline-versus-block comparison. The empirical share of these placebo blocks reaching a likelihood-ratio statistic at least as large as the observed PLS and PCA blocks provides a data-calibrated reference, reported in Section 5.4.4 alongside the conventional asymptotic p-value. This calibration establishes only whether the tested block can be distinguished from an arbitrary direction in the same feature space, not whether the feature space as a whole carries information beyond the baseline.

Direct comparison on the expert-scored subset

The data were narrowed to the 145 intersections that experts had actually scored. A likelihood ratio test asked whether each assessment measure, the raw expert judgement, the CV score, and its volume-residualised version (Section 3.7.2), added anything beyond the six conventional controls. Each measure's marginal association with crashes, without the six controls, was estimated alongside the incremental test, to show whether a standalone association survives their addition. Comparing the CV score against the raw human judgement on the same subset tests whether the result depends on the CV-generalisation step, rather than being a property of the underlying human judgement itself.

Together, these checks identify whether the CV-score coefficient is consistent across reasonable modelling alternatives, and whether the result depends on the CV-generalisation step rather than on the underlying human judgement.

3.7. Phase 5: Validation analysis (RQ5)

Phase 4 tested whether the visual score improves crash prediction in a statistically significant and robust sense. Phase 5 validates the result along a theoretical and practical dimension. The theoretical dimension asks what the CV score represents beyond a statistical estimate. The signal decomposition examines whether the score's relationship with crashes reflects an exposure gradient besides a visual signal, and the attention map analysis examines which image regions the model attends to. The practical dimension asks whether the result changes anything for practice. The Empirical Bayes ranking comparison tests whether adding the score reorders intersection priority, and the expert validation session tests whether the model's reasoning is recognisable to a practitioner.

3.7.1. Empirical Bayes estimation and ranking comparison

To assess whether the addition of the CV score changes network-wide prioritisation, Empirical Bayes (EB) expected crash counts were computed for every intersection under both the baseline and the enriched Negative Binomial model. Ranking intersections directly on the SPF's raw predicted count treats every intersection with the same measured characteristics as equally risky, leaving no room for a site's own crash history to reflect risk factors the model's predictors do not capture. It also leaves the ranking vulnerable to regression to the mean, the tendency for a site's short-term crash count to fluctuate around its true underlying risk by chance, so that the raw count alone is an unreliable signal of that risk. The EB estimate is the standard correction for this limitation in the SPF literature, introduced in Section 2.1 (Ghadi & Török, 2019; Montella, 2010; Persaud & Lyon, 2007), and combines the model's site-specific prediction with the site's own observed crash count as a weighted average:

$$EB_i = w_i \cdot \hat{Y}_i + (1 - w_i) \cdot Y_i \quad (3.4)$$

where the weight is $w_i = 1/(1 + \alpha \cdot \hat{Y}_i)$, with $\hat{Y}_i$ the SPF-predicted crash count at intersection i , Y_i the observed count, and α the Negative Binomial dispersion parameter estimated in Section 5.4.1. As the predicted count $\hat{Y}_i$ grows, w_i shrinks and the estimate leans more heavily on the observed count. At intersections with a small predicted count, the estimate leans more on the model prediction instead. The EB estimate pulls such observations back toward the model's prediction rather than treating them as evidence of elevated risk. This weighted sum produces more stable site-specific estimates than either the raw predicted count or the observed count alone. The EB estimate was computed separately for the baseline and the enriched model, yielding two site-specific rankings.

These two rankings were compared in two ways. A Spearman rank correlation measured overall agreement between them, providing a global summary of how much the rankings diverge. A tier analysis assigned intersections to priority tiers, specifically the top 10% and the top 25% by EB-estimated expected crashes, and recorded the share of intersections that moved between tiers when the CV score was added. This tier-movement rate is the policy-relevant quantity. It directly captures how many intersections a municipality would inspect or intervene at differently if it used the enriched model rather than the baseline.

3.7.2. Signal decomposition: traffic volume and speed entanglement

Traffic volume channel

Whether the marginal association between the CV score and crashes reflects an exposure gradient besides a visual signal was examined at two levels: in the crash model directly, and in the CV training pipeline that produced the score. At the crash-model level, intersections were grouped into quartiles of the CV score, and mean crash counts and the covariate profile were compared across quartiles, to check whether a rising score tracks a parallel rise in traffic volume, speed, and priority regulation. The CV score itself, not crashes, was regressed on the six baseline predictors to measure how much of its own variation those predictors already explain. The residual of that regression, the part of the score unrelated to the conventional variables, was then correlated with crashes to test whether a purely visual remainder still tracks crash risk. The enriched model's conditional CV-score coefficient (Section 3.6.2) was further re-estimated as the same four predictor blocks used in Appendix G.2 were added sequentially, to locate at which stage of that build-up the coefficient shifts from a positive marginal relationship to

a negative conditional one. This connects directly to the leave-one-out finding reported in Section 3.6.3, where excluding major-road AADT alone reversed the coefficient's sign.

At the level of the CV training pipeline, an analysis was conducted to characterise what part of the visual score constitutes genuine free visual signal and what part reflects information that the survey context introduced. This distinction matters because the survey displayed a traffic volume band alongside each photograph as contextual information, intended to help respondents control for exposure and focus their judgement on intersection design rather than on the volume of traffic they expected to see there. However, the stratification dimensions that structured the pair schedule, namely junction type and priority regime, correlate structurally with traffic volume. Because it cannot be verified how consistently respondents applied the mental correction, the resulting Bradley-Terry scores may partly reflect perceived traffic volume rather than purely design-based visual safety. A model trained on those labels therefore has the opportunity to re-encode traffic volume as a visual cue rather than learning a purely compositional visual signal.

To isolate the free visual component, a volume-residualised target was constructed by subtracting from each Bradley-Terry score the mean score of its traffic volume band. A second CV head was then trained to predict this residualised score rather than the full score. Comparing the validation performance of the full-score head with that of the residualised-score head reveals how much of the headline skill depends on the designed traffic volume attribute versus on purely visual information.

Speed channel

A related but distinct concern is that the visual features the model uses to identify dangerous-looking intersections are associated not only with high traffic volume but also with higher speed limits. The volume-residualisation analysis addresses the traffic volume channel. The following test addresses the speed channel. The volume-residualised CV score and a binary indicator for the higher speed-limit regime were each residualised against the two traffic-volume terms, and the two residuals were then correlated, to test whether the part of the score left over once traffic volume is removed still carries a speed signature. The same procedure was repeated against the priority-regime, junction-type, and centrality indicators, to locate which structural attribute, if any, the residual score aligns with most closely. As a direct check on the crash model itself, the enriched NB2 specification was also re-estimated separately on intersections at or below, and above, the 30 km/h speed limit threshold, to test whether the conditional CV-score effect is present in both speed regimes or concentrated in one.

Interaction and principal component checks

A continuous interaction test extended the binary speed-threshold stratification above across all six baseline predictors. An interaction term between the CV score and each predictor was added to the enriched NB2 specification in turn, and evaluated with a likelihood ratio test and the resulting change in AIC. This tests whether the conditional CV-score effect is uniform across the network or varies systematically with traffic volume, speed regime, or priority regulation.

A principal component analysis then provided a second, decorrelation-based check. Applied jointly to the CV score and the two AADT terms, the three variables implicated in the suppression pattern, it produced three mutually orthogonal components. These were entered as NB2 predictors in place of the original variables, to test whether the CV score's association with crashes survives once its shared variance with traffic volume is removed by construction rather than by regression-based partialling alone.

3.7.3. Attention map analysis

To understand which image regions drove the CV model's predictions, this analysis relies on attention maps: visualisations showing which spatial regions of a photograph the model weighted most heavily when forming its prediction. Because DINOv2 was pretrained self-supervised, the appropriate explainability method is the attention from its CLS token, the token whose final representation the model uses to produce its output, to the individual image patches, taken from the final transformer block. This choice is supported by Caron et al. (2021), who show that in self-supervised vision transformers trained with the DINO objective, the CLS token's attention to image patches concentrates on semantically meaningful scene regions without any explicit supervision for this behaviour. The attention maps are overlaid

on the original photographs to show which spatial regions the model weighted most heavily when producing its prediction. The maps are used for identifying where a model "looks." They are not used for why that region predicts a certain label.

The maps serve two purposes. First, they inform the interview questions used in the expert validation session described below, which probe whether the cues the engineer reports attending to are consistent with the regions the model's attention emphasises. Second, they provide a qualitative check on model coherence. If attention consistently falls on traffic-relevant regions such as intersection geometry, lane markings, pedestrian facilities, and signal heads in high-risk images, this is consistent with the model having learned something about visual safety. If attention falls predominantly on peripheral elements such as vegetation or building facades, such regions could function as visual proxies for enclosure or restricted sightlines, or the model could instead be drawing on cues unrelated to safety. The maps alone leave that open.

3.7.4. Expert validation session

The expert validation session was a semi-structured interpretive session with a single traffic engineer from the Municipality of Rotterdam, lasting 60 minutes. Its purpose was not to validate the statistical model, but to assess whether the model's visual reasoning was recognisable to practitioners and to interpret the crash-model result in light of how engineers reason about intersection risk in practice.

The session consisted of three parts. A plain-language framing of the pipeline and null result opened the session, without regression notation and without steering toward a conclusion. A photo round followed, presenting four to six of the anchor photograph pairs used in the pairwise comparison survey. The engineer described what was visible and whether the attended features related to volume or to independent design characteristics. A final discussion addressed two questions: whether a CV visual layer would add anything to current network screening practice, and if the engineer reasoned from absolute crash counts or from crash probability per vehicle passage, a distinction that connects to the sub-proportional major-road elasticity ($\beta_1 = 0.693$) and the question of whether an offset specification would have been more policy-relevant.

Session output was processed into a thematic summary along five themes: traffic volume entanglement, visual assessment cues, comparing intersections beyond traffic volume, crash counts versus per-vehicle risk, and practical usability for priority ranking. No statistical analysis was performed. The session findings are reported in Section 5.5.4, with the full session guide and thematic summary given in Appendix J, and are referenced in the Discussion where relevant. A single participant does not provide statistical generalisability. This was an exploratory member check that added face-validity and practical grounding, not a representative survey of expert opinion.

4

Data sources and availability

4.1. Overview of data sources

All primary data sources required for this study are either openly available or have been secured through institutional access. Table 4.1 provides an overview of each source, its content, availability status, and role in the study.

Table 4.1: Data sources, content, availability, and role in the study

Data source	Content	Availability	Role in study
NWB (Nationaal Wegenbestand)	Road network geometry, junction locations, road attributes including speed limits and road type	Open data, Rijkswaterstaat, updated monthly	Intersection population definition, spatial delineation
VRI (Verkeersregelinstallatie) data	Traffic-signal installation locations, Rotterdam	Provided by the municipality	Used for the VRI locations
NWB intensiteiten	Modelled AADT per road segment per direction, Zuid-Holland, 2024 annual average	Provided by the municipality	Traffic volume for survey stimulus balancing and covariates in NB-model
Cyclomedia street-level imagery	Panoramic images at eye level	Municipal licence via Municipality of Rotterdam	Image source for the expert survey stimulus set and CV model training
BRON (Bestand geRegistreerde Ongevallen in Nederland)	Police-registered injury and fatal crashes, geocoded and coupled to the NWB, period 2020 to 2024	Open data, Rijkswaterstaat, updated annually	Dependent variable: crash counts per intersection in the NB model
Expert pairwise comparison responses	Binary pairwise safety judgements from 26 traffic engineers across 678 pairwise-comparison tasks	Self-generated via survey	Input to Bradley-Terry estimation
Bradley-Terry safety scores	Continuous latent safety score per intersection	Derived from expert responses	Training labels for the CV model in Phase 3
CV-derived safety scores	Predicted visual safety score per intersection across the full Rotterdam network	Derived from CV model inference	Additional predictor in the enriched NB model in Phase 4

The Bradley-Terry scores and the CV-derived scores are not raw inputs but data products generated within the study, in Phase 2 and Phase 3 respectively. Their construction is described in Chapter 3 and their distributions are reported in Chapter 5.

The crash outcome was deliberately restricted to injury and fatal crashes involving at least one motor vehicle. BRON is known to underregister lighter crashes and crashes involving only non-motorised road users, because the police are not called to every crash and minor crashes are often not recorded. Restricting the outcome to motor-vehicle injury crashes targets the segment of the registration that is most completely recorded and therefore carries less reporting noise. This limitation does not undermine the central test of the study. The expert visual safety score is structurally independent of the crash registration process and is therefore orthogonal to this bias.

4.2. Quantitative characterisation of the data sources

Table 4.2 reports, for each source in Table 4.1, the size of the original extract, the principal selection or matching step, and the population ultimately used in the study. Full methodological justification for each step is given in Chapter 3. Distributions and reliability diagnostics for the derived scores are reported in Chapter 5.

Table 4.2: Data source scope and selection outcomes

Source	Initial extract	Main selection step	Final population used
NWB (road network)	40.115 segments	aggregation to junctions ≥ 3 legs, ring-clip, deduplication	4.715 intersections
VRI (signal installations)	308 installations	type filter (VRI only), 30 m match to intersections	186 / 4.715 signalised
NWB intensiteiten (traffic volume)	342.511 segment records	Rotterdam match, junction aggregation, k-NN imputation	4.474 (96.8% direct coverage, 100% after imputation)
Cyclomedia (street-level imagery)	617.579 images	spatial filter around intersections, leg matching	12.677 / 13.957 legs matched (90.8%)
BRON (crash registrations)	600.218 crashes (NL, 2020-2024)	junction match, motor-vehicle + injury/fatal filter	1.058 crashes / 617 intersections; n = 4.474 modelling set
Expert pairwise comparisons	26 experts	678 pairwise tasks	145 intersections compared
Bradley-Terry scores	145 intersections	BT estimation (Section 5.2)	mean 0.568, SD 0.222
CV-derived scores	145 training labels	model inference (Section 5.3)	4.611 intersections scored, mean 0.516, SD 0.107

Table 4.3 summarises the two continuous exposure covariates carried into the Negative Binomial model on the modelling set.

Table 4.3: Modelling-set covariate summary (n = 4.474)

Variable	Mean	SD	Max
Traffic volume, major approach (vehicles/day)	3.114	4.091	28.550
Traffic volume, minor approach (vehicles/day)	1.286	2.494	23.700

The right-skewed shape of both traffic volume variables, visible in Figure 4.1, motivates the $\log(V + 1)$ transformation applied in Section 5.4.

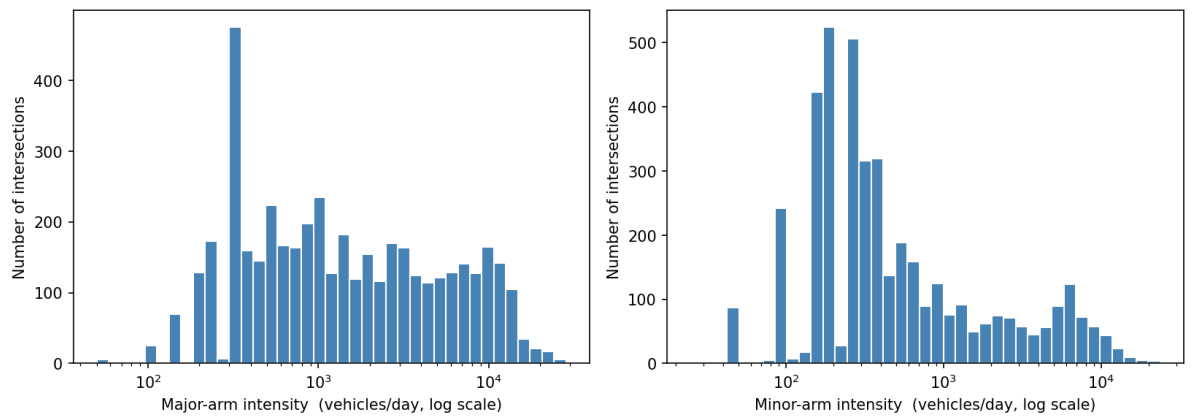


Figure 4.1: Distribution of major- and minor-approach traffic volume across the modelling set (n = 4.474), shown on a log-scaled x-axis.

The crash outcome shows the pronounced overdispersion and zero-inflation typical of intersection-level safety performance data (Table 4.4, Figure 4.2), which motivates the Negative Binomial specification adopted in Section 5.4 over a standard Poisson model.

Table 4.4: Crash count distribution (n = 4.474, injury/fatal, motor vehicle, 2020-2024)

Crashes	N intersections	%
0	3.857	86.2%
1	390	8.7%
2	128	2.9%
3+	99	2.2%
Mean	0.237	–
Variance-to-mean ratio	~2.6	–

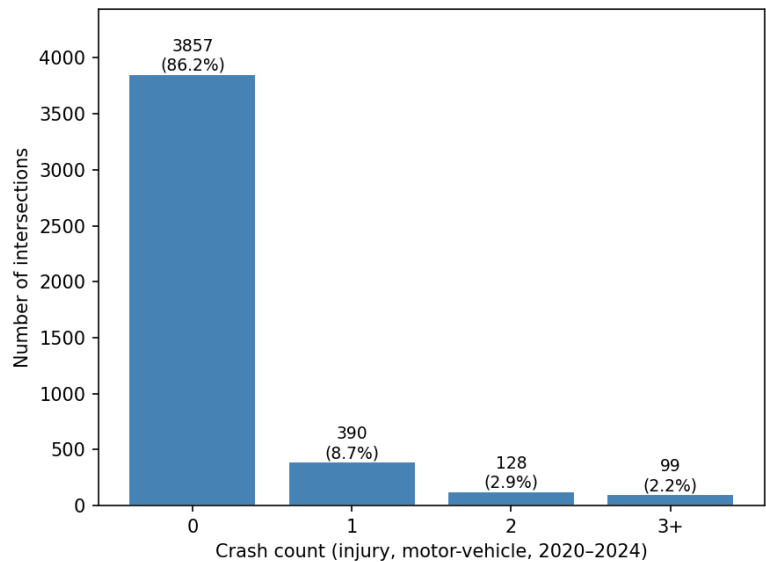


Figure 4.2: Frequency distribution of crash counts per intersection (n = 4.474), truncated at 3+ for readability.

5

Results

5.1. Intersection operationalisation (RQ1)

This section reports the outcomes of Phase 1: the definition of the intersection population, its stratification and the construction of the photographic dataset.

5.1.1. Population definition and stratification outcomes

The operationalisation described in Chapter 3 produced a study population of 4.715 distinct intersections within the Rotterdam ring road. This population follows from filtering the NWB to municipally managed driveable road types and retaining junctions with three or more connected segments, after which duplicate network representations of the same physical junction were merged. Two kinds of duplicates were resolved: dual-carriageway pairs, where a road with a central reservation is stored as two parallel segments and produces two junction points (merged within 25 metres when at least one connecting road carries the multiple-carriageway attribute FOW (Function Of Way) = 2), and bayonet artefacts, where administrative splits in the network create closely spaced junction pairs that are not genuine intersections (merged within 6 metres).

Each resulting intersection was assigned to one of six strata, formed by crossing the number of legs (T for three-leg junctions and 4+ for junctions with four or more legs) with the priority regime (uncontrolled, priority-controlled, or signalised). Table 5.1 reports the resulting distribution.

Table 5.1: Stratum distribution of the Rotterdam intersection population (N = 4.715)

Stratum	N	%
4+-leg, signalised	168	3.6%
4+-leg, uncontrolled	699	14.8%
4+-leg, priority	384	8.1%
T-junction, signalised	18	0.4%
T-junction, uncontrolled	2.894	61.4%
T-junction, priority-controlled	552	11.7%

The population is strongly imbalanced. Uncontrolled T-junctions alone account for 2.894 intersections, roughly 61 percent of the population, while signalised intersections of both types combined account for only 186 locations, about 4 percent. This imbalance is the reason the survey sample was drawn under an equal-allocation strategy rather than proportionally, as detailed below.

5.1.2. Photo matching outcomes

Across the network, 13.957 approach legs were identified across the 4.715 population intersections, an average of 3.0 legs per intersection. Of these, 12.677 legs (90.8%) were successfully matched to

a photograph. At the intersection level, 3,741 intersections (79.3%) achieved complete leg coverage, meaning every approach leg was matched and 100 intersections achieved no coverage at all. This network-wide matched photo set forms the basis from which the expert elicitation sample and CV model inputs described in later sections were drawn.

Matched photographs were captured at a mean distance of 20.3 metres from the intersection centroid (median 20.2 m), consistent with the 20-metre target distance. Figure 5.1 shows the distribution of capture distances. 96.7% of matches fell within the intended 15 to 30 metre range. A minority of legs (44.8%) were matched at distances below 20 metres. This was predominantly a consequence of the ranking criterion rather than limited photo availability. For 89.2% of these shorter matches, a candidate photograph at 20 metres or beyond existed but was not selected because a closer candidate lay nearer the target distance.

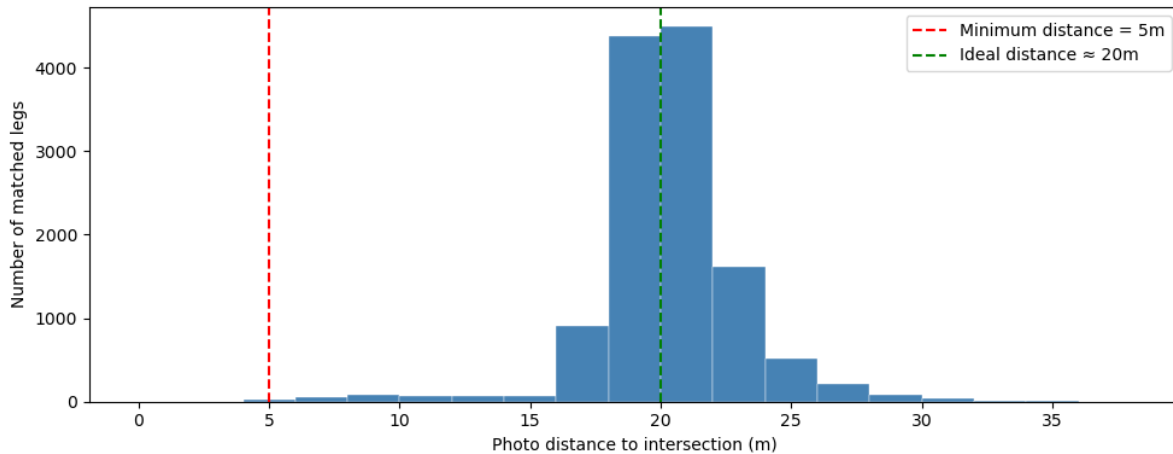


Figure 5.1: Distribution of matched photo capture distances

5.2. Expert elicitation and scoring (RQ2)

5.2.1. Preparatory interview outcomes

The preparatory interview yielded four outcomes relevant to the elicitation.

First, current municipal practice centres on an annual black spot analysis of the ten highest-risk intersections, supported by collision diagrams and manoeuvre analyses, and relies substantially on the tacit knowledge of experienced staff. The dedicated predictive model is recent and not yet routine, and prioritisation requires at least two of three signals: black spot status, model output, and external reports.

Second, visual comparison of intersections from photographs was judged feasible but complex, with traffic volume identified as the main confounder and rule-compliant behaviour required as a baseline assumption.

Third, the existing categorical classification by type, priority regime, and signalisation was endorsed as a sound basis for stratification.

Fourth, a single standardised image per intersection was considered adequate, provided it shows multiple road user types, with an accompanying traffic volume class regarded as useful context rather than a direct risk indicator.

5.2.2. Graph coverage

Twenty-six respondents completed the survey, together providing 678 pairwise comparisons across the 145 intersections. Figure 5.2 shows the resulting comparison graph, in which node size reflects the number of comparisons received.

The median intersection received 5 comparisons, while the mean was 9.35 and the maximum 93,

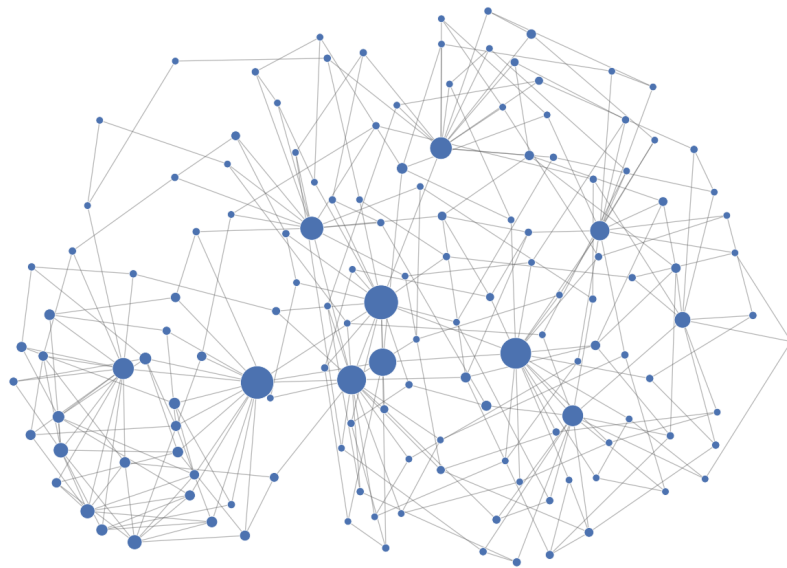


Figure 5.2: Comparison graph of the 145 sampled intersections

reflecting uneven coverage across the sample. This skew is visible in the figure as a small number of large nodes surrounded by many smaller ones. Fifteen intersections received three comparisons or fewer, and eleven received twenty or more. The implications of this skew for score reliability are examined in Section 5.2.4.

5.2.3. Bradley-Terry scores and score distribution

Across the 145 scored intersections the normalised safety scores had a mean of 0.568, a median of 0.590 and a standard deviation of 0.222 (Figure 5.3). The distribution was wide and slightly left-skewed, with marginally more intersections on the dangerous side of the scale. The small gap between mean and median indicates that this skew is mild.

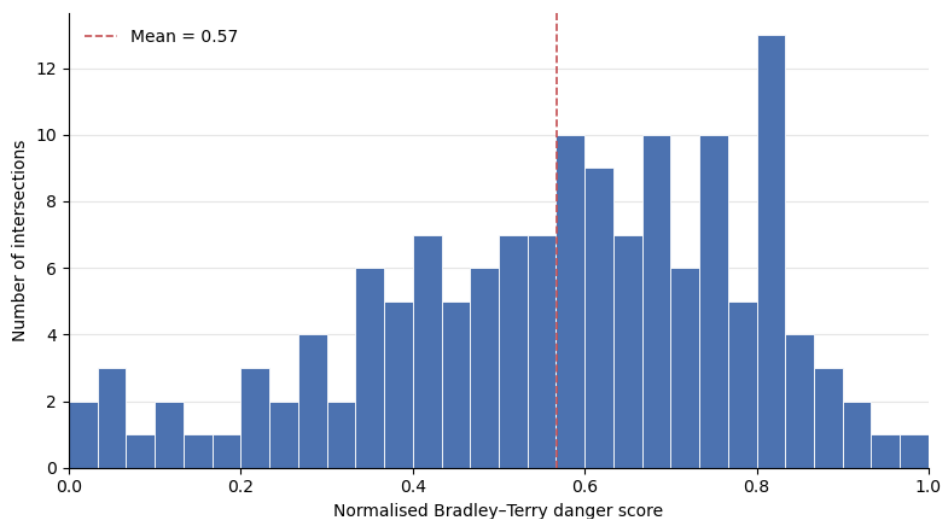


Figure 5.3: Distribution of normalised Bradley-Terry safety scores across the 145 scored intersections

The wide spread across the scale is favourable for the downstream regression that uses these scores

as continuous labels.

5.2.4. Score reliability and coverage

The two Bradley-Terry estimators corresponded closely (Pearson $r = 0.88$, Spearman $\rho = 0.89$), confirming that the intersection ranking is largely insensitive to the estimation method. The ranking was also stable across regularisation strengths from 0.01 upward (Spearman $\rho = 0.96$ to 1.00). The residual departure from perfect estimator agreement reflects the sparse comparison graph, in which thinly compared intersections are less sharply determined. Coverage implications are discussed in Section 5.2.2.

None of the 26 respondents produced a circular triad, indicating high within-rater consistency. This is read as the absence of any indication of inconsistency rather than as proof of consistency, since the sparse and partial schedule gave most raters limited opportunity to contradict themselves.

Between-rater agreement on the seven anchor pairs was mixed. On five of the seven, a clear majority of 78 to 90 per cent of raters selected the same intersection as more dangerous, while the remaining two pairs were evenly split. Mean agreement was 73.2 per cent. Fleiss' κ (-0.015) and Kendall's W (0.249) were low, but in a forced-choice design the chance-expected agreement is fixed at 0.5, making κ uninformative on near-equal pairs. The percent-agreement pattern is therefore treated as the primary indicator. The rank-consensus measures are reported as exploratory given the small and partly incomplete panel. On balance the anchors did not indicate a failure of shared expert judgement, and the labelling procedure was retained. The full table with percentage agreement and the corresponding pictures is given in Appendix E

5.3. Score generalisation (RQ3)

5.3.1. Architecture comparison

Four backbone architectures were compared on the task of predicting the Bradley-Terry safety score from a single reprojected street-level image: the supervised convolutional networks ResNet-50 and EfficientNet-B4, the contrastive language-image transformer CLIP ViT-B/16, and the self-supervised vision transformer DINOv2-Small. Each was evaluated as a frozen feature extractor under two families of downstream mapping. The first trained a neural regression head directly on the frozen features, tested across architectures from a single linear layer to two-layer multilayer perceptrons. The second reduced the feature set with principal component analysis and mapped the result to a safety score with Ridge or Partial Least Squares (PLS) regression; this route was applied to ResNet-50, DINOv2-Small, and CLIP ViT-B/16, but not to EfficientNet-B4, which was evaluated only through the neural-head route. All configurations were evaluated against a null model predicting the training-set mean (MAE = 0.181, RMSE = 0.222).

Table 5.2 reports the best-performing head for each of the four backbones, together with the second-best head where a backbone had two competitive routes: DINOv2-Small, CLIP ViT-B/16, and ResNet-50 are each shown with both a PCA-based route and the linear-head route, ordered by the best MAE achieved within each group. Full per-head results, alpha-weighting variants, and the complete PCA grid search are in Appendix F.

Table 5.2: Head-to-head comparison of backbone and route combinations on the validation set

Model	MAE	RMSE	R^2	Spearman ρ
DINOv2-Small, PCA + Ridge	0.159	0.181	0.329	0.554
DINOv2-Small, linear head	0.168	0.201	0.173	0.571
CLIP ViT-B/16, PCA + Ridge	0.165	0.189	0.273	0.488
CLIP ViT-B/16, linear head	0.176	0.215	0.081	0.497
ResNet-50, PCA + PLS	0.168	0.201	0.180	0.395
ResNet-50, linear head	0.180	0.216	0.047	0.302
EfficientNet-B4, linear head	0.177	0.211	0.112	0.502
Null model (predicts mean)	0.181	0.222	0.000	—

Two patterns are visible. First, the backbone mattered more than the head. DINOv2-Small reached a

significant rank correlation of 0.55 to 0.57, ResNet-50's linear head only about 0.30, with EfficientNet-B4 and CLIP near 0.50. Larger feature dimension and more pretraining data did not improve ranking, as both CLIP and EfficientNet trailed the smaller DINOv2 backbone. Second, fitting PCA on the full feature set rather than the labelled subset alone lowered the absolute error. The PCA and Ridge pipeline on DINOv2 features achieved the lowest validation error while retaining a near-best rank correlation, and was carried forward to the full network in Section 5.3.3.

5.3.2. Model performance on the validation set

The PCA and Ridge pipeline on DINOv2-Small features was selected as the final model, achieving the lowest validation error across all tested configurations (Table 5.3). It was subsequently retrained on all 137 labelled intersections before network-wide inference.

Table 5.3: Validation performance of the selected model, with five-fold cross-validation as a stability check

	MAE	RMSE	R^2	Spearman ρ
Selected model (single fold)	0.159	0.181	0.329	0.554
Five-fold cross-validation (mean $\pm$ SD)	0.164 $\pm$ 0.021	—	—	0.528 $\pm$ 0.041

The rank correlation of 0.554 ($p = 0.002$) corresponds to roughly 69 per cent of intersection pairs ordered correctly by assessed safety. The cross-validation figures track the single-fold result closely, indicating that performance is not an artefact of a favourable split. The root mean squared error of 0.181 carries into the crash model as a source of coefficient attenuation, addressed in Section 5.4.4.

A volume-residualisation diagnostic was run alongside the primary model. Predicting the volume-residualised score rather than the full Bradley-Terry score yielded a rank correlation of 0.278 ($R^2 = 0.068$), against 0.554 for the primary model. These figures are reported in Appendix F (Table F.6), their interpretation is taken up in Section 5.5.2. The full-score prediction is carried forward as the primary output for Phase 4.

5.3.3. Predicted score distribution across the full network

The selected model was applied to all Rotterdam intersections with an available reprojected photograph, producing one predicted safety score per intersection. Of the 4.615 intersections selected city-wide, approximately 4.611 were scored, with the remainder lacking a usable photograph.

Figure 5.4 shows the spatial distribution of predicted scores across the network, coloured by quartile. Scores vary throughout the network rather than concentrating in any single area.

The distribution of predicted scores is shown in Figure 5.5, with summary statistics in Table 5.4. The network mean of 0.516 sits slightly below the training-label mean of 0.568, and the spread is markedly narrower than that of the training labels ($SD = 0.107$ against 0.222). This contraction toward the centre is expected of a model that explains only about a third of the variance, and it implies that the predicted scores discriminate between intersections more weakly than the original expert labels do. The ranking is nonetheless preserved, which is what the crash model in Section 5.4 relies on.

Table 5.4: Summary statistics of the predicted safety scores across the scored network

Statistic	Value
N scored	4.611
Mean	0.516
Median	0.491
SD	0.107
Min	0.295
Max	0.897

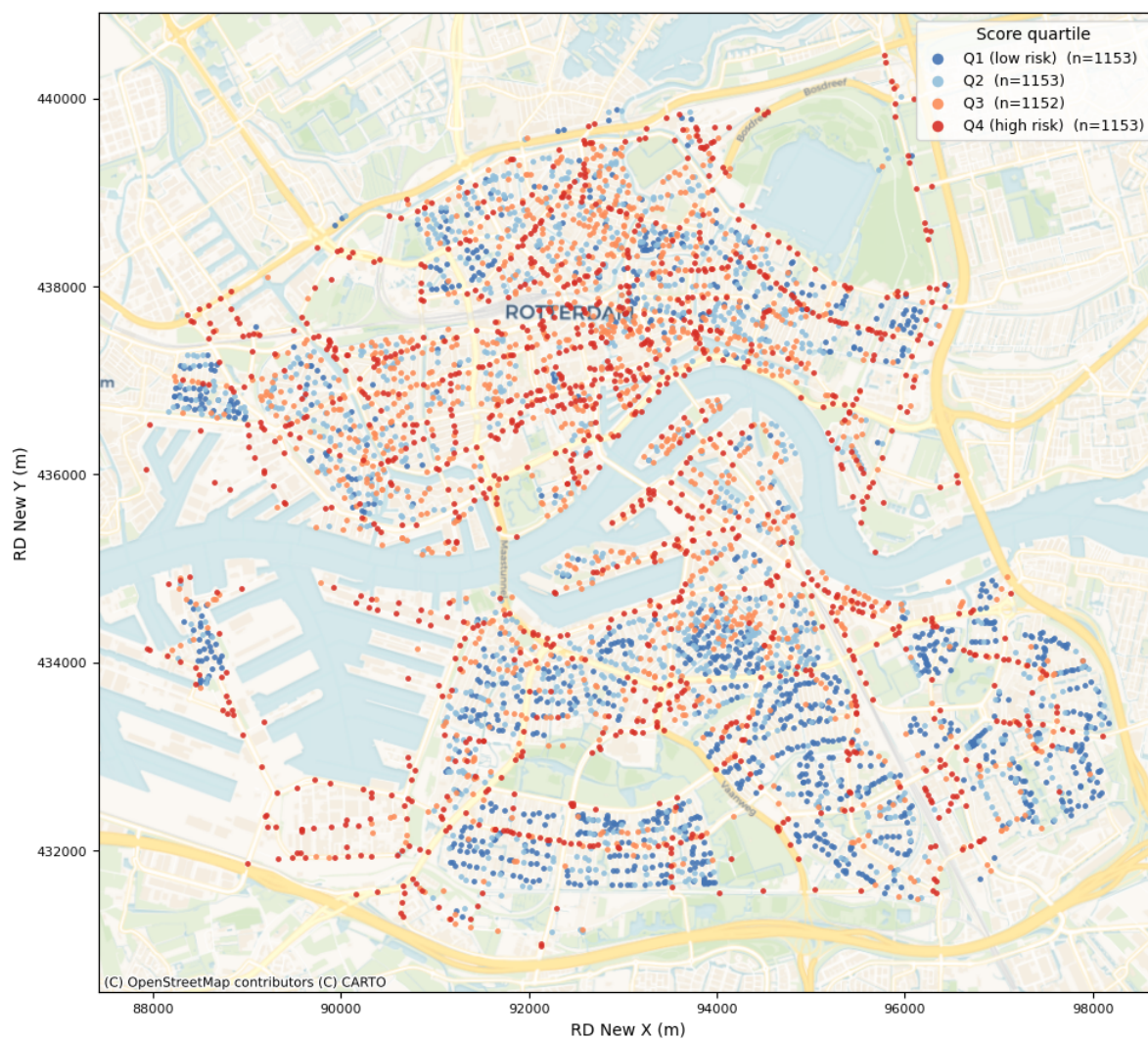


Figure 5.4: Spatial distribution of predicted safety scores across the Rotterdam intersection network (N = 4.611; DINOv2-Small backbone), coloured by quartile of the predicted CV-score

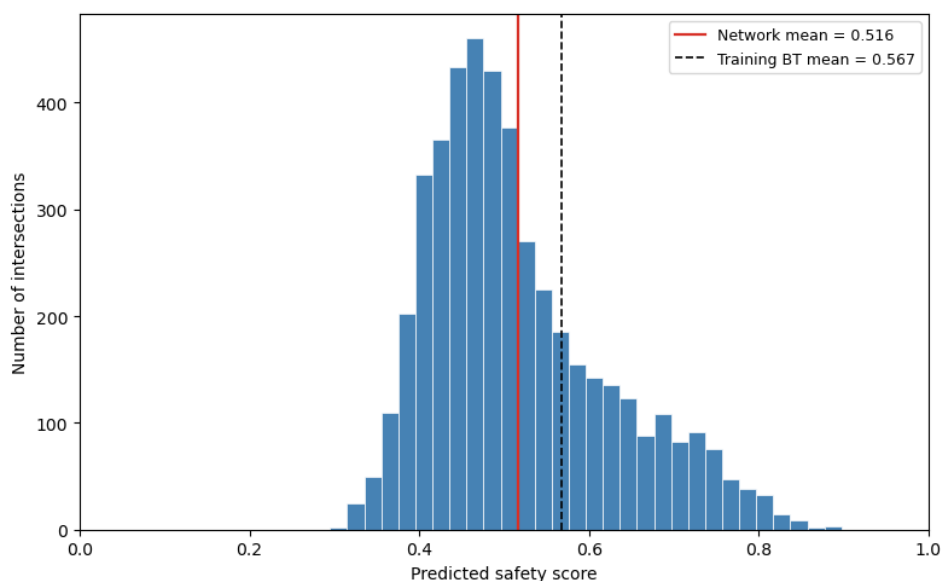


Figure 5.5: Distribution of predicted safety scores across the scored network (N = 4,611), with the network mean and the training Bradley-Terry mean indicated

5.4. Crash model and enrichment (RQ4)

5.4.1. Negative Binomial baseline: model fit and variable significance

The baseline model was estimated on the 4,474 intersections carrying a predicted CV score, with the 109 training and 28 validation intersections withheld to avoid leakage (Section 3.5.3). The outcome is the count of injury or fatal crashes involving at least one motor vehicle, 2020 to 2024. Six of eight candidate predictors were retained. The continuous speed limit and leg count were dropped before estimation as near-collinear with the 50 km/h and four-leg dummies ($r = 0.99$ and 0.78). The retained specification has variance inflation factors below 2.6 and an observation-to-parameter ratio of 639. AADT enters as $\log(V + 1)$, and categorical terms are referenced against a three-leg junction, no formal priority, and a 30 km/h regime.

Overdispersion is pronounced (Cameron-Trivedi estimate 1.015, $p < 0.001$; fitted dispersion 0.95), so NB2 is retained as the primary specification. Coefficients are reported in Table 5.5. The four significant terms act in the expected direction, with predicted crash counts rising in approach AADT, at four-leg junctions, and under a 50 km/h regime. Neither priority term is significant. The major-road elasticity of 0.69 lies within the 0.5 to 0.8 range typical of the SPF literature, whereas the minor-road elasticity of 0.14 falls below it.

Table 5.5: Negative Binomial (NB2) baseline coefficients

Term	β	SE	IRR	95% CI	p
Intercept	-8.720	0.462	—	—	<0.001 ***
Four-leg junction	1.245	0.095	3.47	[2.88, 4.19]	<0.001 ***
Log major AADT	0.693	0.065	2.00	[1.76, 2.27]	<0.001 ***
Log minor AADT	0.136	0.035	1.15	[1.07, 1.23]	<0.001 ***
50 km/h regime	0.294	0.117	1.34	[1.07, 1.69]	0.012 *
Signalised priority (VRI)	0.156	0.173	1.17	[0.83, 1.64]	0.366
Right-of-way priority	-0.102	0.132	0.90	[0.70, 1.17]	0.440
Dispersion α	0.950	0.116	—	—	<0.001 ***

5.4.2. Model enrichment: likelihood-ratio test and fit statistics

The enriched model adds the predicted CV score to the six baseline predictors on the same 4,474 intersections, and its incremental value is assessed by a likelihood ratio test against the baseline alongside

AIC and BIC (Table 5.6). The CV score does not significantly improve fit ($LR = 3.01$, $df = 1$, $p = 0.083$). AIC falls by only 1.0, below the conventional two-point threshold, and BIC rises by 5.4, favouring the baseline.

Table 5.6: Fit of the baseline and CV-enriched NB2 models

Specification	LL	AIC	BIC	LR	p
Baseline	-1935.53	3887.06	3938.30	—	—
Enriched	-1934.02	3886.04	3943.70	3.01	0.083

Figure 5.6 visualises this near-equivalence from two complementary angles. Panel (a) bins each model's own fitted values into deciles and plots the mean observed against the mean fitted crash count per decile. The baseline and enriched curves track each other closely across all deciles, both deviating similarly from perfect calibration at the lowest-risk end, so neither model systematically over- or under-predicts risk relative to the other. Panel (b) compares the two models directly at the intersection level, plotting each intersection's enriched fitted count against its baseline fitted count. The density concentrates almost entirely on the diagonal (Pearson $r = 0.997$, $R^2 = 0.995$), showing that adding the CV score leaves most individual intersections' predicted counts essentially unchanged rather than shifting them in offsetting directions. Together, the two panels support the marginal fit result in Table 5.6, showing that the near-zero aggregate improvement reflects a genuinely small effect rather than a cancellation of larger case-level changes.

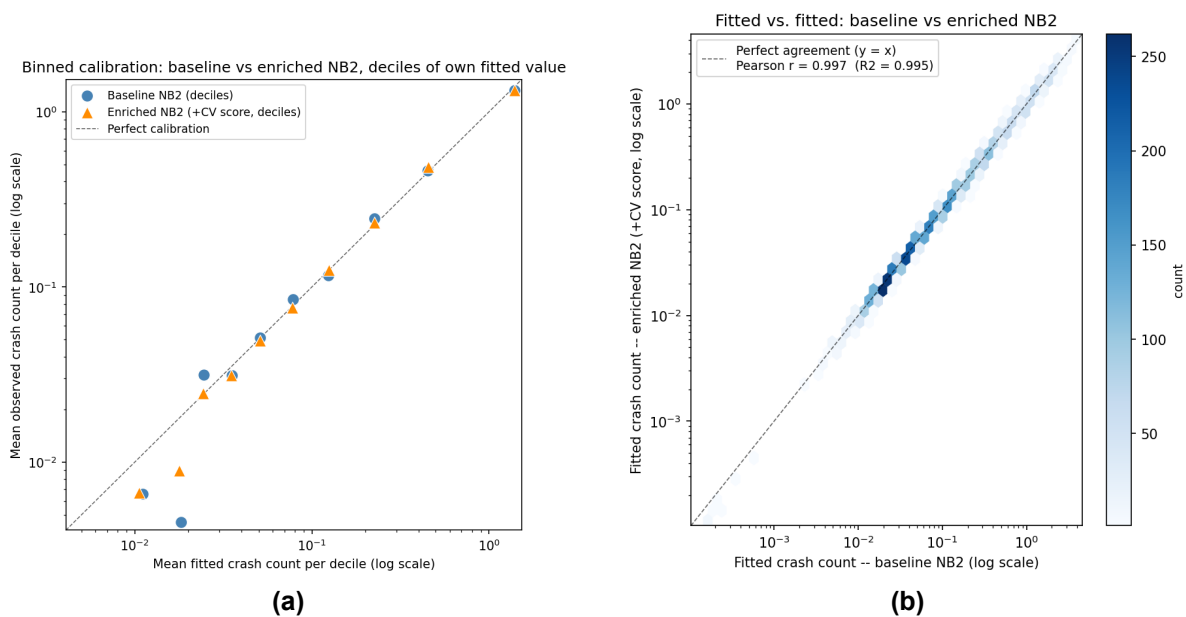


Figure 5.6: Calibration and agreement between the baseline and CV-enriched NB2 models ($n = 4,474$). (a) Binned calibration: mean observed against mean fitted crash count per decile of each model's own fitted value (log-log scale). (b) Fitted crash counts, baseline versus enriched, at the individual-intersection level (log-log scale, hexbin density).

5.4.3. Coefficient comparison and the CV-score effect

Beyond the aggregate fit statistics in Table 5.6, it is informative to compare the full coefficient set of the baseline and enriched models directly. Table 5.7 reports both specifications side by side. The six baseline coefficients retain their sign, magnitude, and significance pattern after the CV score is added. The largest shift is in log major AADT (β from 0.693 to 0.727), and the two priority terms remain non-significant in both specifications. The dispersion parameter α is essentially unchanged (0.950 versus 0.946), indicating that the overdispersion in the outcome is not explained away by the addition of the CV score.

Table 5.7: NB2 baseline versus CV-enriched model, full coefficient comparison

Term	β (base)	IRR (base)	95% CI (base)	p (base)	β (enr.)	IRR (enr.)	95% CI (enr.)	p (enr.)
Intercept	-8.720	0.0002	[0.0001, 0.0004]	<0.001 ***	-8.819	0.0001	[0.0001, 0.0004]	<0.001 ***
Four-leg junction	1.245	3.47	[2.88, 4.19]	<0.001 ***	1.250	3.49	[2.89, 4.21]	<0.001 ***
Log major AADT	0.693	2.00	[1.76, 2.27]	<0.001 ***	0.727	2.07	[1.81, 2.36]	<0.001 ***
Log minor AADT	0.136	1.15	[1.07, 1.23]	<0.001 ***	0.138	1.15	[1.07, 1.23]	<0.001 ***
50 km/h regime	0.294	1.34	[1.07, 1.69]	0.012 *	0.337	1.40	[1.11, 1.77]	0.005 **
Signalised priority (VRI)	0.156	1.17	[0.83, 1.64]	0.366	0.218	1.24	[0.88, 1.76]	0.217
Right-of-way priority	-0.102	0.90	[0.70, 1.17]	0.440	-0.075	0.93	[0.71, 1.20]	0.571
CV score (predicted)	—	—	—	—	-0.522	0.59	[0.33, 1.07]	0.083 .
Dispersion α	0.950	—	—	<0.001 ***	0.946	—	—	<0.001 ***

The CV-score enters on a MinMax-scaled [0, 1] range (mean 0.365, SD 0.177 on the modelling set), so a one-unit change spans the full observed range. The incidence rate ratio of 0.593 corresponds to a min-to-max reduction in predicted crash count of approximately 41 percent, with a 0.1 increment on this scale corresponding to a factor of 0.949. The estimate is negative but imprecise: the 95% confidence interval spans 1.0 and the coefficient is marginal ($p = 0.083$).

5.4.4. Robustness and measurement-error checks

As described in Section 3.6.3, five checks qualify the CV-score coefficient here.

Measurement error correction

Because the CV score entering the model is a prediction rather than a direct observation, its coefficient is biased toward zero by measurement error. A SIMEX correction for this bias ($\sigma_e = 0.18$) roughly doubles the magnitude of the CV-score coefficient ($\beta = -1.10$, Table 5.8), directionally consistent with the uncorrected estimate and with a simpler naive attenuation-factor correction ($\beta = -1.07$; Section G.3).

Spatial dependence check

The model treats each intersection as an independent observation, an assumption that would be violated if nearby intersections shared unmodelled risk factors. Deviance residuals from both models showed weak but significant spatial dependence (Moran's $I = 0.038$ baseline, 0.036 enriched, both $p = 0.001$; Appendix G.3). A cluster-robust refit of the enriched model left the CV-score coefficient unchanged ($\beta = -0.52$) but widened its standard error (0.301 to 0.366, $p = 0.083$ to 0.154), confirming the non-significant result under both specifications (Table 5.8).

Specification robustness

Reasonable alternative modelling choices, in how the score is coded, which controls are included, and how the outcome is defined, can also shift a coefficient's sign or size. Across six such checks, the sign remained negative in five (Table 5.8). Re-expressing the score as ranks or a binary indicator reduces the magnitude and raises the p-value. Excluding each of the six baseline predictors in turn leaves the coefficient negative and short of conventional significance, except when major-road AADT is excluded, which reverses the sign to positive ($\beta = +0.43$, Table H.1). This exception is interpreted in Section 5.5.2. The coefficient also weakens under the two broader crash-outcome definitions.

Block operationalisation and calibration check

The block operationalisation described in Section 3.6.3 was tested against the baseline by the same likelihood-ratio logic as the scalar enrichment above. Under the conventional asymptotic $\chi^2(5)$ reference, both blocks appeared significant. The PLS block reached LR = 40.54 ($p < 0.0001$) and the PCA block reached LR = 43.53 ($p < 0.0001$). The calibration check gave a different picture. Of the 200 placebo blocks, 24 reached a likelihood-ratio statistic at least as large as the PLS block's and 15 reached at least as large as the PCA block's, giving empirical p-values of approximately 0.12 and 0.075 respectively. Under the standard, uncalibrated model output, two of the five PLS components and three of the five PCA components were individually significant, and the PLS component most correlated with the Bradley-Terry score was not among them. These results are interpreted in Section 6.1.

Direct comparison on the expert-scored subset

This check compares the CV-generalised score directly against the human judgement it was trained to reproduce, to see whether any weakness in the result originates in the CV-generalisation step itself. On the 145 expert-scored intersections, a likelihood ratio test compared the raw human judgement, the CV score, and its volume-residualised version (Section 3.7.2) against the six conventional controls in turn (Table 5.8). Each measure is significantly associated with crashes on its own, with marginal incidence rate ratios ranging from 5.53 to 27.4 ($p \leq 0.020$, Table H.2), but none remains significant once the six controls are added. The raw judgement stands at $\beta = +0.06$ ($p = 0.931$), the CV score at $\beta = -1.72$ ($p = 0.096$), and the residualised score at $\beta = -0.45$ ($p = 0.538$). This absence of incremental significance already appears in the raw human judgement, so the result does not depend on the CV-generalisation step.

Table 5.8: Robustness of the CV-score coefficient

Robustness check	β	p
Primary specification (continuous)	-0.52	0.083
Measurement error (SIMEX)	-1.10	dir. only
Spatial dependence: cluster-robust SE	-0.52	0.154
Functional form: rank	-0.28	0.23
Functional form: binary, safest quartile	-0.06	0.72
Predictor exclusion: drop major AADT	+0.43	0.15
Outcome: injury, all road users	-0.29	0.30
Outcome: all crashes	-0.26	0.20
Bootstrap 95% CI (continuous)	[-1.13, +0.02]	incl. 0
Expert-scored subset: human judgement (n = 145)	+0.06	0.931
Expert-scored subset: CV score (n = 145)	-1.72	0.096
Expert-scored subset: residualised CV score (n = 145)	-0.45	0.538

5.5. Validation analysis (RQ5)

Phase 4 tested whether the visual score improves crash prediction in a statistically significant and robust sense. Phase 5 reports that result along a theoretical and practical dimension. The theoretical dimension, covering what the CV score represents beyond a statistical estimate, is addressed by the signal decomposition and the attention map analysis. The practical dimension, covering whether the result changes anything for practice, is addressed by the Empirical Bayes ranking comparison and the expert validation session.

5.5.1. Empirical Bayes-based ranking comparison

Adding the CV-derived visual safety score to the Negative Binomial model produces a near-identical intersection ranking to the baseline. The Empirical Bayes expected crash counts of the two models correlate at Spearman $\rho = 0.9992$ across all 4.611 scored intersections.

The tier analysis confirms this at the priority thresholds that govern actual resource allocation. At the top 10% tier, three intersections move up and three move down when the enriched model replaces the

baseline, yielding a tier-shift rate of 0.13%. At the top 25% tier, 21 intersections are promoted and 21 are demoted, a shift rate of 0.91%. In both cases, intersections that change tier do so by one position at the margin of the tier boundary.

5.5.2. Signal decomposition results: traffic volume entanglement and speed proxy

Traffic volume channel

Because the survey displayed a traffic-volume band alongside each photograph, the CV score may partly reflect traffic volume rather than a purely visual signal, and this channel tests how much of its association with crashes that volume confound explains. Table 5.9 shows mean crash counts rising roughly twelvefold from the safest to the most hazardous-looking quartile, accompanied by a parallel rise in traffic volume, speed, and priority regulation.

Table 5.9: Crash count and covariate profile by CV-score quartile

Quartile	Mean crashes	Share 50 km/h	Share priority	Median major AADT
Q1 (safest)	0.049	0.03	0.02	450
Q2	0.085	0.09	0.04	700
Q3	0.198	0.28	0.18	1.576
Q4 (most hazardous-looking)	0.614	0.79	0.54	7.000

A second test asks how much of the CV score itself, rather than crashes, the six baseline predictors already explain. Regressing the CV score on those predictors leaves little unexplained ($R^2 = 0.546$), and correlating that unexplained residual, the part of the score unrelated to the conventional variables, with crashes reduces the association to near zero (Spearman $\rho = -0.001$): whatever is left of the CV score once traffic volume, speed, and priority are accounted for carries essentially no independent link to crashes.

A third test asks whether the CV score's own coefficient in the crash model survives as controls are added one at a time. Re-estimating the conditional CV-score coefficient as the four baseline blocks are added sequentially traces a clear sign-flip path (full block-by-block detail in Table H.4): strongly positive with no controls, it collapses once exposure and four-leg junction are added, then moves increasingly negative as speed limit category and priority regulation complete the specification. Excluding major-road AADT alone, reported in Section 5.4.4, reverses the coefficient's sign.

When the traffic volume attribute embedded in the survey context is removed by predicting the residualised target instead, validation performance drops sharply (Table 5.10).

Table 5.10: Validation performance of the full and volume-residualised CV-score heads

Score head	Spearman ρ	R^2
Full CV score	0.554	0.329
Volume-residualised CV score	0.278	0.068

The coefficient stays negative and similarly marginal after this residualisation ($\beta = -0.48$, $p = 0.07$, $n = 4.474$, bootstrap CI including zero), and weakens further to $\beta = -0.41$ ($p = 0.12$) once additional structural controls are added. The survey buckets used for this residualisation retain substantial within-bucket variation in the underlying traffic volume (23.97 per cent of total variance), but the residual target itself shows no meaningful correlation with that remaining variation (Pearson $r = -0.01$, $p = 0.92$, $n = 132$).

Across the full network of 4.474 intersections, the CV score's raw correlation with traffic volume falls from $\rho \approx 0.64$ to $\rho \approx 0.33$ once the volume-residualised score is used instead.

Speed channel

A related concern is that the visual features driving the score may track higher speed limits rather than traffic volume alone, so this channel tests whether the coefficient's negative sign is concentrated in higher-speed environments. Net of traffic volume, the volume-residualised CV score correlates most strongly with the speed limit category (partial Spearman +0.29) and far less with priority or geometry. Stratifying the crash model by speed limit category shows that the conditional negative coefficient is absent at intersections with a speed limit of 30 km/h or below ($\beta = +0.05$, $p = 0.93$, $n = 3.140$) and present only at intersections above 30 km/h ($\beta = -0.74$, $p = 0.032$, $n = 1.334$).

Interaction and principal component checks

An interaction analysis extends this binary stratification to a continuous, joint test across all six baseline predictors, tested without a global multiple-comparison correction (Table 5.11). Four of the six interactions reach significance, all with a negative sign, log major AADT, log minor AADT, the 50 km/h regime, and right-of-way priority, indicating that the conditional CV-score effect is consistently more negative where traffic volume, speed, or priority regulation is higher. The interactions with four-leg junction type and with signalised priority do not reach significance, and the four significant results should be read as exploratory rather than confirmatory.

Table 5.11: Interaction between the CV score and each baseline predictor (NB2, $n = 4.474$)

Interaction	β	SE	p	LR p	ΔAIC
CV score $\times$ four-leg junction	+0.35	0.46	0.443	0.427	-1.37
CV score $\times$ log AADT major	-0.83	0.26	0.002	0.002	+7.92
CV score $\times$ log AADT minor	-0.43	0.16	0.005	0.009	+4.90
CV score $\times$ 50 km/h regime	-1.39	0.57	0.015	0.016	+3.81
CV score $\times$ Signalised priority (VRI)	+1.33	0.97	0.171	0.172	-0.13
CV score $\times$ right-of-way priority	-1.30	0.56	0.020	0.020	+3.39

As an independent check on the suppression pattern, principal component analysis was applied jointly to the CV score and the two AADT terms, and the three resulting orthogonal components were entered as NB2 predictors in place of the original variables. The first component (69.6% of variance), on which all three variables load positively, represents a shared exposure axis. On the second and third components the CV score loads negatively (-0.66 and -0.50) while the AADT terms load with mixed sign. All three components predict crashes positively (Table 5.12).

Table 5.12: Principal component decomposition of the CV score and the two AADT terms, with components entered as NB2 predictors ($n = 4.474$)

Variable	PC1	PC2	PC3
log AADT major	+0.62	-0.06	+0.78
log AADT minor	+0.54	+0.75	-0.37
CV score	+0.56	-0.66	-0.50
Variance explained	69.6%	20.3%	10.1%
NB2 coefficient (β , IRR, p)	+1.02, 2.78, <0.001	+0.53, 1.70, <0.001	+0.49, 1.63, <0.001

5.5.3. Explainability: CLS-to-patch attention analysis

CLS-to-patch attention maps were generated for a stratified sample of eight intersections spanning the full predicted score range, including two high-residual cases from the training split (see Appendix I, Figure I.1), and for a separate high-versus-low comparison set of six intersections (Figure 5.7).

Across high-scoring intersections, attention concentrates predominantly on the open road surface extending toward the horizon, rather than resolving onto specific infrastructure elements.



Figure 5.7: CLS attention overlay for the three highest- and three lowest-scoring intersections. Backbone: DINOv2-ViT-S14

Across low-scoring intersections, the attended regions shift markedly. Activation falls predominantly on building facades, dense roadside vegetation, and parked vehicles, rather than on carriageway or intersection geometry. These regions correspond to environments that are visually enclosed and characterised by lower traffic exposure, consistent with the speed-proxy finding.

For the two high-residual cases, attention is distributed broadly across the frame without a coherent focus on traffic-relevant regions. This diffuse pattern is consistent with the model's difficulty in accurately ranking these intersections, and with the modest overall R^2 of 0.33 reported in Section 5.3.

5.5.4. Expert validation session

The expert validation session was conducted with a single traffic engineer from the Municipality of Rotterdam. The session lasted 60 minutes and followed the protocol described in Section 3.7.4.

On the relationship between visual danger and traffic volume, the engineer indicated that it is difficult to assess the one independently of the other. He also added that risk is understood as a probability of an event and that one therefore always wants to relate danger to traffic volume. The features described as most salient, including the number of lanes, the signal infrastructure, and the scale of the conflict zone, were characterised as closely associated with how busy a location looks.

In the photo round, the cues reported as most salient were the overall complexity of the situation, whether the layout matched its function and use, and whether the intersection was legible and clear to the road user, together with sight lines and the location of conflict points. Softer judgements, such as whether priority and sub-conflicts would be clear to every road user, were described as central to the assessment but hard to read reliably from a single static photograph. For one complex case the engineer noted that video would have been preferable to a still image.

On the question of whether they reason from absolute crash counts or from crash probability per vehicle passage, the engineer described current practice as anchored in absolute recorded crashes at a location. This starting point structurally foregrounds high-volume intersections. The busiest locations recur at the top of the ranking, but because their exposure cannot be reduced, attention then shifts to the tier of locations behind them. A per-passage normalisation was regarded as conceptually defensible but difficult to apply in practice, and the engineer stressed that exposure cannot be equalised across the network, since not every road can be brought to a 30 km/h regime.

6

Discussion

6.1. Interpretation of the main finding

Adding the CV-derived visual safety score does not significantly improve crash prediction at Rotterdam intersections beyond the conventional infrastructure and traffic variables. The coefficient does not reach statistical significance under the primary specification ($p = 0.083$) or across nearly all robustness and sensitivity checks. The following sections unpack these estimates and the broader pattern of findings behind them.

The coefficient's shift as controls are added is a suppression pattern. The CV score correlates strongly with traffic volume ($\rho \approx 0.64$; Section 5.5.2), so when both enter the same regression, they compete for much of the same variance. If most of the CV score's variation is really just a stand-in for traffic volume, then once volume is already in the model, only a small, volume-independent sliver of the score is left to explain crashes. That sliver need not point in the same direction as the raw, unconditional correlation, and can even reverse it. Before accounting for traffic volume, the CV score and crash counts correlate strongly, with mean crash counts rising twelvefold from the safest to the most hazardous-looking quartile. Once traffic volume is added to the model, this association is no longer significant, because traffic volume explains most of the CV score's own variance and leaves little independent signal to correlate with crashes. Removing the major-road AADT term as a diagnostic reverses the estimate to positive, though still short of significance, confirming traffic volume as the specific source of the suppression (Section 5.5.2). An independent principal component decomposition corroborates this finding. The CV score loads negatively on the two components orthogonal to the shared exposure axis, while all three components remain significant positive predictors of crashes (Table 5.12). This same suppression mechanism plausibly explains why the coefficient reaches significance specifically among higher-speed intersections. Traffic volume and speed limit carry the most predictive weight in that subgroup, so partialling them out reassigns more of the CV score's shared variance away from it there, intensifying rather than departing from the same suppression pattern.

The correlation between the CV score and traffic volume is explained by what the score visually captures. In this network, dangerous-looking intersections are largely busy intersections. The visual cues that signal one largely signal the other. This pattern is not unique to the present data. Studies isolating individual streetscape features report it from the opposite direction, with visual openness associated with higher crash risk and enclosed, visually obstructed environments with lower risk (Chen et al., 2025). The model's own attention maps show a related pattern. At the lowest-scoring intersections attention falls on building facades, roadside vegetation, and parked vehicles rather than the carriageway, while at the highest-scoring intersections it spreads toward the open horizon rather than resolving onto specific design features. This is consistent with openness as a distinguishing visual feature. What separates a hazardous-looking from a safe-looking intersection may have less to do with any single design feature than with how open or enclosed its surroundings look.

A second, distinct mechanism can compound this entanglement. The Bradley-Terry aggregation reduces every pairwise judgement to a single scalar per intersection, so only one axis of discrimination

survives regardless of how many distinct visual cues entered a rater's decision. Experts plausibly still discriminate between intersections of comparable busyness on other grounds, but that finer discrimination competes for the same single axis as the coarse busy-versus-quiet contrast, where almost any rater picks the busier intersection. Once aggregated across the whole network, the coarse contrast dominates that axis and the finer signal is largely absorbed. This aggregation effect explains why compressing the judgement into one scalar could absorb any volume-independent signal that finer discrimination might carry into the dominant axis. The present analysis therefore bounds the incremental value of the visual impression at network scale, without testing whether a finer signal survives within exposure strata.

The compression account above is corroborated by retaining the visual signal as a richer, unreduced representation instead of one scalar (Section 5.4.4). A label-guided (PLS) and a label-blind (PCA) decomposition both appear significant under the standard asymptotic test, but neither is distinguishable from 200 randomly constructed alternatives under a calibration check (empirical $p \approx 0.12$ and 0.075), a match rate too high to attribute to pure noise, though it could partly reflect the added predictors' own correlation rather than genuine signal. What is clearer is that whatever diffuse association survives is not tied to the expert-elicited construct specifically. The label-blind decomposition outperforms the label-guided one on both the likelihood ratio and the number of significant components, and the one label-guided component correlated with the label is also the one most correlated with the baseline controls, and is itself non-significant, the same suppression pattern already seen in the scalar score.

On the 145 expert-scored intersections, the raw human Bradley-Terry judgement, tested directly against the six conventional controls under NB2, fails to reach significance, just as the CV score and its volume-residualised version do. Because a CV model cannot exceed the information contained in the label it was trained on, this rules out the CV-generalisation step as the source of the result. The limitation instead lies in the underlying construct itself, already present in the expert judgements the model was trained to replicate. The configural and sight-relational information discussed in Section 2.3 would be expected to produce substantial overlap between an integrated judgement and the coded attributes, alongside a residual, non-overlapping part. A score containing only noise and an intensity proxy would produce the same near-complete overlap found here, without any configural signal at all, so this result cannot distinguish between the two. What this test can establish is that the residual became entangled with the exposure component under the chosen operationalisation. The result therefore bounds the test rather than the hypothesis, leaving the question of configural information open for a different operationalisation to resolve.

This result appears to contradict a growing body of literature reporting that street-level visual features are associated with crash risk in predictive models (Yue, 2024), including a causal estimate that visual openness raises crash risk (Chen et al., 2025). This is not a genuine disagreement about the underlying effect, but a difference in what each study tests. Those studies test individual segmented features on their own predictive merit, without a conventional baseline of infrastructure and traffic variables against which the added value of the visual signal could be judged. This includes Kwon and Cho (2020)'s intersection-level analysis. The present study, by contrast, tests a holistic expert impression against that full baseline. This is the stricter test, under which the visual signal is largely absorbed by the conventional variables.

Taken together, this suggests a broader reading of what an integrated visual judgement actually measures. An integrated, network-scale visual judgement functions less as an additional measure of intersection design and more as an intuitive readout of the same broad risk factors a conventional model already captures, in this network chiefly exposure and speed. Retaining the visual signal in a richer, unreduced form does not change this picture. Any diffuse association a richer representation adds is not specifically tied to the expert-elicited construct either. This plausibly reflects a structural feature of integrated risk judgement, which forms under the time pressure of a single glance and prioritises the most perceptually salient cues over subtler configural or sight-relational ones. Read this way, the null result is informative beyond this network's data. It suggests that compressing many individual comparisons into a single score may preserve the same gross risk factors the conventional variables already capture, while absorbing the finer configural or sight-relational distinctions any one comparison might otherwise carry.

6.2. Implications for network screening practice and policy

For the Municipality of Rotterdam, a practical question is whether adding a CV visual layer would change which intersections are prioritised for inspection or intervention. The Empirical Bayes-based ranking comparison addresses this directly. Adding the CV score leaves the intersection ranking essentially unchanged, consistent with the coefficient's non-significance (Section 5.5.1). Because the CV score does not reliably shift the fitted crash count, it does not meaningfully reorder intersections by predicted risk either.

This finding is informative for policy. Where an established crash-and-exposure-based model already exists, the negligible tier-shift indicates that this citywide CV-based visual scoring layer is not worth adding for ranking purposes. Scarce safety budget is better directed at the established indicators than at building and maintaining a visual layer whose data-collection and modelling costs are not recovered in ranking improvement. One over-reading of this result should nonetheless be avoided. The absence of incremental ranking value is bounded by the construct and network tested here, and is not a general verdict that visual information is irrelevant to intersection safety.

This result carries a broader methodological implication that extends beyond the CV visual layer tested here. Before conventional variables were added, the marginal correlation between the CV score and crash counts was strong, and taken alone would have looked like a clear case for adoption. The incremental test against the full conventional baseline overturned that reading. The same distinction applies to any model that adds an image-derived or expert-elicited signal on top of an existing model built on standard variables, not to visual safety scoring specifically. Organisations evaluating such additions should require an incremental-value test against the full existing baseline before adoption, rather than relying on the new signal's marginal association with the outcome alone. Section 6.4 makes the corresponding point for the research literature.

The scores and pipeline produced by this study retain value independent of their ranking performance. The predicted safety scores are, in themselves, a new visually grounded data source for the municipality, useful for internal discussion among traffic engineers and policy staff. The underlying setup, expert elicitation feeding a CV model that generalises the judgement at scale, is also a first instance of a broader method that turns an expert visual judgement into a standing variable. Such a variable could in principle be added to other network-wide models as a fixed predictor, or built from other kinds of photographs to generalise a different expert judgement across a network.

A third, more forward-looking implication follows from the same entanglement with traffic volume that limits the score's value where traffic volume is already measured. The same correlation could make the score informative in its own right precisely where the conventional baseline cannot be computed. One such setting is a network where traffic counts are incomplete or unavailable, whether due to partial local coverage or, more substantially, because systematic traffic volume registration is largely absent, as is often the case in low- and middle-income contexts. There, the score could support a first-pass exposure estimate ahead of formal traffic counting, which could, in turn, inform where network screening or count infrastructure should be prioritised. A second setting is a newly built or substantially redesigned intersection, where neither traffic counts nor a crash history exist yet, regardless of how complete the underlying registration system otherwise is. Because the CV score only requires a photograph, or even an architectural rendering of a planned design, it could provide a temporary risk indicator ahead of construction, or before enough data has accumulated for the conventional model to apply.

These implications are untested beyond what this study measured. The CV model was validated against expert-elicited safety scores, not against measured traffic volume or crash outcomes directly, and has not been validated outside the Rotterdam network. Using it as an exposure or crash-history proxy in practice would require a dedicated validation step, testing how well the score predicts measured traffic volume or eventual crash outcomes directly, in the way Chapter 5 tested how well it predicts crashes under known traffic volume, and whether that relationship transfers to networks the model was not trained on.

A further, more technical implication concerns the intersection-consolidation pipeline built to define the population in the first place. Reconciling the NWB's inconsistent encoding of physical junctions, dual-carriageway direction splits and digitisation artefacts alike, into a clean, one-node-per-intersection population is a recurring data-preparation step for any network-level analysis, not one specific to visual

safety scoring. The empirically grounded merge rules developed here, and the manual quality checks behind them, could be reused directly by other studies working from the same network data, whatever their eventual research question.

A further, data-oriented implication concerns the crash-registration system underlying this and any future SPF work in Rotterdam. Supplementing BRON with linked ambulance or hospital records would directly improve the completeness of the crash outcome available for such work. This matters because BRON systematically underreports less severe and cyclist-only or single-vehicle crashes, a constraint that is not specific to this study and applies to any future Negative Binomial or other crash-frequency model built on BRON alone. Because this registration gap sits beyond what any single study can fix, Rotterdam is well placed to use its political weight, together with regional and national safety partners, to press for more complete crash registration as a shared data priority.

6.3. Limitations and mitigations

Construct validity

A limitation of this study concerns construct validity. Namely, an inherent entanglement between visual danger and other roadway variables that this study could not fully isolate. This study set out to measure a visual danger signal, defined as how dangerous an intersection looks, independent of how busy it is. Danger and traffic volume are not just statistically related, they are visually intertwined in the photograph itself. A rater looking at a busy scene cannot help but register how much traffic it carries while judging how dangerous it looks.

This entanglement also showed up at the measurement stage. Experts compared intersections pairwise, each represented by a single photograph, and judged which of the two looked more dangerous. Section 6.1 showed that the resulting scores are highly correlated with traffic volume. The survey design had anticipated this risk, showing each intersection's traffic volume class alongside the photographs specifically so respondents could mentally set exposure aside and judge the design itself (Section 3.4.2), but the entanglement persisted regardless. Visual danger is, ultimately, a latent construct that cannot be measured directly, and any attempt to elicit it requires choosing some operationalisation. The one used here turned out not to separate cleanly from traffic volume. What remains open is whether a different way of measuring visual danger would separate the two, and produce a different result.

Data pipeline and model assumptions

Beyond construct validity, further limitations concern the data pipeline that produced the intersection population and its measurements. There is no single canonical definition of what counts as one intersection in road network data. Standard databases such as the NWB encode some physical conflict points as multiple topology nodes, particularly where carriageway directions are stored separately (Boeing, 2025). Resolving this required choosing a merge threshold, and a stricter or looser choice would have produced a modestly different population and, in turn, a different sample and network-wide count. This is a boundary inherent to studying intersections as discrete units in network data, not a flaw specific to this pipeline.

The photo-matching procedure that produced the images entering both the survey and the CV model was itself imperfect. Network-wide, 90.8 per cent of approach legs were matched to a usable photograph, and 100 intersections received no usable photograph at all (Section 5.1). The matching protocol and manual quality control (Section 3.3.3) reduce, but cannot eliminate, the chance that an occasional selected photograph is not the most representative view of an intersection's geometry. This adds an unquantified source of noise upstream of the expert elicitation and CV-model stages discussed below, on top of the coverage gap itself.

The pipeline underlying the CV score compounds several sources of uncertainty. The expert elicitation stage carries label noise from the panel's uneven comparison coverage. The CV model itself carries its own prediction error, which attenuates the crash-model coefficient. One of several robustness checks reported in Chapter 5 corrects for the latter. It does not change which conclusion the data support. Because this correction addresses only the most precisely measured stage in the chain, the true uncertainty around the coefficient is plausibly larger than this single correction implies.

The Negative Binomial model treats intersections as independent observations. A direct test on the model residuals found weak but statistically significant spatial dependence between nearby intersections. Refitting the coefficient's standard error under this dependence widened it without changing the coefficient or the substantive conclusion. The assumption's violation therefore does not alter how the result should be read, and the check is reported in Section 5.4.4 as an additional robustness check.

Survey design choices

Beyond the data pipeline, two further limitations concern choices made in the survey design itself. The entanglement between visual danger and traffic volume, discussed above, is compounded by drawing pairwise comparisons across the full range of traffic volumes in the network, rather than restricting them to intersections of similar volume. This followed from the study's ambition to construct a network-wide measure, not a within-stratum one. Restricting comparisons to similar-volume pairs would have reduced the entanglement, but at the cost of the network-wide comparability the study was designed to achieve.

A related limitation concerns the elicitation stimulus itself. In the expert validation session, the interviewed traffic engineer described legibility and sub-conflicts as difficult to read reliably from a single static photograph (Section 5.5.4). Section 2.3 identifies exactly these aspects as the material an integrated judgement should add beyond coded attributes, which makes this a substantive constraint rather than a general shortcoming of images as a representational form. The limitation lies in the stimulus format, a single static image from one angle, not in a principled inability of imagery to carry the relational structure a photograph can in principle convey.

Sample size and statistical power

A more practical limitation is the small expert panel underlying that construct test. Twenty-six respondents provided 678 comparisons across 145 intersections, with uneven coverage. The median intersection received five comparisons and fifteen received three or fewer. Thinly compared intersections carry uncertain Bradley-Terry scores that enter the downstream chain as label noise. The hub-spoke pairing structure and shared anchor pairs mitigated this by keeping the comparison graph connected, and regularisation shrinks thin estimates toward the centre, but the residual uncertainty cannot be removed. The honest implication is that the expert construct tested here rests on a sparse and skewed empirical base, which limits how strongly the study can claim a stable, generalisable expert safety impression.

The scale available for CV-model training was similarly limited, since only a subset of the surveyed intersections carried a usable photo and could be split into training and validation sets (Section 3.5.3). A regression model trained and validated on so few images is unlikely to generalise as reliably as one trained on a substantially larger labelled set, and the model's moderate validation performance (Section 5.3.2) is consistent with this constraint. A larger labelled set would let the CV model reproduce the expert construct more precisely, independent of whether that construct itself proves informative for crash prediction.

The direct construct test, whether any assessment measure adds incremental value over the conventional controls, ran on only 145 intersections and returned no significant result. At that sample size the result is no evidence for added value rather than evidence of absence. The study is too underpowered on the expert-scored subset to distinguish a genuine null from an effect that simply cannot be detected. This is the principal limit on how strongly the null can be stated.

Crash-data quality and outcome scope

Police-registered crash data such as BRON systematically underreport certain crash types, particularly less severe crashes and those involving only cyclists or pedestrians. The crash outcome was therefore restricted to injury and fatal crashes involving at least one motor vehicle, the most completely registered segment of the data. This restriction is a deliberate trade-off rather than an oversight. BRON does not register cyclist-only and pedestrian-only crashes completely enough to support a reliable outcome on its own, and no substantially more complete alternative source was available for this study. The consequence is nonetheless substantive. The analysis and its conclusions say nothing about the risk of crashes involving only cyclists or pedestrians, a segment of road safety that matters greatly in a cycling-oriented city like Rotterdam. If the underreporting that remains within the motor-vehicle

outcome is not random with respect to the visual score, it could also bias the estimated coefficient in either direction. The robustness checks in Chapter 5 test broader, less completely registered outcome definitions as a direct check on this concern. The effect weakens rather than strengthens under those broader definitions, the opposite of what would be expected if underregistration were masking a real signal. A separate geographic boundary applies alongside this data-quality limitation. The analysis covers Rotterdam intra-ring intersections only, excluding roundabouts and interchanges, and no claim is made about transferability to other cities or network types.

6.4. Directions for future research

A next step is a larger and denser expert panel. A power analysis calibrated on the present effect size would set the intersection count and comparison budget required for a decisive test, since the current sample cannot distinguish a genuine null from an effect too small to detect. Both dimensions of that budget matter. More intersections would broaden network coverage, while more comparisons per intersection would address the coverage skew that currently limits Bradley-Terry score reliability. A larger labelled set would also give the CV model more training examples to learn the expert construct from, beyond what the current scale allows.

The second direction concerns the exposure entanglement established earlier in this chapter. In this study, the visual impression is correlated with traffic volume, and the survey design already attempted to reduce this by showing raters each intersection's traffic volume class so they could judge design separately from exposure. Weakening this coupling further suggests two possible strategies. One is to split the construct along the two dimensions identified in Section 2.3, legibility and sightline adequacy, rather than eliciting them as the single combined impression used here. These could be scored separately, since the two dimensions may be entangled with traffic intensity to different degrees, or a study could narrow its scope to just one of them. Either way, each score would remain a holistic rating of its own dimension rather than a decomposition into discrete attributes such as lane count, preserving the configural and sight-relational information that attribute-level coding cannot capture. A second is to retain the full holistic construct and instead weaken the entanglement through the comparison design itself, for instance by matching compared intersections more tightly on traffic volume. How best to do this while preserving network-wide comparability is unclear. Whether a purely visual construct carries crash-relevant information beyond exposure remains an open question that this study could not resolve with photographic stimuli alone.

The third direction is testing whether this study's results hold beyond Rotterdam. One need is the validation that the two forward-looking use cases identified in Section 6.2 would require. For low- and middle-income networks where systematic traffic volume registration is largely absent, this means testing how well the score predicts measured traffic volume directly, rather than crash counts conditional on known traffic volume. For newly built or redesigned intersections evaluated from a photograph or design rendering before any traffic or crash data exists, it means testing how well the score predicts eventual crash outcomes once enough data has accumulated to check it against. A second, broader need is to test the pipeline on a different city or network entirely, ideally with a different crash-registration system than BRON, to establish whether the entanglement between visual danger and traffic volume, and the resulting null result, is specific to Rotterdam and its data or generalises more broadly.

The fourth direction concerns the elicitation stimulus itself rather than the sample or the modelling chain. A single static photograph could not always convey the relational cues the interviewed traffic engineer identified as central to judging danger, such as whether priority and sub-conflicts read clearly from the design, or how sightlines change across an approach. For one complex case discussed in the expert validation session, the engineer stated that video would have been preferable to a still image (Section 5.5.4). Future work could test whether short video clips, or a small sequence of frames spanning an approach, let raters form a more complete judgement of this kind. Such a study would also need to establish whether the added visual information changes the entanglement with traffic volume documented in this thesis, rather than assuming it necessarily reduces it.

A fifth and final recommendation follows from the literature positioning developed in Section 6.1. Studies such as Kwon and Cho (2020) evaluate street-view visual features against crash or risk outcomes without benchmarking against a full set of infrastructure and traffic variables. This study's suppression

pattern shows how closely visual and exposure cues are entangled. Without that baseline, such a test risks crediting the visual feature with added value that is at least partly the shared exposure signal instead. Benchmarking visual features against a full baseline, as done here, would let the field distinguish a genuine visual contribution from one that mostly reflects unmeasured exposure.

Conclusion

This thesis asked whether formalising the visual safety impression of traffic engineers into a scalable model improves crash prediction at Rotterdam intersections beyond conventional infrastructure and traffic variables. The answer is no, not robustly, but the more informative reading is why. The visual impression tracks crashes closely yet adds little to their prediction, for two related reasons. Danger and traffic volume are visually intertwined in the photograph itself, so a rater judging how dangerous an intersection looks cannot help but also register how busy it is. Compounding this, aggregating every pairwise judgement into a single Bradley-Terry scalar could absorb any volume-independent signal that finer expert discrimination might carry into that same dominant contrast.

The computer vision model trained on expert pairwise comparison judgements generalises those judgements across the network with moderate skill. Its unconditional correlation with crashes runs in the expected direction, confirming the pipeline is internally consistent. However, when the resulting score is introduced into a safety performance function, it adds no significant incremental predictive value beyond the conventional variables. It does not reach statistical significance in the primary specification and in nearly all robustness checks, and weakens further under alternative functional forms and broader outcome definitions. Excluding major-road traffic volume reverses its sign. This is a suppression pattern, consistent with shared variance between the CV score and traffic volume rather than evidence of a substantive directional effect. Critically, the same result appears when the raw human Bradley-Terry judgements are tested directly on the expert-scored subset. A CV model cannot be expected to outperform its own training signal, so this points to a limitation in the underlying construct rather than the vision architecture. Expert visual safety impression, as elicited and formalised here, largely re-encodes what exposure and speed already capture.

For Rotterdam, the finding has practical value. The municipality's existing prioritisation already captures the same exposure and speed cues that a visual impression tracks, and the negligible ranking change produced by the visual layer does not justify the data-collection and modelling investment required to produce it. The conventional approach remains leading for network screening. The pipeline infrastructure and city-wide visual dataset produced by this study retain value for non-ranking purposes. Beyond Rotterdam, the same entanglement with exposure points to a second, more exploratory use. Because the visual impression tracks traffic volume, the score may hold value as a low-cost proxy for exposure where traffic volume data is unavailable. This is a possibility this thesis identifies but does not test and that would require dedicated validation against measured traffic volume before it could inform practice.

The methodological contribution is a validated pipeline for testing whether a visual safety predictor adds incremental value over conventional crash-model variables. The study offers a replicable blueprint for researchers making a similar claim, one that controls for exposure and verifies that the visual signal does not merely re-encode a channel the conventional variables already cover. A further, more general contribution is the intersection-consolidation methodology developed to define the population, reusable by any study that needs a clean, one-node-per-intersection population from road network data.

AI Statement

Generative artificial intelligence, specifically Anthropic's Claude, including Claude Code for programming tasks, supported this thesis throughout three phases: designing the study, implementing the analysis code, and writing and revising the manuscript. This statement describes how AI was used in each phase, how its output was checked, and how academic integrity was maintained throughout.

Use of AI in designing the study

During the design phase, AI was used as a sparring partner to think through methodological choices before they were finalised. This included things such as: discussing how to elicit expert visual judgements or weighing pairwise comparison against direct rating. AI proposed options and trade-offs. These were then verified online and checked. Which approach to eventually adopt, and why, remained a decision made by the author.

Use of AI in implementing the analysis code

AI, primarily through Claude Code, assisted with writing, debugging, and reviewing the code underlying the coding pipeline. AI-suggested code was run, inspected, and tested by the author before being incorporated. Where output did not match expectations, the underlying logic was checked independently and verified for correct working.

Use of AI in writing and revising the manuscript

During writing, AI supported the conversion of the manuscript to LaTeX and a systematic review of the text for internal consistency. This included checking that abbreviations were defined at their first use and consistently thereafter, for example standardising how NWB, BRON, and VRI were introduced, verifying that numeric claims matched across the chapters in which they appeared, and flagging inconsistent hyphenation and terminology. AI also proposed alternative phrasings for sentences that were repetitive or weirdly phrased. Proposed rewrites were checked against the intended meaning before being accepted, revised further, or rejected.

Validation and quality control

Across all three phases, AI output was treated as a proposal rather than a final answer. Findings and suggested changes were reported before being applied, factual and numeric claims were checked against the underlying data and text rather than taken at face value, and code changes were verified by running them. Several AI-identified issues were incomplete on a first pass, and were only caught and corrected because the author carefully verified the results.

Academic integrity

AI-generated content was reviewed before being incorporated into the manuscript. Where AI proposed text, it was checked against the source material. The interpretation of the results, the methodological decisions, and the conclusions presented in this thesis are my own. AI's role throughout was to support brainstorming, implementation, and revision, not to originate the thesis's claims on its own.

References

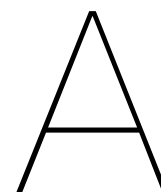
- Aarts, L. T. (2018). *Prestatie-indicatoren voor verkeersveiligheid (SPI's): Een overzicht van de kennis op dit moment over SPI's als basis voor risicogestuurd beleid* (tech. rep. No. R-2018-19). SWOV.
- Aarts, L., & van Schagen, I. (2006). Driving speed and the risk of road crashes: A review. *Accident Analysis & Prevention*, 38(2), 215–224. <https://doi.org/10.1016/j.aap.2005.07.004>
- Ambros, J., Borsos, A., & Sipos, T. (2017). Exploring an alternative approach to iRAP star rating validation. *96th Annual Meeting of the Transportation Research Board*.
- American Association of State Highway and Transportation Officials. (2010). *Highway safety manual* (1st, tech. rep.). AASHTO. Washington, D.C.
- Asian Development Bank. (2022). *CAREC road safety engineering manual 5: Star ratings for road safety audit* (tech. rep.). <https://www.adb.org/sites/default/files/publication/814761/carec-rse-manual-5-star-ratings-road-safety-audit.pdf>
- Bax, C., Eenink, R., Commandeur, J., & Loenis, B. (2017). *Een lichte variant van ProMeV toegepast in twaalf provincies* (tech. rep. No. R-2017-7A). SWOV. https://www.researchgate.net/publication/321379151_Een_lichte_variant_van_ProMeV_toegepast_in_twaalf_provincies
- Boeing, G. (2017). OSMnx: New methods for acquiring, constructing, analyzing, and visualizing complex street networks. *Computers, Environment and Urban Systems*, 65, 126–139. <https://doi.org/10.1016/j.compenvurbsys.2017.05.004>
- Boeing, G. (2025). Topological graph simplification solutions to the street intersection miscount problem. *Transactions in GIS*, 29(3), e70037. <https://doi.org/10.1111/tgis.70037>
- Bradley, R. A., & Terry, M. E. (1952). Rank analysis of incomplete block designs: i. The method of paired comparisons. *Biometrika*, 39(3/4), 324–345. <https://doi.org/10.2307/2334029>
- Caliendo, C., Guida, M., & Parisi, A. (2007). A crash-prediction model for multilane roads. *Accident Analysis & Prevention*, 39(4), 657–670. <https://doi.org/10.1016/j.aap.2006.10.012>
- Caron, M., Touvron, H., Misra, I., Jégou, H., Mairal, J., Bojanowski, P., & Joulin, A. (2021). Emerging Properties in Self-Supervised Vision Transformers. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 9650–9660. <https://arxiv.org/abs/2104.14294>
- Charlton, S. G., & Starkey, N. J. (2017). Driving on urban roads: How we come to expect the “correct” speed. *Accident Analysis & Prevention*, 108, 251–260. <https://doi.org/10.1016/j.aap.2017.09.010>
- Chen, H., Han, T., Chen, S., Guo, Z., Chen, Y., & Wu, M. (2025). Semantic4Safety: Causal Insights from Zero-shot Street View Imagery Segmentation for Urban Road Safety. *Proceedings of the 8th ACM SIGSPATIAL International Workshop on AI for Geographic Knowledge Discovery*, 146–156. <https://doi.org/10.1145/3764912.3770830>
- Cheng, W., & Washington, S. P. (2005). Experimental evaluation of hotspot identification methods. *Accident Analysis & Prevention*, 37(5), 870–881. <https://doi.org/10.1016/j.aap.2005.04.015>
- Colson, A. R., & Cooke, R. M. (2018). Expert elicitation: using the classical model to validate experts’ judgments. *Review of Environmental Economics and Policy*, 12(1), 113–132. <https://doi.org/10.1093/reep/rex022>
- Cook, J. R., & Stefanski, L. A. (1994). Simulation-Extrapolation Estimation in Parametric Measurement Error Models. *Journal of the American Statistical Association*, 89(428), 1314–1328. <https://doi.org/10.1080/01621459.1994.10476871>
- Cooke, R. M. (1991). *Experts in uncertainty: Opinion and subjective probability in science*. Oxford University Press.
- CROW. (n.d.). CROW Kennisbank. https://kennisbank.crow.nl/public/WOBI/ASVV_2021/Wegvakvoor_zieningen_op_erftoegangswegen/113086
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Hounsby, N. (2021). An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/2010.11929>

- Dubey, A., Naik, N., Parikh, D., Raskar, R., & Hidalgo, C. A. (2016). Deep learning the city: Quantifying urban perception at a global scale. *Computer Vision – ECCV 2016*, 196–212. https://doi.org/10.1007/978-3-319-46448-0_12
- Elvik, R., & Bjørnskau, T. (2017). Safety-in-numbers: A systematic review and meta-analysis of evidence. *Safety Science*, 92, 274–282. <https://doi.org/10.1016/j.ssci.2015.07.017>
- Elvik, R., Høye, A., Vaa, T., & Sørensen, M. (2009). *The Handbook of Road Safety Measures* (2nd). Emerald Group Publishing Limited. <https://doi.org/10.1108/9781848552517>
- Elvik, R., Vadeby, A., Hels, T., & van Schagen, I. (2019). Updated estimates of the relationship between speed and road safety at the aggregate and individual levels. *Accident Analysis & Prevention*, 123, 114–122. <https://doi.org/10.1016/j.aap.2018.11.014>
- Federal Highway Administration. (2017). *Road Safety Fundamentals: Concepts, Strategies, and Practices that Reduce Fatalities and Injuries on the Road, Unit 4: Solving Safety Problems* (tech. rep. No. FHWA-SA-18-003). U.S. Department of Transportation. <https://highways.dot.gov/safety/learn-safety/road-safety-fundamentals-html-version/unit-4-solving-safety-problems>
- Fleischmann, M., Vybornova, A., Gaboardi, J. D., Brázdová, A., & Dančejová, D. (2026). Adaptive continuity-preserving simplification of street networks. *Computers, Environment and Urban Systems*, 123, 102354. <https://doi.org/10.1016/j.compenvurbsys.2025.102354>
- Gemeente Rotterdam. (2019). *Verkeersveiligheidsmodel* (tech. rep.). https://verkeersveiligheidscoalitie.nl/wp-content/uploads/2019/09/Verkeersveiligheidsmodel_gemeente_Rotterdam.pdf
- Ghadi, M., & Török, Á. (2019). A comparative analysis of black spot identification methods and road accident segmentation methods. *Accident Analysis & Prevention*, 128, 1–7. <https://doi.org/10.1016/j.aap.2019.03.002>
- Hamim, O. F., & Ukkusuri, S. V. (2024). Towards safer streets: A framework for unveiling pedestrians' perceived road safety using street view imagery. *Accident Analysis & Prevention*, 195, 107400. <https://doi.org/10.1016/j.aap.2023.107400>
- Hammond, K. R., Hamm, R. M., Grassia, J., & Pearson, T. (1987). Direct comparison of the efficacy of intuitive and analytical cognition in expert judgment. *IEEE Transactions on Systems, Man, and Cybernetics*, 17(5), 753–770. <https://doi.org/10.1109/TSMC.1987.6499282>
- Han, L., Abdel-Aty, M., Joo, Y.-J., Zhai, S., & Wang, D. (2025). Intersection crash frequency analysis considering visual environment features using random parameter negative Binomial-Lindley model. *Transportation Research Record*, 2679(12), 578–598. <https://doi.org/10.1177/03611981251353186>
- Hardin, J. W., Schmiediche, H., & Carroll, R. J. (2003). The Simulation Extrapolation Method for Fitting Generalized Linear Models with Additive Measurement Error. *The Stata Journal*, 3(4), 373–385. <https://www.stata.com/merror/simex.pdf>
- Harpe, S. E. (2015). How to analyze Likert and other rating scale data. *Currents in Pharmacy Teaching and Learning*, 7(6), 836–850. <https://doi.org/10.1016/j.cptl.2015.08.001>
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition. *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778. <https://doi.org/10.1109/cvpr.2016.90>
- Hoerl, A. E., & Kennard, R. W. (1970). Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1), 55–67. <https://doi.org/10.1080/00401706.1970.10488634>
- Hunter, D. R. (2004). MM algorithms for generalized Bradley-Terry models. *The Annals of Statistics*, 32(1), 384–406. <https://doi.org/10.1214/aos/1079120141>
- Hwang, H., Kwon, S., Kim, Y., & Kim, D. (2024). Is it Safe to Cross? Interpretable Risk Assessment with GPT-4V for Safety-Aware Street Crossing. *2024 21st International Conference on Ubiquitous Robots (UR)*, 281–288. <https://doi.org/10.1109/UR61395.2024.10597464>
- Intini, P., Berloco, N., Cavalluzzi, G., Lord, D., Ranieri, V., & Colonna, P. (2021). The variability of urban safety performance functions for different road elements: an Italian case study. *European Transport Research Review*, 13(1), 30. <https://doi.org/10.1186/s12544-021-00490-6>
- iRAP. (2013). *iRAP Methodology Fact Sheet* (tech. rep.). <https://www.southnorfolkandbroadland.gov.uk/asset-library/imported-assets/cd-8.25-irap-factsheet-7.pdf>
- iRAP. (2023, April). EuroRAP RPS 1.0. <https://irap.org/rap-tools/light-ratings/eurorap-rps-1-0/>
- iRAP. (2025, October). CycleRAP. <https://irap.org/nl/cyclerap/>
- Janssen, S. T. M. C. (2004). *Veiligheid op kruisingen van verkeersaders binnen de bebouwde kom: Vergelijking van ongevalleerisico's* (tech. rep. No. R-2003-36). SWOV.

- Jolliffe, I. T., & Cadima, J. (2016). Principal component analysis: A review and recent developments. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 374(2065), 20150202. <https://doi.org/10.1098/rsta.2015.0202>
- Kaćan, M., Oršić, M., Šegvić, S., & Ševrović, M. (2020). Multi-Task Learning for iRAP Attribute Classification and Road Safety Assessment. *2020 IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC)*, 1–6. <https://doi.org/10.1109/itsc45102.2020.9294305>
- Kaćan, M., Ševrović, M., & Šegvić, S. (2024). Dynamic Loss Balancing and Sequential Enhancement for Road-Safety Assessment and Traffic Scene Classification. *IEEE Transactions on Intelligent Transportation Systems*, 25(11), 15628–15640. <https://doi.org/10.1109/TITS.2024.3456214>
- Kennisnetwerk SPV. (2023a). *Factsheet Risico-indicatoren* (tech. rep.). <https://aanpakspv.nl/wp-content/uploads/2024/11/Publicatie-Risico-indicatoren-2023.pdf>
- Kennisnetwerk SPV. (2023b). *Veilige kruispunten* (tech. rep.). <https://aanpakspv.nl/wp-content/uploads/2024/11/Publicatie-Factsheet-Veilige-Kruispunten.pdf>
- Khattak, M. W., Pirdavani, A., De Winne, P., Brijs, T., & De Backer, H. (2021). Estimation of safety performance functions for urban intersections using various functional forms of the negative binomial regression model and a generalized Poisson regression model. *Accident Analysis & Prevention*, 151, 105964. <https://doi.org/10.1016/j.aap.2020.105964>
- Kilani, O., Gouda, M., Weiß, J., & El-Basyouny, K. (2021). Safety assessment of urban intersection sight distance using mobile LiDAR data. *Sustainability*, 13(16), 9259. <https://doi.org/10.3390/su13169259>
- Klein, G. (1998). *Sources of power: How people make decisions*. MIT Press.
- Kwon, J.-H., & Cho, G.-H. (2020). An examination of the intersection environment associated with perceived crash risk among school-aged children: using street-level imagery and computer vision. *Accident Analysis & Prevention*, 146, 105716. <https://doi.org/10.1016/j.aap.2020.105716>
- Larkin, J. H., & Simon, H. A. (1987). Why a diagram is (sometimes) worth ten thousand words. *Cognitive Science*, 11(1), 65–100. <https://doi.org/10.1111/j.1551-6708.1987.tb00863.x>
- Li, H., & Chen, L. (2025). Traffic accident risk prediction based on deep learning and spatiotemporal features of vehicle trajectories. *PLoS ONE*, 20(5), e0320656. <https://doi.org/10.1371/journal.pone.0320656>
- Lord, D., & Mannering, F. (2010). The statistical analysis of crash-frequency data: A review and assessment of methodological alternatives. *Transportation Research Part A: Policy and Practice*, 44(5), 291–305. <https://doi.org/10.1016/j.tra.2010.02.001>
- Luce, R. D. (1977). The choice axiom after twenty years. *Journal of Mathematical Psychology*, 15(3), 215–233. [https://doi.org/10.1016/0022-2496\(77\)90032-3](https://doi.org/10.1016/0022-2496(77)90032-3)
- Maystre, L., & Grossglauser, M. (2015). Fast and accurate inference of Plackett–Luce models. *Advances in Neural Information Processing Systems*, 28, 172–180. <https://proceedings.neurips.cc/paper/2015/hash/2a38a4a9316c49e5a833517c45d31070-Abstract.html>
- McInerney, R., & Fletcher, M. (2013, January). *Relationship between Star Ratings and crash cost per kilometre travelled: the Bruce Highway, Australia* (tech. rep.). iRAP. https://www.academia.edu/68454551/Relationship_between_Star_Ratings_and_crash_cost_per_kilometre_travelled_the_Bruce_Highway_Australia
- Ministerie van Infrastructuur en Waterstaat. (2018). Strategisch Plan Verkeersveiligheid 2030 – Veilig van deur tot deur. <https://aanpakspv.nl/kenniscatalogus/strategisch-plan-verkeersveiligheid-2030-veilig-van-deur-tot-deur/>
- Montella, A. (2010). A comparative analysis of hotspot identification methods. *Accident Analysis & Prevention*, 42(2), 571–581. <https://doi.org/10.1016/j.aap.2009.09.025>
- Moran, P. A. P. (1950). Notes on continuous stochastic phenomena. *Biometrika*, 37(1/2), 17–23. <https://doi.org/10.2307/2332142>
- Naik, N., Philipoom, J., Raskar, R., & Hidalgo, C. (2014). Streetscore: Predicting the perceived safety of one million streetscapes. *2014 IEEE Conference on Computer Vision and Pattern Recognition Workshops*, 793–799. <https://doi.org/10.1109/CVPRW.2014.121>
- NDW. (n.d.). Wegvakken - NDW Docs. <https://docs.ndw.nu/handleidingen/nwb/nwb-basisstructuur/wegvakken/>

- Odiijk, M., & Van Petegem, J. (2024). *Naar ongevalmodellen voor Nederlandse wegtypen: Introductie, gebruik en ontwikkeling van CPM's, SPF's en CMF's* (tech. rep. No. R-2024-25). SWOV. Den Haag. <https://swov.nl/sites/default/files/bestanden/downloads/R-2024-25.pdf>
- O'Hagan, A. (2019). Expert knowledge elicitation: subjective but scientific. *The American Statistician*, 73(sup1), 69–81. <https://doi.org/10.1080/00031305.2018.1518265>
- Oquab, M., Darcet, T., Moutakanni, T., Vo, H., Szafraniec, M., Khalidov, V., Fernandez, P., Haziza, D., Massa, F., El-Nouby, A., Assran, M., Ballas, N., Galuba, W., Howes, R., Huang, P.-Y., Li, S.-W., Misra, I., Rabbat, M., Sharma, V., ... Bojanowski, P. (2024). DINOv2: Learning Robust Visual Features without Supervision. *Transactions on Machine Learning Research*. <https://arxiv.org/abs/2304.07193>
- Perez, J., & Fusco, G. (2025). Streetscape Analysis with Generative AI (SAGAI): Vision-language assessment and mapping of urban scenes. *Geomatica*, 77(2), 100063. <https://doi.org/10.1016/j.geomat.2025.100063>
- Persaud, B., & Lyon, C. (2007). Empirical Bayes before–after safety studies: Lessons learned from two decades of experience and future directions. *Accident Analysis & Prevention*, 39(3), 546–555. <https://doi.org/10.1016/j.aap.2006.09.009>
- Phelps, A. S., Naeger, D. M., Courtier, J. L., Lambert, J. W., Marcovici, P. A., Villanueva-Meyer, J. E., & MacKenzie, J. D. (2015). Pairwise comparison versus Likert scale for biomedical image assessment. *American Journal of Roentgenology*, 204(1), 8–14. <https://doi.org/10.2214/AJR.14.13022>
- Pollitt, A. (2012). The method of Adaptive Comparative Judgement. *Assessment in Education: Principles, Policy & Practice*, 19(3), 281–300. <https://doi.org/10.1080/0969594x.2012.665354>
- Pomerantz, J. R., & Portillo, M. C. (2011). Grouping and emergent features in vision: Toward a theory of basic Gestalts. *Journal of Experimental Psychology: Human Perception and Performance*, 37(5), 1331–1349. <https://doi.org/10.1037/a0024330>
- Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., & Sutskever, I. (2021). Learning Transferable Visual Models from Natural Language Supervision. *Proceedings of the 38th International Conference on Machine Learning*, 139, 8748–8763. <https://proceedings.mlr.press/v139/radford21a.html>
- Salesses, P., Schechtner, K., & Hidalgo, C. A. (2013). The Collaborative Image of the City: Mapping the Inequality of Urban Perception. *PLoS ONE*, 8(7), e68400. <https://doi.org/10.1371/journal.pone.0068400>
- Shbeeb, L. (2022). Road safety performance index: A tool for crash prediction. *Cogent Engineering*, 9(1), 2124637. <https://doi.org/10.1080/23311916.2022.2124637>
- Sørensen, M., & Mosslemi, M. (2009). *Subjective and objective safety: the effect of road safety measures on subjective safety among vulnerable road users* (tech. rep. No. 1009/2009). Institute of Transport Economics (TØI). <https://www.semanticscholar.org/paper/Subjective-and-Objective-Safety-The-Effect-of-Road-Sorenson-Mosslemi/76714ce1f3007a319fcb82c567f0e16f7c34bc73>
- SWOV. (2012, February). *Fact sheet Subjective safety in traffic* (tech. rep.). SWOV. https://swov.nl/system/files/publication-downloads/fs_subjective_safety_archived.pdf
- SWOV. (2020). How does the Dutch number of road deaths compare to the numbers elsewhere? <https://swov.nl/en/fact/intperspective-how-does-dutch-number-road-deaths-compare-numbers-elsewhere>
- SWOV. (2022, June). *Rotondes en andere kruispunten* (SWOV-factsheet). SWOV. Den Haag. <https://swov.nl/nl/factsheet/rotondes-en-andere-kruispunten>
- SWOV. (2023). *De Staat van de Verkeersveiligheid 2023* (tech. rep. No. R-2023-12). <https://swov.nl/sites/default/files/bestanden/downloads/R-2023-12.pdf>
- SWOV. (2025). SWOV en SWECO willen Crash Prediction Model doorontwikkelen. <https://swov.nl/nl/nieuws/swov-en-sweco-willen-crash-prediction-model-doorontwikkelen>
- Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. *Proceedings of the 36th International Conference on Machine Learning*, 97, 6105–6114. <https://proceedings.mlr.press/v97/tan19a.html>
- Theeuwes, J. (2021). Self-explaining roads: What does visual cognition tell us about designing safer roads? *Cognitive Research: Principles and Implications*, 6(1), 15. <https://doi.org/10.1186/s41235-021-00281-6>

- Theeuwes, J., & Godthelp, H. (1995). Self-explaining roads. *Safety Science*, 19(2-3), 217–225. [https://doi.org/10.1016/0925-7535\(94\)00022-u](https://doi.org/10.1016/0925-7535(94)00022-u)
- Thurstone, L. L. (1927). A law of comparative judgment. *Psychological Review*, 34(4), 273–286. <https://doi.org/10.1037/h0070288>
- Van Cranenburgh, S., & Garrido-Valenzuela, F. (2025). Computer vision-enriched discrete choice models, with an application to residential location choice. *Transportation Research Part A: Policy and Practice*, 192, 104300. <https://doi.org/10.1016/j.tra.2024.104300>
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, 5998–6008. <https://arxiv.org/abs/1706.03762>
- Verhavert, S., Bouwer, R., Donche, V., & De Maeyer, S. (2019). A meta-analysis on the reliability of comparative judgement. *Assessment in Education: Principles, Policy & Practice*, 26(5), 541–562. <https://doi.org/10.1080/0969594x.2019.1602027>
- Washington, S., Afghari, A. P., & Haque, M. M. (2018, April). Detecting High-Risk accident locations. In *Safe mobility: Challenges, methodology and solutions* (pp. 351–382, Vol. 11). Emerald Publishing Limited. <https://doi.org/10.1108/s2044-994120180000011018>
- Yao, Y., Liang, Z., Yuan, Z., Liu, P., Bie, Y., Zhang, J., Wang, R., Wang, J., & Guan, Q. (2019). A human-machine adversarial scoring framework for urban perception assessment using street-view images. *International Journal of Geographical Information Science*, 33(12), 2363–2384. <https://doi.org/10.1080/13658816.2019.1643024>
- Ye, X., Li, S., Gong, W., Li, X., Li, X., Dadashova, B., Li, W., Du, J., & Wu, D. (2025). Street view imagery in Traffic Crash and Road Safety Analysis: a review. *Applied Spatial Analysis and Policy*, 18(2), 50. <https://doi.org/10.1007/s12061-025-09653-7>
- Yue, H. (2024). Investigating the influence of streetscape environmental characteristics on pedestrian crashes at intersections using street view images and explainable machine learning. *Accident Analysis & Prevention*, 205, 107693. <https://doi.org/10.1016/j.aap.2024.107693>
- Zhang, F., Salazar-Miranda, A., Duarte, F., Vale, L., Hack, G., Chen, M., Liu, Y., Batty, M., & Ratti, C. (2024). Urban Visual Intelligence: Studying Cities with Artificial Intelligence and Street-Level Imagery. *Annals of the American Association of Geographers*, 114(5), 876–897. <https://doi.org/10.1080/24694452.2024.2313515>



Junction deduplication

The NWB road network database encodes the geometry of divided roads and complex junctions in ways that produce multiple topology nodes at what is physically a single intersection. Two distinct encoding patterns were identified in the Rotterdam network and addressed by the deduplication pipeline described in Section 3.3.1. The figures below illustrate one example of each pattern.

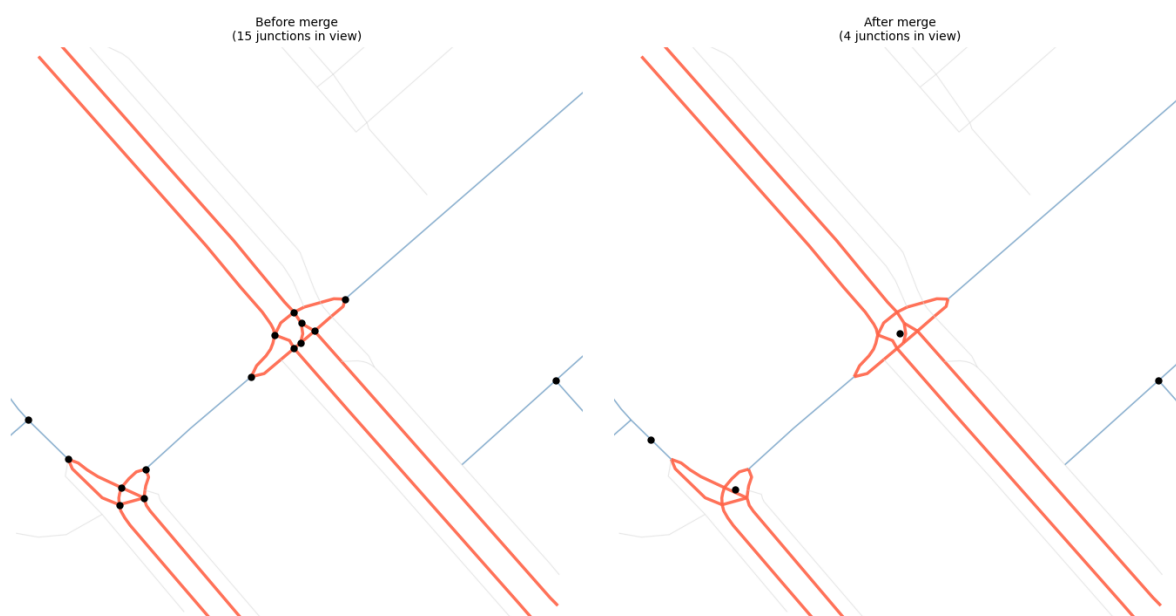


Figure A.1: Dual-carriageway (FOW=2) junction merge, one of the four largest such clusters (cluster size 8)

Figure A.1 shows a divided road. In the left panel, eight NWB junction nodes (black dots) are present within a small area where two dual-carriageway roads (red lines, FOW=2) intersect. Each carriageway direction is stored as a separate road segment, and each segment endpoint is registered as an independent junction. The result is a cluster of nodes that collectively represent one physical conflict point. In the right panel, the deduplication pipeline has resolved these eight nodes into a single canonical junction by identifying all pairs within 25 metres where at least one node had two or more FOW=2 connections, resolving transitive chains via connected components, and replacing the cluster with its centroid. Two additional nearby nodes visible in both panels fall outside the merge threshold and are correctly retained as separate intersections.

Figure A.2 shows a single-carriageway intersection (blue lines, FOW=3) in which the NWB has digitised one physical junction as two separate nodes placed within 6 metres of each other. This pattern, referred to here as a bayonet intersection, arises from digitisation artefacts rather than from the dual-carriageway

encoding that produces FOW=2 clusters. Because the two nodes are not connected to any FOW=2 segment, the FOW=2 signal would not detect this pair. The bayonet signal, based on pure proximity without a FOW filter, correctly identifies and merges them. In the right panel, the two overlapping dots have been replaced by a single canonical node at their centroid. The surrounding junctions are unaffected. This example is a cluster of size 2 (canonical JTE_ID 187264105), representative of the 18 bayonet-only merges identified across the Rotterdam network.

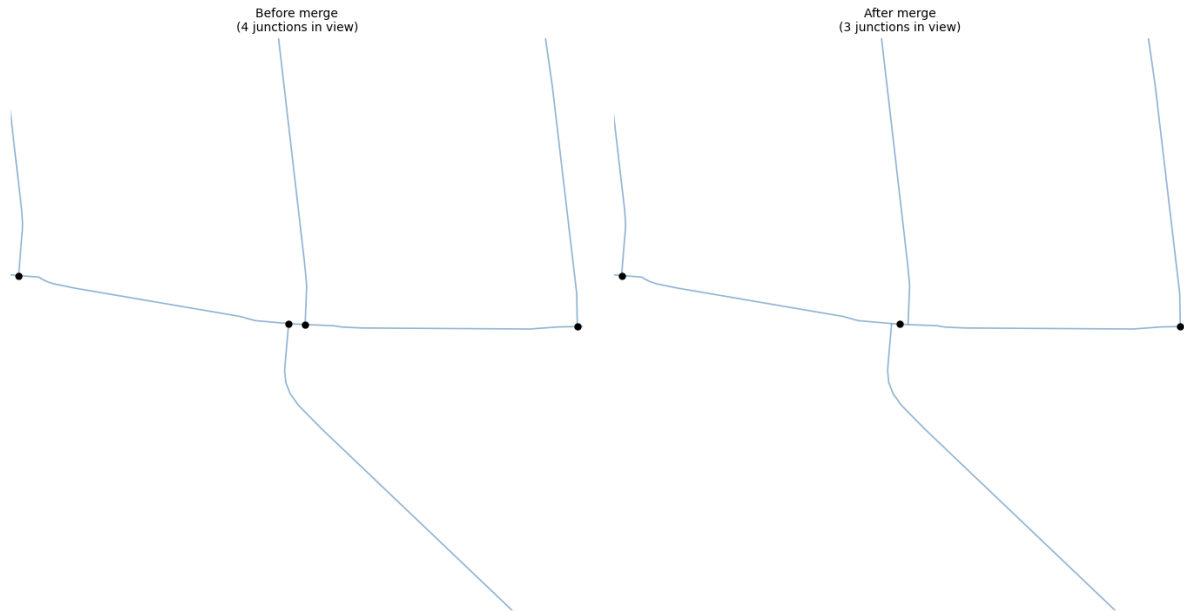


Figure A.2: Bayonet junction merge, where closely spaced administrative junction pairs are merged into a single node

B

Approach leg photo selection

The photo selection procedure matched each approach leg across the network to a single Cyclomedia photograph, based on three sequential criteria: capture distance to the intersection centroid, alignment between the candidate photograph's viewing direction and the leg's approach bearing, and containment within a leg-specific road corridor that excluded photographs on nearby parallel roads. Candidates satisfying all three criteria were ranked by proximity to a target capture distance of approximately 20 metres. Where two legs of the same intersection matched photographs looking at the intersection from near-identical directions, the less well-positioned of the two was treated as redundant and removed. Table B.1 reports the exact distance, bearing, and corridor thresholds governing this procedure.

Table B.1: Matching parameters for approach leg to photograph assignment

Parameter	Value	Role in the matching procedure
Minimum capture distance	5 m	Lower bound on candidate photo distance to the intersection centroid
Maximum capture distance	55 m (fallback; leg-specific corridors may extend further via chaining)	Upper bound on candidate photo distance to the intersection centroid
Target capture distance	20 m	Distance candidates are ranked against; closest-to-target candidate is selected
Bearing alignment threshold	45°	Maximum allowed deviation between a candidate photo's viewing direction and the leg's approach bearing
Road corridor width	3 m	Width of the leg-specific corridor a candidate must fall within, excluding photos on nearby parallel roads
Stitch threshold	1°	Maximum angular offset from a single camera direction beyond which two adjacent frames are stitched
Post-match deduplication threshold	20°	Bearing difference below which two matched photos for the same intersection are treated as redundant, the one farther from the target distance is dropped

The stitch threshold of 1° was set deliberately close to zero. At this value, a candidate photograph's viewing direction almost never falls within the threshold of the leg's approach bearing, so stitching was applied in effectively every match rather than being reserved for occasional edge cases. This departs from an earlier, higher threshold that was tested during development. The lower value was retained because it produced consistently better-aligned synthetic views, and it is reported here explicitly to avoid the impression that stitching was an exceptional rather than a routine step.

The figure below illustrates this procedure for one representative intersection from the Rotterdam network.

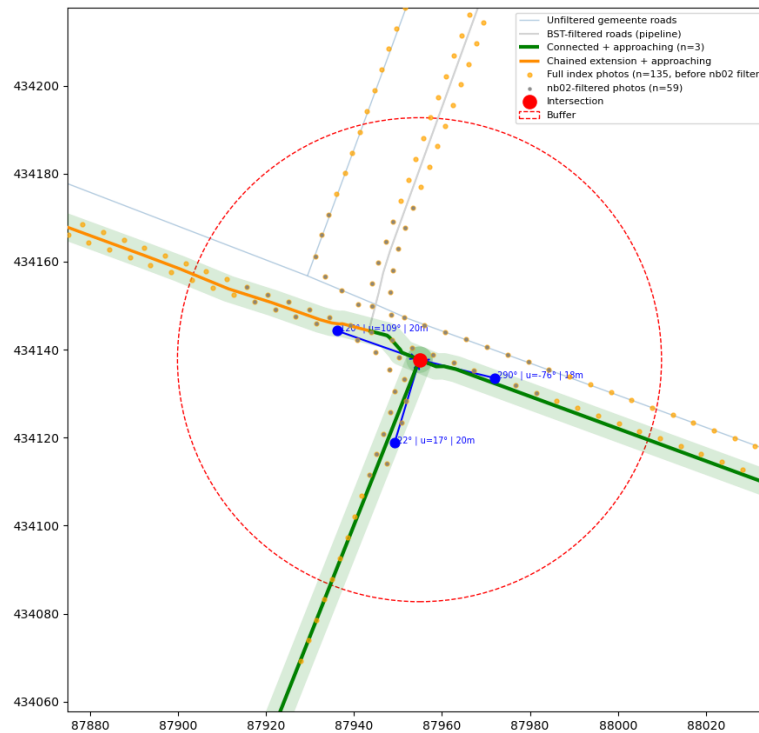
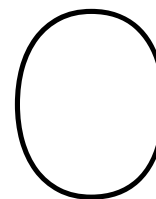


Figure B.1: Approach leg photo selection for intersection 175268012

Figure B.1 shows the photo selection outcome for a three-leg intersection 175268012 in RD New coordinates (EPSG:28992). The red dot marks the intersection centroid. The green segments indicate the NWB approach legs classified as connected and approaching. Where present, an orange segment shows a chained extension used to establish the approach bearing where the direct leg was too short to determine a reliable direction independently. The red dashed circle marks the outer search buffer within which candidate photographs were considered. Small dots show the Cyclomedia photo index within this buffer, before and after the directional pre-filter (135 photographs before, 59 retained after). For each approach leg, the selected photograph is marked with a blue dot, connected to the intersection centroid by a blue line, annotated with the approach bearing, the camera bearing of the selected frame, and the capture distance in metres. Network-wide matching outcomes, including overall coverage and the distribution of capture distances, are reported in Section 5.1.2.

This example is representative of the 3.741 intersections (79.3% of the population) for which complete leg coverage was achieved.



Preparatory interview

Interview guide and thematic summary

C.1. Interview guide

The semi-structured interview was organised around five thematic clusters. Questions were open-ended and used as a starting point for discussion rather than a fixed script.

Safety assessment practice

- How do you assess the safety of intersections in your day-to-day work?
- Do you use the traffic safety model, and if so, how?
- What role do black spot analyses and collision diagrams play in your assessments?

Behavioural factors

- To what extent can traffic behaviour be incorporated into intersection design?
- Are there design elements that actively invite unsafe behaviour?

Comparability of intersections

- Do you perceive differences between intersections when assessing them visually?
- Is it possible to compare the safety of intersections on the basis of photographs?
- What factors complicate such a comparison (e.g. traffic volume, intersection size)?

Photograph framing and survey design

- What do you think of the current cropping of the photographs?
- Would one photograph per intersection be sufficient, or is a per-approach perspective preferable?
- Should additional information such as traffic volumes be provided alongside the photographs?

Practical survey logistics

- How much time can respondents realistically dedicate to the survey?
- What is the most practical format and timing for distribution?

C.2. Thematic summary of interview outcomes

Safety assessment practice and the traffic safety model

The Municipality of Rotterdam employs a structured safety assessment process centred on an annual black spot analysis, in which the ten most dangerous intersections are identified and prioritised for

intervention. For individual intersections, a formal analysis protocol is used that incorporates collision diagrams and manoeuvre analyses. A dedicated traffic safety model has been in use for only a few years and is not yet routinely applied. The assessment process relies heavily on tacit knowledge held by experienced professionals, and a living document exists per intersection that consolidates all relevant data. For an intersection to be prioritised, it must satisfy at least two of three criteria: black spot classification, output from a predictive model, or signals from external reports.

The role of traffic behaviour

The professional indicated that traffic behaviour is difficult to incorporate directly into intersection design, as unsafe behaviour can occur regardless of the physical layout. Nevertheless, certain design elements were identified as actively inviting unsafe behaviour. The signal timing is one such element. Earlier two-phase traffic light systems, which released all east-west and north-south traffic simultaneously, have been replaced by more segmented signal plans, which have measurable effects on driver behaviour. Cycle times are capped at a maximum of two minutes. More generally, the professional noted that the relationship between design and behaviour is bidirectional and that an oversized or underused intersection can itself generate unsafe behaviour. Drone footage was mentioned as a desirable tool for capturing real behavioural patterns at intersections, and the professional noted that inspection visits already include video recordings for this purpose.

Comparability of intersections

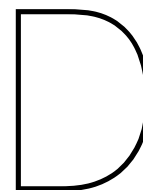
Comparing intersections visually was considered feasible but inherently complex. Traffic volume is a key confounding factor. A very large or complex intersection may produce careful behaviour simply because its complexity is self-evident, whereas the same design logic does not apply to smaller intersections. The professional emphasised that any comparison instrument would likely need to assume compliant behaviour as a baseline, since violations cannot be reliably predicted from physical layout alone. The existing categorical classification of intersections (by type, priority regulation, and VRI status) was considered a reasonable basis for stratification.

Photograph framing and information provision

The professional considered a single photograph per intersection a reasonable starting point, but stressed that the image should capture multiple road users, including cyclists and pedestrians, to reflect the full range of conflict types. A per-approach perspective (one photograph per leg) was seen as potentially more informative but also more demanding for respondents. The suggestion to include traffic volume data alongside photographs was discussed. The professional noted that volume figures could help contextualise the assessment but that experienced professionals often have an implicit sense of traffic volume from the image itself. The framing, distance to the intersection, and angle relative to the centre point were identified as relevant parameters for standardisation.

Practical survey logistics

A realistic time commitment per respondent was estimated at approximately fifteen minutes. The survey population was identified as approximately seventeen intersection designers, to be approached via their team coordinator. An online format distributed over a set period was considered the most practical approach. It was suggested that the survey be introduced during a regular design team meeting to maximise uptake.



Survey task interface

Figure D.1 reproduces the instructions presented to respondents prior to the pairwise comparison task. The instructions covered three elements.

The assessment criterion required respondents to indicate which of two intersections carries a higher probability of involvement in an injury or fatal road crash, including single-vehicle crashes and crashes involving at least one motor vehicle. Pure property damage was excluded.

Respondents were instructed to assume normal, rule-compliant behaviour and to base their judgement solely on what was visible in the photograph, disregarding any prior knowledge of the location. Three guiding questions were offered: the clarity of priority communication, the adequacy of sightlines, and the predictability of road user behaviour. The traffic volume class shown beneath each photograph was to be used to assess whether the visible layout was proportionate to the expected load, not as a direct risk indicator.

Practically, no tie option was available and respondents were always required to select one intersection, even when the perceived difference was small. No explanation of the choice was required.

Taakinstructies

Wat u gaat doen

U krijgt een reeks kruispuntparen te zien, elk weergegeven door één straatfoto. Geef bij elk paar aan **bij welk kruispunt de kans hoger is dat een auto betrokken raakt bij een letsel- of dodelijk verkeersongeval**.

Beschouw zowel enkelvoudige ongevallen (zoals controleverlies) als botsingen op de rijbaan waarbij minstens één motorvoertuig betrokken is (inclusief met fietsers, voetgangers of andere weggebruikers). Puur materiële schade telt niet mee.

Hoe te beoordelen

Ga bij uw beoordeling uit van normaal, regelconform rijgedrag — het gaat om de inherente veiligheid van de inrichting, niet om incidenteel afwijkend gedrag van weggebruikers.

Baseer uw beoordeling uitsluitend op wat zichtbaar is in de foto. Eventuele voorkennis over de locatie of het kruispunt telt niet mee.

De verkeersintensiteitsklasse wordt ter referentie getoond en betreft de tak die zichtbaar is in de foto. Gebruik deze om te beoordelen of de zichtbare inrichting passend is bij de belasting, niet als directe risicomaatstaf.

Klasse	Drempelwaarde
Laag	< 1.000 voertuigen/dag
Middelmatig	1.000 – 6.000 voertuigen/dag
Hoog	> 6.000 voertuigen/dag

Let daarbij op de algehele indruk van de inrichting:

- Hoe duidelijk communiceert de lay-out de voorrangrelaties?
- Hoe adequaat zijn de zichtlijnen?
- Hoe voorspelbaar is het verwachte gedrag van weggebruikers?
- U hoeft geen afzonderlijke kenmerken te scoren of uw keuze toe te lichten.
- Reageer op basis van uw eerste professionele indruk.

Praktische opmerkingen

- Er is geen gelijkspeloptie; kies altijd het kruispunt dat u risicovoller acht, ook als het verschil klein lijkt.
- U kunt korte pauzes nemen tussen de vergelijkingen

Klik op **Doorgaan als u klaar bent om te beginnen**.

Figure D.1: Screenshot of the pairwise comparison task interface

Rater agreement

E.1. Percentage agreement

Figures E.1 and E.2 show the percentage agreement per anchor pair across the 26 respondents, with the corresponding photographs, underlying the summary statistics reported in Section 5.2.4.



Figure E.1: Rater agreement, part one



Figure E.2: Rater agreement, part two

Full computer vision results

Tables F.1 through F.4 report validation performance for every regression head tested on each frozen backbone, underlying the head-to-head comparison in Section 5.3.1. Tables F.5 and F.6 report the PCA-and-Ridge configurations and the volume-residualised-score head respectively.

Table F.1: ResNet-50, all regression heads ($n_{\text{train}} = 109$, $n_{\text{val}} = 28$).

Head	MAE	RMSE	R^2	Spearman ρ	p-value
linear	0.180	0.216	0.047	0.302	0.119
mlp_128	0.180	0.219	0.021	0.314	0.104
mlp_32	0.181	0.220	0.010	0.306	0.113
mlp_2	0.180	0.218	0.031	0.307	0.113

Table F.2: DINOv2-Small, all regression heads ($n_{\text{train}} = 109$, $n_{\text{val}} = 28$).

Head	MAE	RMSE	R^2	Spearman ρ	p-value
linear	0.168	0.201	0.173	0.571	0.0015
mlp_128	0.173	0.208	0.122	0.569	0.0016
mlp_32	0.173	0.209	0.114	0.567	0.0017
mlp_2	0.170	0.206	0.136	0.576	0.0013

Table F.3: EfficientNet-B4, all regression heads ($n_{\text{train}} = 109$, $n_{\text{val}} = 28$).

Head	MAE	RMSE	R^2	Spearman ρ	p-value
linear	0.177	0.211	0.112	0.502	0.009
mlp_128	0.179	0.217	0.061	0.470	0.015
mlp_32	0.180	0.219	0.049	0.454	0.020
mlp_2	0.180	0.221	0.029	0.516	0.007

Table F.4: CLIP ViT-B/16, all regression heads ($n_{\text{train}} = 109$, $n_{\text{val}} = 28$).

Head	MAE	RMSE	R^2	Spearman ρ	p-value
linear	0.176	0.215	0.081	0.497	0.010
mlp_128	0.180	0.221	0.031	0.478	0.013
mlp_32	0.180	0.222	0.023	0.504	0.009
mlp_2	0.181	0.223	0.015	0.486	0.012

Table F.5: PCA-based configurations (Ridge and PLS)

Configuration	MAE	RMSE	R^2	Spearman ρ	p-value
DINOv2-Small, $k = 5$, $\alpha = 0.1$ (NB03b grid)	0.156	0.181	0.335	0.507	0.006
DINOv2-Small, $k = 5$ (NB04 re-evaluation)	0.159	0.181	0.329	0.554	—
CLIP ViT-B/16, $k = 10$, $\alpha = 0.01$	0.165	0.189	0.273	0.488	0.008
ResNet-50, PLS, $k = 15$, L2-normalised	0.168	0.201	0.180	0.395	0.037

The two DINOv2 rows differ because the NB03b grid search and the NB04 re-evaluation applied slightly different fits to the same validation set. The NB04 figure is the one used for the head-to-head comparison in Section 5.3.1 and for full-network inference.

Table F.6: Residualised-score head (NB03e), validation metrics ($n_{\text{val}} = 28$).

Metric	Value
MAE	0.149
RMSE	0.179
R^2	0.068
Spearman ρ	0.278

The MAE and RMSE for the residualised-score head are expressed on the residualised-target scale rather than on the $[0, 1]$ score scale, and are therefore not directly comparable with the absolute-error figures in the preceding tables. Only the Spearman ρ is comparable across tables. The role of this head is discussed in Section 5.3.2.

G

Baseline crash model, predictor selection and diagnostics

This appendix documents the predictor selection and goodness-of-fit checks underlying the Negative Binomial baseline reported in Section 5.4.1.

G.1. Candidate predictors and collinearity

Eight conventional candidates were screened before estimation. Two were removed as near-collinear with retained dummies: the continuous speed limit correlated at 0.99 with the 50 km/h regime, and the leg count at 0.78 with the four-leg dummy (Figure G.1). The six retained predictors show no remaining collinearity of concern, with all variance inflation factors below 2.6 (Table G.1).

Table G.1: Variance inflation factors for the six retained predictors

Predictor	VIF
Four-leg junction	1.21
Log major AADT	2.55
Log minor AADT	1.66
50 km/h regime	2.03
Signalised priority (VRI)	1.41
Right-of-way priority	2.15

G.2. Incremental specification

The specification was built in four nested blocks, each tested with a likelihood ratio test and AIC (Table G.2). Exposure, four-leg junction, and speed each improve fit ($\Delta\text{AIC} > 2$, $\text{LRT } p < 0.05$). The priority block does not ($\Delta\text{AIC} +0.53$, $\text{LRT } p = 0.104$). It is retained for specification completeness rather than fit.

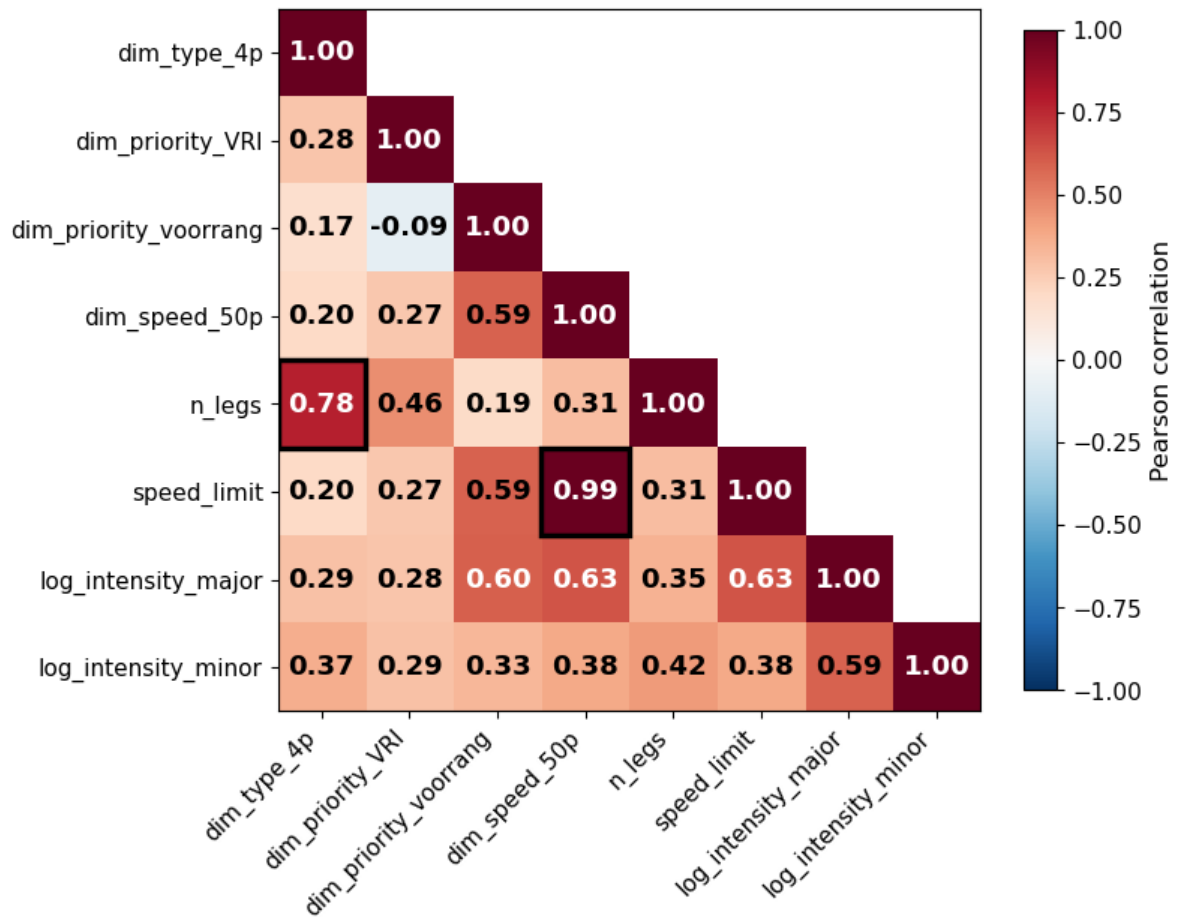


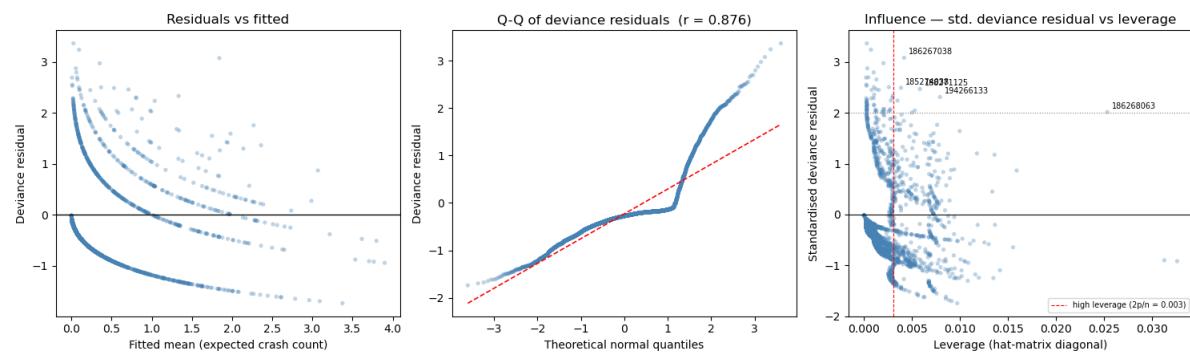
Figure G.1: Pairwise correlations among the eight candidate predictors. The two boxed pairs drove exclusion: leg count with the four-leg junction ($r = 0.78$), and speed limit with the 50 km/h regime ($r = 0.99$)

Table G.2: Incremental block-building of the baseline model

Block	Predictors	Log-likelihood	AIC	Δ AIC	LRT χ^2	df	p
M1 — Exposure	2	−2024.93	4057.87	—	—	—	—
M2 — + Four-leg junction	3	−1940.99	3891.98	+165.89	167.89	1	<0.001
M3 — + Speed	4	−1937.79	3887.59	+4.40	6.40	1	0.011
M4 — + Priority	6	−1935.53	3887.06	+0.53	4.53	2	0.104

G.3. Residual diagnostics

Residual behaviour is summarised in Figure G.2. The deviance residuals show no systematic trend against the fitted mean. The banding in panel (a) is the expected signature of low integer counts rather than misspecification. The normal Q-Q plot (panel b) is approximately linear through the centre ($r = 0.876$), with the heavier tails typical of discrete count data. The influence diagnostic (panel c) flags a small number of intersections above the high-leverage threshold ($2p/n = 0.003$) or with standardised deviance residuals near three, most notably JTE_ID 186267038 (largest residual) and 186268063 (highest leverage). These are retained rather than removed, as they represent genuinely high-crash locations rather than data errors.

**Figure G.2:** Baseline NB2 residual diagnostics

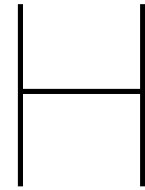
Spatial dependence.

Coordinates for all 4,474 modelled intersections were joined from the intersection geometry data, and a k -nearest-neighbour spatial weights matrix ($k = 5$, row-standardised) was constructed in projected coordinates (EPSG:28992). Moran's I on the deviance residuals was 0.038 for the baseline model and 0.036 for the enriched model, against an expectation under spatial randomness of essentially zero. Both were significant under a 999-permutation test ($p = 0.001$ in each case, the lowest attainable value at this permutation count). The magnitude of both statistics is small relative to the $[-1, 1]$ range Moran's I can take, indicating dependence that is detectable at this sample size rather than pronounced.

Because the test indicated dependence, the enriched model was additionally refitted with cluster-robust standard errors, grouping intersections into a 500 by 500 metre spatial grid (271 clusters, an average of 16.5 intersections per cluster), as a robustness check alongside the primary specification. The coefficient itself is unaffected by this change, since cluster-robust estimation adjusts only the standard error, not the point estimate. Table G.3 compares the two specifications. The standard error widens by about a fifth and the p -value moves further from conventional significance, confirming the marginal, non-significant result reported in Section 5.4.3.

Table G.3: CV-score coefficient under nonrobust and cluster-robust standard errors

Specification	β	SE	p
Nonrobust (primary)	-0.522	0.301	0.083
Cluster-robust (500m grid)	-0.522	0.366	0.154



Coefficient robustness and signal decomposition

H.1. Coefficient robustness detail

This section reports the full detail behind the robustness and direct-comparison checks summarised in Section 5.4.4 and Table 5.8. It includes the leave-one-out exclusion of each baseline predictor in turn, the measurement-error correction detail, and the full expert-scored-subset comparison.

Table H.1: Conditional CV-score coefficient with each baseline predictor excluded in turn

<i>Predictor excluded</i>	β	<i>IRR</i>	<i>95% CI</i>	<i>p</i>	ΔAIC
None (full model)	-0.52	0.59	[0.33, 1.07]	0.083	+1.01
Four-leg junction	-0.41	0.66	[0.36, 1.22]	0.187	-0.26
Log major AADT	+0.43	1.53	[0.85, 2.76]	0.154	+0.02
Log minor AADT	-0.48	0.62	[0.35, 1.12]	0.112	+0.54
50 km/h regime	-0.35	0.70	[0.39, 1.26]	0.234	-0.58
Signalised priority (VRI)	-0.45	0.64	[0.36, 1.14]	0.130	+0.29
Right-of-way priority	-0.54	0.58	[0.32, 1.05]	0.070	+1.30

This leave-one-out comparison spans all six baseline predictors, not only major-road AADT. Excluding any control other than major-road AADT leaves the coefficient negative and, in every case, short of conventional significance. The right-of-way exclusion comes closest ($p = 0.070$) without crossing the threshold. The sign reversal is therefore specific to major-road AADT, since no other single control, when removed, produces a positive or a significant coefficient. This rules out the possibility that the reversal reflects a general sensitivity to any one baseline term. The interpretation of this exception is developed in Section 5.5.2.

The measurement-error correction (SIMEX, quadratic extrapolation, $\sigma_\varepsilon = 0.18$, attenuation factor $\lambda = 0.49$) returns $\beta = -1.10$ against a naive correction of -1.07 . Both preserve the sign, but neither is treated as a point estimate, as the underlying conditional effect is itself fragile. σ_ε is the Phase 3 validation RMSE on the original Bradley-Terry score scale, applied directly to a predictor that was subsequently MinMax-rescaled before entering the crash model; on the rescaled predictor's own scale, the measurement error, and hence the correction, would be larger. The correction should therefore be read as conservative in magnitude, consistent with the direction-only interpretation adopted above.

Table H.2 reports the full expert-scored-subset comparison summarised in Section 5.4.4, giving both the marginal and incremental association for each assessment measure on the 145 expert-scored intersections.

Table H.2: Marginal versus incremental association for each assessment measure, on the expert-scored subset (n = 145)

Sample	Signal	Marginal IRR (p)	Incremental IRR (p)
Subset (145)	Human (raw BT)	5.53 (0.020)	1.06 (0.931)
Subset (145)	CV score	27.4 (<0.001)	0.18 (0.096)
Subset (145)	Residualised CV score	9.61 (0.003)	0.64 (0.538)

H.2. Signal decomposition detail

This section reports the full detail behind the signal-decomposition mechanism described in Section 5.5.2. It includes the quartile profile and partialling regression that establish the marginal-to-conditional shift, the sequential block-by-block build-up, the speed-stratum breakdown, the comparison between the full and volume-residualised CV score on the full network, and the independent interaction and PCA checks.

Table H.3 gives the full quartile breakdown behind the exposure-gradient reading in Section 5.5.2. Figure H.1 shows the coefficient moving from strongly positive to negative as the exposure controls are added in sequence.

Table H.3: Crash count and covariate profile by CV-score quartile

Quartile	Mean crashes	Share 50 km/h	Share priority	Median major AADT
Q1 (safest)	0.049	0.03	0.02	450
Q2	0.085	0.09	0.04	700
Q3	0.198	0.28	0.18	1.576
Q4 (most hazardous-looking)	0.614	0.79	0.54	7.000

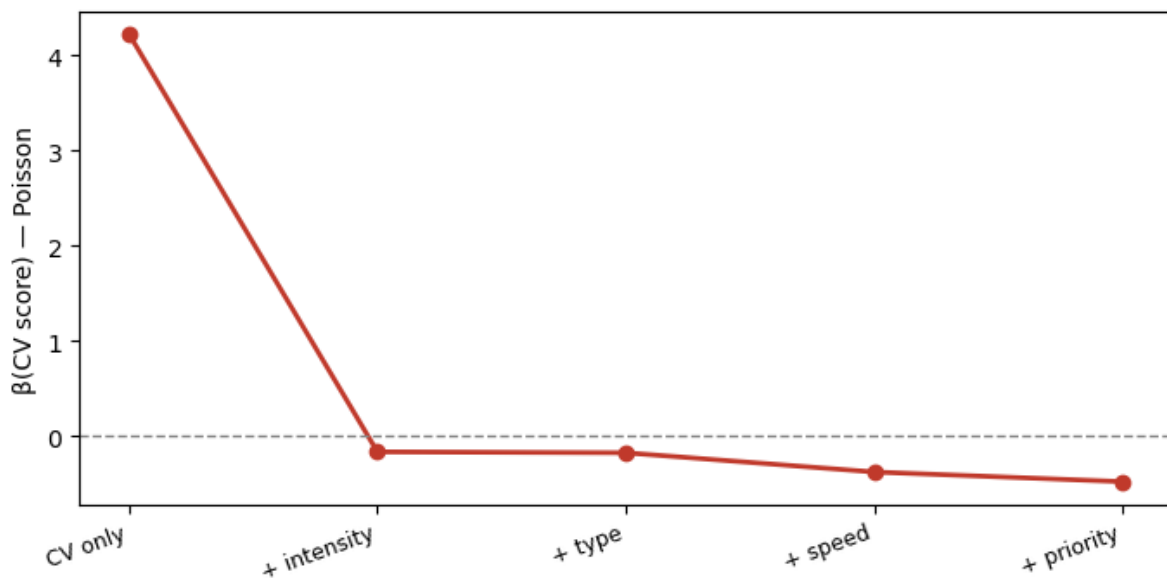


Figure H.1: CV-score coefficient (Poisson scale, chosen for numerical stability across the nested fits) as exposure and non-exposure controls are added sequentially, illustrating the sign reversal from marginal to conditional. The primary NB2 specification shows the same pattern (Table H.4).

The AADT-specific reversal established in Section H.1 raises a further question. A second check asks at which point in the baseline build-up the coefficient's magnitude diverges from zero. The conditional

coefficient was re-estimated as the same four predictor blocks used in Appendix G.2 were added sequentially (Table H.4). With no controls, the coefficient is strongly positive and highly significant (LR $p < 0.001$). Once the two exposure terms are included, it drops to small and statistically indistinguishable from zero. Adding the four-leg junction term shifts it further negative without reaching significance. It is only the subsequent addition of speed limit category that moves the coefficient most of the way to the value reported in the primary specification, reached once priority regulation completes the specification. Exposure alone therefore does not generate the negative coefficient. The shift in magnitude is driven predominantly by the inclusion of speed limit category, which is consistent with the speed-proxy reading developed in Section 5.5.2.

Table H.4: Conditional CV-score coefficient as baseline blocks are added sequentially

Block	k	β	LR	LR p	ΔAIC
None (marginal)	0	+4.96	413.00	<0.001	+411.00
+ Exposure	2	-0.10	0.12	0.732	-1.88
+ Four-leg junction	3	-0.21	0.53	0.466	-1.47
+ Speed	4	-0.43	2.08	0.149	+0.08
+ Priority (full)	6	-0.52	3.01	0.083	+1.01

Table H.5 gives the full speed-stratum breakdown behind the speed-channel finding in Section 5.5.2. The effect is significant only on the higher-speed stratum.

Table H.5: CV-score coefficient by speed stratum (NB2)

Stratum	n	β	IRR	95% CI	p
Full model	4.474	-0.52	0.59	[0.33, 1.07]	0.083
≤ 30 km/h	3.140	+0.05	1.05	[0.33, 3.32]	0.929
> 30 km/h	1.334	-0.74	0.48	[0.24, 0.94]	0.032

Table H.6: Marginal versus incremental association for the CV score and its volume-residualised version, on the full network (n = 4.474)

Sample	Signal	Marginal IRR (p)	Incremental IRR (p)
Full (4.474)	CV score	142.0 (<0.001)	0.59 (0.083)
Full (4.474)	Residualised CV score	24.4 (<0.001)	0.62 (0.073)

To test whether the conditional CV-score effect is uniform across the network, interaction terms between the CV score and each baseline predictor were added to the NB2 model.

Table 5.11, reported alongside Section 5.5.2 in the main text, gives the full interaction results. This is consistent with, and extends, the binary speed stratification reported in the same section and Table 5.8.

As an independent check on the suppression-as-collinearity reading, principal component analysis was applied jointly to the CV score and the two AADT terms, and the resulting orthogonal components were entered as NB2 predictors in place of the original variables. Table 5.12, reported alongside Section 5.5.2 in the main text, gives the component loadings and NB2 coefficients.

This decomposition is a robustness check on the suppression-as-collinearity reading, independent of the partialling and SIMEX approaches reported in Section 5.4.4 and Section 5.5.2. Because PC1 absorbs the large majority of shared variance between the CV score and traffic volume, the positive coefficients on PC2 and PC3, combined with the CV score's negative loading on both, are consistent with a CV score that is, net of the shared exposure axis, associated with marginally lower rather than higher crash risk.



CLS attention maps

Figure I.1 shows CLS-to-patch attention overlays for the eight intersections discussed in Section 5.5.3, including the two high-residual cases and the separate high-versus-low comparison set.

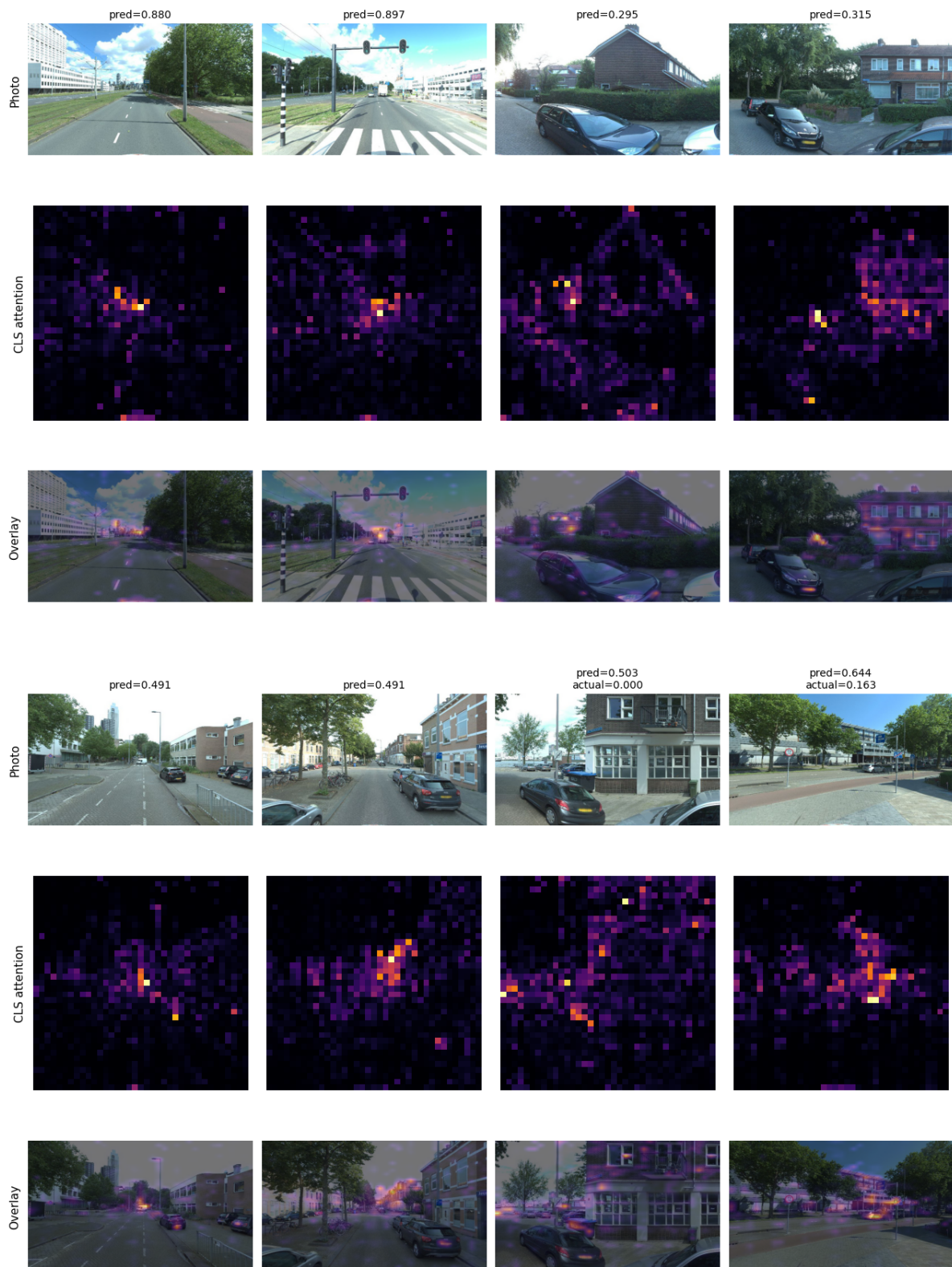


Figure I.1: CLS attention: stratified sample across score strata and high-residual cases. Backbone: DINOv2-ViT-S14



Expert validation session

Session guide and thematic summary. The session was an exploratory member check with a single traffic engineer of the Municipality of Rotterdam. It is reported to add face validity and practical grounding to the interpretation of the null result, not as a representative survey of expert opinion.

J.1. Session guide

The 60-minute session was semi-structured and organised in three parts. The prompts below served as a starting point for discussion rather than a fixed script.

Framing

- A short, plain-language recap of what was done: the elicitation of expert visual safety judgements, the computer vision model generalising them across the network, and their introduction into the crash model.

Photo round

- Four to six anchor photograph pairs were shown.
- How do you assess these situations? How complex is the situation? Does the road layout match its function and use? Is it understandable and clear to the road user?
- What do you look at, and does that coincide with traffic volume?
- How did you experience scoring the photographs?

Practice

- Do you reason from crash probability per vehicle, from crash counts per intersection, or both?
- Would a visual layer add anything to current network screening practice?

J.2. Thematic summary of session outcomes

Traffic volume entanglement

The engineer recognised the entanglement immediately. Risk is understood as a probability of an event and is therefore, in this reasoning, inherently coupled to exposure. The point was made that one always wants to relate danger to traffic volume, that a purely relative (per-exposure) view is conceptually possible but hard to apply, and that exposure cannot be equalised across the network because not every road can be brought to a 30 km/h regime.

Visual assessment cues: function, form, and use

The assessment was framed around the interplay of function (the role of the intersection in the network, such as a distributor road), form (how it is laid out), and use (how road users actually behave). Legibility

and overview were emphasised as central: whether the layout is self-explanatory, whether conflict points are well separated, and whether sight lines are adequate. Unsafe behaviour was described as difficult to anticipate from physical layout alone, and an oversized or underused intersection was noted to invite unsafe behaviour in its own right.

Comparing intersections beyond traffic volume

Comparing intersections on design alone, with traffic volume held aside, was described as comparing unlike cases. For a complex example, an entrance construction on a busy urban through-road, the engineer noted that a great deal must be assessed simultaneously and that video would be preferable to a still image. For a large intersection, the discussion centred on sight lines and conflict points. The recurring criterion was whether priority and sub-conflicts would be clear to every road user.

Crash counts versus per-vehicle risk

Current prioritisation was described as anchored in absolute recorded crashes, which foregrounds the busiest locations. Because little more can be done to reduce exposure at those sites, attention moves to the next tier of locations. A per-passage normalisation was regarded as conceptually defensible but difficult to apply in practice, consistent with the view that danger is inherently a matter of probability given exposure.

Practical usability for priority ranking

A CV visual layer was not expected to substantially change how intersections are prioritised for inspection, given that the visual impression largely tracks traffic volume. Potential value was noted for stakeholder communication and for screening locations that lack a usable crash history. The engineer observed that the municipal prioritisation target had deliberately been widened beyond recorded crashes, so that ranking already rests on a broader basis than crash history alone, and described the established workflow, examining what is happening at a flagged location, characterising the crash types, and then verifying on site through an inspection, as the standard against which any new layer would be judged.